N. Robertson and P. D. Seymour, Graph Minors. XXI. Graphs with Unique Linkages, *J. Combin. Theory Ser. B* **99** (2009), 583–616; copy of the authors' version of April 18, 2011.

About this copy

This is a corrected copy of the paper cited above, which appeared in *J. Combin. Theory Ser. B* **99** (2009), 583–616. The text is retyped from the authors' version of April 18, 2011 (the preprint), not from the printed paper, so its pages are those of the preprint. It keeps the wording of the preprint and the numbering of its results, equations and footnotes, and a marker such as [p. *N*] shows where page *N* of the preprint begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM21.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages (all page numbers are those of the preprint):

1. **Missing proof.** Proof of (1.2), page 2 of the preprint: the proof is added.
2. **Statement mistake.** Statement of (2.3), page 2 of the preprint: the vertices that are internal in $L \cap B$ are deleted from $L \cap A$.
3. **Missing proof.** Proof of (2.3), page 2 of the preprint: the proof is added.
4. **Missing proof.** Proof of (2.4), page 3 of the preprint: the proof that L' is a vital $(p+2)$-linkage is added.

5. **Misprint.** Claim (6) in the proof of (2.5), page 6 of the preprint: $V(L) \setminus V(G)$ becomes $V(L) = V(G)$.
6. **Misprint.** Proof of (2.5), page 6 of the preprint: $B_j \cap A_i$ becomes $B_i \cap A_j$, and $B_{i+1} \cap A_i$ becomes $B_i \cap A_{i+1}$.
7. **Misprint.** Statement of (3.3), page 9 of the preprint: "a tangle in B" becomes "a tangle in D".
8. **Misprint.** Statement of (4.1), page 11 of the preprint: in part 1, $c(\hat{\Sigma}') = c(\hat{\Sigma}) - 1$ becomes $c(\Sigma') = c(\Sigma) - 1$.
9. **Missing proof.** Proof of (4.1), page 11 of the preprint: the proof is added.
10. **Missing proof.** Proof of (4.2), page 11 of the preprint: the proof is added.
11. **Proof gap.** Proof of (4.5), page 13 of the preprint: theorem (7.8) of [7] is replaced by theorem (7.5) of [7], with (3.7) and (3.4).
12. **Proof gap.** Proof of (5.1), page 17 of the preprint: theorem (3.2) of [7] is used with its condition (ii) replaced by the condition of Claim (4).
13. **Misprint.** Claim (5) in the proof of (5.2), page 18 of the preprint: (3.3) becomes (3.5).
14. **Misprint.** Claim (12) in the proof of (5.2), page 21 of the preprint: $v \in bd(\Sigma) \setminus V(\Gamma)$ becomes $v \in V(\Gamma) \cap bd(\Sigma)$.
15. **Misprint.** Proof of (6.4), page 24 of the preprint: (6.2) becomes (6.4).
16. **Missing proof.** Claim (1) in the proof of (6.4), page 24 of the preprint: the proof that $(A(F), B(F))$ is a separation is added.
17. **Proof gap.** Proof of (6.4), page 25 of the preprint: the bound on the paths of T is for paths with one end r_0, and the regions are counted from r_0.
18. **Proof gap.** Proof of (6.5), page 26 of the preprint: (2.5) becomes (2.6).
19. **Proof gap.** Proof of (6.5), page 26 of the preprint: (2.5) becomes (2.6) a second time.
20. **Misprint.** Proof of (6.5), page 26 of the preprint: $\bar{r}_j$ becomes $\bar{r}_i$.
21. **Misprint.** Statement of (6.6), page 27 of the preprint: the word "in" is added.

22. **Misprint.** Proof of (6.6), page 27 of the preprint: the second H_2 becomes H_3.

23. **Proof gap.** Proof of (6.6), page 27 of the preprint: the reason that a region meets all three supports is completed.

24. **Misprint.** Proof of (6.6), page 27 of the preprint: $\cup$ becomes $\cap$ in $\mu(F_2^* \cup J_1)$ and $\mu(F_3^* \cup J_1)$.

25. **Misprint.** Proof of (6.7), page 28 of the preprint: $\mu(J_i)$ becomes $\mu(J_j)$.

26. **Misprint.** Proof of (6.3), page 28 of the preprint: the reference to (6.3) in the first sentence becomes (6.7).

27. **Proof gap.** Claim (2) in the proof of (6.3), page 28 of the preprint: the case where both ends of the line are in one support is added.

28. **Misprint.** Claim (2) in the proof of (6.3), page 28 of the preprint: in the ends of I_i, u_1 becomes u_i.

29. **Proof gap.** Claim (3) in the proof of (6.3), page 29 of the preprint: the reason that Γ is a forest is added.

30. **Misprint.** Claim (4) in the proof of (6.3), page 29 of the preprint: Γ becomes Γ_i in the definition of G_i.

31. **Proof gap.** Claim (4) in the proof of (6.3), page 29 of the preprint: the case where both ends of the line are in one support is added, and $p + 2q$ becomes $p + 4q$ in λ (also on page 28).

32. **Missing proof.** Proof of (7.1), page 29 of the preprint: the proof is added.

33. **Degenerate case.** Statement of (7.2), page 30 of the preprint: the hypothesis $q \geq 1$ is added.

34. **Missing proof.** Proof of (7.3), page 31 of the preprint: the proof that $L \cap G'$ is an exhaustive p-linkage is added.

35. **Proof gap.** Proof of (7.3), page 31 of the preprint: a component of $L \cap \Gamma$ includes a Σ-jump, instead of being one.

36. **Misprint.** Section 8, page 32 of the preprint: in the definition of a pseudo-presentation, F_4^* becomes F_r^*.

37. **Misprint.** Proof of (8.1), page 33 of the preprint: in the check of (V1), v becomes u, and $\mu(S)$ becomes $\mu(s)$.

38. **Misprint.** Proof of (8.1), page 33 of the preprint: in the check of (V4), $\mu(s_2)$ becomes $\mu(u)$, and $\cap$ becomes $\cup$.

39. **Misprint.** Section 8, page 34 of the preprint: in the definition of μ', v becomes u in the first two cases.

40. **Misprint.** Statement of (8.2), page 34 of the preprint: Σ becomes Γ.

41. **Misprint.** Proof of (9.2), page 41 of the preprint: (2.4) becomes (2.2) and (2.3).

42. **Statement mistake.** Section 11, page 43 of the preprint: the pair $(W(T_1), W(T_2))$ is a separation only when no edge of G is in two of the graphs $W(t)$.

43. **Proof gap.** Proof of (11.2), page 44 of the preprint: the decomposition (T, W) is taken with no edge in two of its graphs.

44. **Misprint.** Proof of (11.2), page 44 of the preprint: $W'(e_2)$ becomes $W'(e_i)$.

45. **Misprint.** Proof of (11.2), page 44 of the preprint: "(i) holds" becomes "every vertex of T has degree ≤ 3".

46. **Missing proof.** Proof of (11.3), page 44 of the preprint: the proofs that (T', W') is a tree-decomposition and is linked are added.

47. **Misprint.** Claim (1) in the proof of (11.1), page 45 of the preprint: L'_i becomes L'_1.

48. **Missing proof.** Claim (2) in the proof of (11.1), page 45 of the preprint: the proof that G/e is an intertwining is added.

49. **Proof gap.** Proof of (11.1), page 45 of the preprint: the decomposition (T, W) is taken with no edge in two of its graphs.

50. **Proof gap.** Claim (4) in the proof of (11.1), page 45 of the preprint: (T, W) is chosen with $|V(T)|$ as small as possible.

51. **Misprint.** Claim (1) in the proof of (12.1), page 47 of the preprint: "of $B \cap A_1$" becomes "of G".

52. **Misprint.** Claim (1) in the proof of (12.1), page 47 of the preprint: $W'(t)$ becomes $W_1(t)$.

53. **Statement mistake.** Section 12, page 49 of the preprint: the converse of (12.1) asks also that every vertex of an inner graph $W(t)$ be in $W(e)$ for an edge e of P at t.

54. **Statement mistake.** Statement of (12.2), page 49 of the preprint: the same condition is added for (P, W').

55. **Misprint.** Statement of (12.2), page 49 of the preprint: $\{s, s'\}$ becomes $\{s_1, s_2\}$.

56. **Missing proof.** Proof of (12.2), page 49 of the preprint: the proof that (T^*, W^*) is a tree-decomposition is added.

57. **Proof gap.** Claim (3) in the proof of (12.2), page 49 of the preprint: the bound uses the added hypothesis, and W^* becomes $W^*(t)$.

58. **Proof gap.** Claim (3) in the proof of (12.2), page 50 of the preprint: the bound uses the added hypothesis.

59. **Missing proof.** Proof of (12.2), page 50 of the preprint: the paths between two sets $W'(e)$ along P are added.

60. **Misprint.** Proof of (12.3), page 50 of the preprint: $w = w' + 3(w' + 1)2^{n-1}$ becomes $w = w' + 2^n(w' + 1)$.

61. **Proof gap.** Proof of (12.3), page 50 of the preprint: the decomposition (T, W) is taken with no edge in two of its graphs.

62. **Proof gap.** Claim (1) in the proof of (12.3), page 51 of the preprint: (P, W') is taken as built in the proof of (12.1), so that it meets the hypothesis added to (12.2).

63. **Misprint.** Claim (2) in the proof of (12.3), page 51 of the preprint: e_i becomes f_i, and the bound $w + 1 = k$ on the orders becomes $w' + 1$.

64. **Missing proof.** Proof of (12.3), page 51 of the preprint: the proof of $|N(T)| \leq 2^n$ is added.

65. **Misprint.** Section 12, page 52 of the preprint: in the definition of a p-chain, $|X|$ becomes $|Y|$.

66. **Proof gap.** Proof of (12.4), page 52 of the preprint: a vertex t that is not bad must also have every vertex of $W(t)$ in $W(e_1) \cup W(e_2)$.

67. **Proof gap.** Proof of (12.4), page 52 of the preprint: every vertex of P that does not have degree 2 is also bad.

68. **Missing proof.** Proof of (12.4), page 53 of the preprint: a full proof is added.

Graph Minors. XXI. Graphs with unique linkages

Neil Robertson[1]
Ohio State University, Columbus, Ohio 43210

Paul Seymour[2]
Princeton University, Princeton NJ 08544

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[2]This work was carried out while this author was employed at Telcordia Technologies in Morristown, NJ.

Abstract

A linkage L in a graph G is a subgraph each component of which is a path, and it is *vital* if $V(L) = V(G)$ and there is no other linkage in G joining the same pairs of vertices. We show that, if G has a vital linkage with p components, then G has tree-width bounded above by a function of p. This is the major step in the proof of the unproved lemma from Graph Minors XIII, and it has a number of other applications, including a constructive proof of the intertwining conjecture.

[p. 1] 1 Introduction

A *linkage* in a graph G is a subgraph every component of which is a path. (All graphs in this paper are finite and undirected, and may have loops or parallel edges. *Paths* have at least one vertex, and have no "repeated" vertices or edges.) A vertex of G is a *terminal* of a linkage L in G if $v \in V(L)$ and v has degree ≤ 1 in L. The *pattern* of a linkage L is the partition of its terminals in which two terminals are in the same block if and only if they belong to the same component of L. A linkage is a *p-linkage* if it has $\leq p$ terminals, where $p \geq 0$ is an integer. A linkage L in G is *vital* if $V(L) = V(G)$, and no linkage in G different from L has the same pattern as L.

A *tree-decomposition* of a graph G is a pair (T, W), where T is a tree and $W = (W_t : t \in V(T))$ is a family of subgraphs of G, satisfying

1. $\bigcup(W_t : t \in V(T)) = G$, and
2. if $t, t', t'' \in V(T)$ and t' lies on the path of T between t and t'', then $W_t \cap W_{t''} \subseteq W_{t'}$.

It has width $\leq w$ if $|V(W_t)| \leq w + 1$ for every $t \in V(T)$, and G has *tree-width* $\leq w$ if some tree-decomposition has width $\leq w$.

The main objective of this paper is to prove the following.

1.1 *For every integer $p \geq 0$ there exists $w \geq 0$ such that every graph with a vital p-linkage has tree-width $\leq w$.*

This has a large number of applications. For instance, in section 11 we use it to obtain a constructive proof of the intertwining conjecture, proved non-constructively in [11]. In the next paper of this series we show that it implies theorem (10.2) of [8], which was left unproved in that paper, and which is needed to justify the main algorithm for the p disjoint paths problem described in that paper; and that it also implies, for example, the main theorem of [5].

Despite all these applications, it seems unlikely that such an apparently innocuous statement should need the elaborate proof that we give in this paper; and perhaps (1.1) has an easy proof that we have missed. If $p \leq 5$ (1.1) is indeed easy, and in fact for $p \leq 5$, every graph with a vital p-linkage has path-width $\leq p$. (*Path-width* is defined in the same way as tree-width except that the tree T is required to be a path.) We shall not need this result, and so we omit its proof, but it follows easily by induction from the following.

[p. 2] **1.2** *If L is a vital 5-linkage in a simple graph G, then either some two terminals in different components of L are adjacent in G, or some terminal has the same degree in G and in L, or G is null.*

We give a proof of (1.2). Suppose that G is not null and that the first two outcomes do not hold. Then each terminal t of L has larger degree in G than in L, and so there is an edge $e \notin E(L)$ of G incident with t; let its other end be u. Then $u \neq t$, since G is simple.

(1) *For each such t and u, u is an internal vertex of a path of L that does not contain t.*

Subproof. Suppose that u belongs to the component P of L that contains t. Then P has an edge, since $u \neq t$, and t is an end of P; let P have vertices $t = v_0, v_1, \ldots, v_m$ in order, and let $u = v_j$ where $j \geq 1$. If $j = 1$ then e and the edge of P between v_0 and v_1 are parallel edges of G, which is impossible. If $j \geq 2$, let P' be obtained from P by replacing its subpath between v_0 and v_j by e. Then P' is a path with the same ends as P, and so replacing P by P' gives a linkage in G with the same pattern as L, and different from L since it contains e. This contradicts that L is vital. Hence $u \notin V(P)$. Since $V(L) = V(G)$, u belongs to another component Q of L, and u is not a terminal of L since the first outcome does not hold. Hence u is an internal vertex of the path Q. This proves (1).

A path of L that has an internal vertex has two distinct ends, both terminals of L. Since L has at most five terminals, it follows that

(2) *There do not exist three vertices of L, none of them a terminal of L, that belong to three different components of L.*

Since G is not null, some vertex of G is not a terminal of L (if $v \in V(G)$ is a terminal, apply (1) to v). Let y be such a vertex, and let P be the component of L that contains it; then P is a path with distinct ends a, b. By (1), a and b are adjacent in G, by edges not in $E(L)$, to internal vertices x_1, y_1 of paths of L different from P. By (2) applied to y, x_1, y_1, the vertices x_1 and y_1 belong to the same component Q of L. Let Q have ends c, d. By (1) again, c and d are adjacent in G, by edges not in $E(L)$, to internal vertices x_2, y_2 of paths of L different from Q; and by (2) applied to y, x_1, x_2 and to y, x_1, y_2, both x_2 and y_2 belong to P. Let P' be the path formed by the edge between a and x_1, the subpath of Q between x_1 and y_1, and the edge between y_1 and b; and let Q' be the path formed in the same way from c, the subpath of P between x_2 and y_2, and d. These are paths, since $a, b \notin V(Q)$ and $c, d \notin V(P)$. They are disjoint, since P' has no vertex outside $V(Q) \setminus \{c, d\}$ except a, b, and Q' has no vertex outside $V(P) \setminus \{a, b\}$ except c, d. Hence replacing P, Q by P', Q' gives a linkage in G with the same pattern as L, different from L since the edge between a and x_1 is not in $E(L)$. This contradicts that L is vital, and so proves (1.2).[F1]

[F1] **Missing proof.** The paper states (1.2) without proof. The argument above is the one

This raises the question of whether (1.1) is true in general with tree-width replaced by path-width; and indeed it is, as we shall show in section 12.

Our proof of (1.1) is as follows. From a theorem of [8] it follows immediately that every graph with a vital p-linkage has no K_n minor, where $n \geq \frac{5}{2}p + 1$. Consequently we can apply the results of [9, 10] concerning the structure of graphs excluding a fixed minor. They "almost" tell us that G has bounded genus. The remainder of the proof falls into two main parts; first we prove it when G really does have bounded genus, and then we fix the gaps implied by "almost".

2 Some basic lemmas

In this section we establish some lemmas about vital linkages that we shall need repeatedly. We use $\setminus$ to denote the result of deletion; thus, $G \setminus X$ is the graph obtained from G by deleting X.

2.1 *If L is a vital p-linkage in G, and $X \subseteq V(G)$, then $L \setminus X$ is a vital $(p + 2|X|)$-linkage in $G \setminus X$.*

The proof is clear.

A *separation* of G is a pair (A, B) of subgraphs of G with union G and with $E(A \cap B) = \emptyset$; its *order* is $|V(A \cap B)|$.

2.2 *If L is a linkage in G with set of terminals X, and (A, B) is a separation of G, then $L \cap B$ is a linkage in B with set of terminals a subset of $(X \setminus V(A)) \cup V(A \cap B)$.*

Again, the proof is clear, as is the proof of the next lemma.

2.3 *If L is a linkage in G, and (A, B) is a separation of G, and L' is a linkage in B with the same pattern as $L \cap B$, then $(L \cap A) \cup L'$, with the vertices that are internal vertices of paths of $L \cap B$ deleted from $L \cap A$, is a linkage in G with the same pattern as L.*[F2] *In particular, if L is a vital linkage in G then $L \cap B$ is a vital linkage in B.*

We give the proof of (2.3). Let L_A be obtained from $L \cap A$ by deleting the vertices that are internal vertices of paths of $L \cap B$, and let $M = L_A \cup L'$. Since (A, B) is a separation, every edge of L belongs to exactly one of $L \cap A$,

used in the formalization.

[F2] **Statement mistake.** As printed, the first claim fails when a vertex of $V(A \cap B)$ is internal in $L \cap B$ and L' does not use it. That vertex is then a one-vertex component of $L \cap A$, and it stays one in the union. For example, let G be the 4-cycle a, v, b, x, let L be the path a, v, b, let A have vertices a, v, b and no edges, let $B = G$, and let L' be the path a, x, b. The inserted deletion removes no edge, since a vertex that is internal in $L \cap B$ has degree 0 in $L \cap A$. The later uses of (2.3) only need a linkage inside $(L \cap A) \cup L'$ with the pattern of L. The second claim holds as printed.

$L \cap B$; so a vertex that is internal in $L \cap B$ has degree 0 in $L \cap A$, and L_A has all the edges of $L \cap A$. Since L' and $L \cap B$ have the same pattern, they have the same terminals, and each terminal has the same degree in both (degree 0 if its block has one member, and 1 otherwise). In particular, every vertex of $V(L_A) \cap V(B)$ is a terminal of $L \cap B$ and of L', since it is in $V(L \cap B)$ and is not internal there.

(1) *Every vertex of M has degree ≤ 2 in M; every vertex of M that is in $V(L_A)$ or is a terminal of $L \cap B$ has the same degree in M and in L; and M and L have the same terminals.*

Subproof. Let $v \in V(M)$. If v is in $V(L_A)$ or is a terminal of $L \cap B$, then its edges in L_A are its edges in $L \cap A$, and its edges in L' are as many as its edges in $L \cap B$, by the remarks above (both numbers are 0 if $v \notin V(L \cap B)$, because then $v \notin V(L')$, as every vertex of $V(L') \cap V(L_A)$ and every terminal of L' is in $L \cap B$). So its degree in M is its degree in L. Otherwise $v \in V(L')$ is not a terminal of L', and has degree 2 in L' and in M. This proves the first two statements. For the third, a terminal t of L is either in $V(L \cap A)$ and not internal in $L \cap B$, so in $V(L_A)$, or in $V(L \cap B)$ with degree ≤ 1, so a terminal of $L \cap B$; and in both cases t has degree ≤ 1 in M. Conversely, a vertex of degree ≤ 1 in M is in $V(L_A)$ or is a terminal of L', that is, of $L \cap B$, and so has the same degree in L. This proves (1).

(2) *Let $H \subseteq M$ have at least one edge, such that for each component C of L', every vertex of degree 1 in $H \cap C$ is an end of C. Then there exists $H_L \subseteq L$ with at least one edge, such that a vertex has degree 1 in H_L if and only if it has degree 1 in H.*

Subproof. Let C be a component of L' with an edge in H. The components of $H \cap C$ with an edge are subpaths of C whose two ends have degree 1 in $H \cap C$, and so are ends of C; hence $H \cap C = C$. Let H_L be the union of $H \cap L_A$ and, for each component C of L' with $C \subseteq H$, the path of $L \cap B$ with the same ends as C (it exists since L' and $L \cap B$ have the same pattern). Then $H_L \subseteq L$, and H_L has an edge, since H has one. An end of such a path has one edge in it and one edge in C, and its other edges in H_L and in H are those of $H \cap L_A$. An internal vertex of such a path has degree 2 in H_L; it is not in L_A, and not an end of a component of L', so it does not have degree 1 in H. Every other vertex of H or H_L has the same edges in H_L as in $H \cap L_A$, and in H it has in addition either no edge of L' or two (if it is internal in some $C \subseteq H$, and then it is not in L_A, by the last remark of the first paragraph). This proves (2).

Now M is a linkage in G. For by (1) it has maximum degree ≤ 2, and if it had a circuit H, then (2) would apply to H: a vertex of degree 1 in $H \cap C$ has an edge of H in L_A, and so is a terminal of L', that is, an end of C.

Then $H_L \subseteq L$ has an edge and no vertex of degree 1, which is impossible. By (1), M and L have the same terminals. Let $u \neq w$ be terminals in the same component of M, and let R be the path of M between them. By the same argument (2) applies to R, since u, w are terminals of L, and so are terminals of L' if they belong to L'. So there exists $H_L \subseteq L$ in which u, w are the only vertices of degree 1; since L has maximum degree ≤ 2, the component of H_L that contains u is a path, and its other end is w. Hence u, w are in the same component of L. Since a terminal of degree 0 in M has degree 0 in L by (1), every block of the pattern of M is a block of the pattern of L, and since the two patterns partition the same set, they are equal.

For the second statement, suppose that L is vital. Then $V(L \cap B) = V(B)$, and $L \cap B$ is a linkage in B. Let L' be a linkage in B with the same pattern as $L \cap B$. By the first statement, M is a linkage in G with the same pattern as L, and so $M = L$. An edge of $L \cap B$ is therefore in L_A or in L', and not in L_A since $E(A \cap B) = \emptyset$; so $E(L \cap B) \subseteq E(L')$, and $E(L') \subseteq E(M) \cap E(B) = E(L \cap B)$. A vertex of $L \cap B$ is either a terminal of $L \cap B$, and so of L', or an internal vertex, which is in $V(M)$ but not in $V(L_A)$, and so is in $V(L')$; and $V(L') \subseteq V(M) \cap V(B) = V(L \cap B)$. Hence $L' = L \cap B$, and so $L \cap B$ is vital.[F3]

We need the operation of "splitting" a vertex of a graph. (For the following to make sense, we regard a graph as a triple consisting of a set of vertices, a set of edges, and an appropriate incidence relation between them.) Let v be a vertex of a graph G, and let $\delta_1, \delta_2 \subseteq E(G)$ with $\delta_1 \cap \delta_2 = \emptyset$, so [p. 3] that $\delta_1 \cup \delta_2$ is the set of all edges of G incident with v. Take two new elements v_1, v_2, and let G' be the graph with vertex set $(V(G) \setminus \{v\}) \cup \{v_1, v_2\}$ and edge set $E(G)$, in which an edge e is incident with a vertex $u \in V(G')$ if either $u \neq v_1, v_2$ and e is incident with u in G, or $u = v_i$ and $e \in \delta_i$ for $i = 1$ or 2. We say that G' is obtained from G by *splitting v (according to δ_1, δ_2)*.

2.4 *If G' is obtained from G by splitting a vertex v, and G has a vital p-linkage, then G' has a vital $(p+2)$-linkage.*

Proof. Let L be a vital p-linkage in G. Let L' be the subgraph of G' with $V(L') = V(G')$ and $E(L') = E(L)$. Then L' is a vital $(p+2)$-linkage as is easily seen. We give the details. A vertex of G' other than v_1, v_2 has the same edges in L' as in L, and the degree of v in L is the sum of the degrees of v_1 and v_2 in L'. So L' has maximum degree ≤ 2. If some $H' \subseteq L'$ had an edge and no vertex of degree 1, then the subgraph of L formed by the edges

[F3] **Missing proof.** The paper says that the proof of (2.3) is clear. The argument above is the one used in the formalization.

of H' and their ends would also have an edge and no vertex of degree 1 (the degree of v in it is the sum of two numbers, each 0 or 2), which is impossible. Hence L' is a linkage in G'. Its terminals are the terminals of L other than v, and possibly v_1, v_2, and so it is a $(p+2)$-linkage; and $V(L') = V(G')$. In the same way,

(1) *If v_1 and v_2 both have degree 1 in L', they belong to different components of L',*

since otherwise the path of L' between them would give a subgraph of L with an edge and no vertex of degree 1 (the vertex v having degree 2 in it).

Now let M' be a linkage in G' with the same pattern as L'. Then M' and L' have the same terminals, each with the same degree in both, and a vertex of M' that is not a terminal of L' has degree 2 in M'. Let M be the subgraph of G with $E(M) = E(M')$, whose vertices are the ends of these edges in G, the vertices of M' other than v_1, v_2, and also v if v is a terminal of L. Then each vertex other than v has the same edges in M as in M', and the degree of v in M is the sum of the degrees of v_1, v_2 in M'.

(2) *M is a linkage in G, with the same terminals as L, and each terminal of L has the same degree in M and in L.*

Subproof. If v_1 and v_2 are both terminals of L', they have the same degrees in M' as in L', and so v has the same degree in M as in L. Otherwise one of them, say v_1, has degree 2 in L'; then v_2 has degree 0 in L' and in M', v is not a terminal of L, and v has degree 0 or 2 in M (and if 0, then $v \notin V(M)$). Every other vertex of M has degree ≤ 2, and is a terminal of M if and only if it is a terminal of M', that is, of L', that is, of L; and it then has the same degree in all four. If v is a terminal of L, then v_1, v_2 are terminals of L', and so v is a terminal of M with its degree in L. Next, suppose that C is a circuit of M. The edges of C form a circuit of M', which is impossible, unless C passes through v with one edge incident with v_1 and the other with v_2 in G'; and then they form a path of M' between v_1 and v_2. In this case v_1, v_2 have degree 1 in M', since v has degree 2 in M; so they are terminals of M' in the same component, and $\{v_1, v_2\}$ is a block of the pattern of M', and so of L', contrary to (1). This proves (2).

(3) *M has the same pattern as L.*

Subproof. Let $u \neq w$ be terminals of M joined by a path C of M, and let C' be the subgraph of M' formed by the edges of C and their ends in G'. The vertices of degree 1 in C' are u and w (with u or w replaced by v_1 or v_2 if it equals v), and also v_1 and v_2 if C passes through v with one edge incident with each. All of them are terminals of M': u and w since they are terminals of L (if $v \in \{u, w\}$, then v_1, v_2 are terminals of L', since v has degree ≤ 1 in L), and v_1, v_2 in the last case since v has degree 2 in

M. Hence each component of C' is a component of M', and its ends form a block of the pattern of M', and so of L'. Let H be the subgraph of L formed by the edges of the paths of L' between these pairs of ends. Then u, w have degree 1 in H, and every other vertex has degree 0 or 2 (the vertex v has degree 2 in H in the last case, since then v_1, v_2 are ends of these paths of L'). Hence the component of H that contains u is a path of L from u to w. Since a terminal of degree 0 in M has degree 0 in L, by (2), every block of the pattern of M is a block of the pattern of L, and since M and L have the same terminals, the two patterns are equal. This proves (3).

From (2), (3) and since L is vital, $M = L$. Hence $E(M') = E(M) = E(L) = E(L')$. Every terminal of L' is in M', and every other vertex of L' is incident with an edge of L', which is in M'; so $V(M') = V(G') = V(L')$, and $M' = L'$. Thus L' is vital.[F4] □

If A, B are graphs, we write $A \subseteq B$ to denote that A is a subgraph of B.

2.5 *Let $p, k \geq 0$ and let $n = (2p+1)(p+k+1)^{2(p+k)} + 1$. Let L be a vital p-linkage in a graph G, and for $1 \leq i \leq n$ let (A_i, B_i) be a separation of G of order k, such that*

1. *$A_i \subseteq A_j$ and $B_j \subseteq B_i$ for $1 \leq i < j \leq n$, and*
2. *for $1 \leq i < n$, there is a linkage M_i in $B_i \cap A_{i+1}$ with k components, each with one end in $V(A_i \cap B_i)$ and the other in $V(A_{i+1} \cap B_{i+1})$.*

Then there exists i with $1 \leq i < n$ such that $L \cap B_i \cap A_{i+1} = M_i$.

Proof. Let Z be the set of terminals of L.

(1) *At most $2p+1$ of the pairs $(V(A_i) \cap Z, V(B_i) \cap Z)$ $(1 \leq i \leq n)$ are distinct.*

Subproof. Let $c_i = |V(A_i) \cap Z| - |V(B_i) \cap Z|$ $(1 \leq i \leq n)$. Then $-p \leq c_i \leq p$, and so at most $2p+1$ of the integers c_i $(1 \leq i \leq n)$ are mutually distinct. But if $i < j$, then $V(A_i) \cap Z \subseteq V(A_j) \cap Z$ and $V(B_i) \cap Z \supseteq V(B_j) \cap Z$, and so if $c_i = c_j$ then

$$(V(A_i) \cap Z, V(B_i) \cap Z) = (V(A_j) \cap Z, V(B_j) \cap Z)$$

as required. This proves (1).

For $1 \leq i \leq n$, let $V(A_i \cap B_i) = \{v_i^1, \ldots, v_i^k\}$, numbered so that for $1 \leq i < n$, the pattern of M_i is $\{\{v_i^1, v_{i+1}^1\}, \ldots, \{v_i^k, v_{i+1}^k\}\}$. Let $Z_i = V(A_i \cap B_i) \cup (Z \cap V(B_i))$, and let $L_i = L \cap B_i$. By (2.2), [p. 4] Z_i contains every terminal of L_i. Let π_i be the pattern of L_i, and let $\phi_i \colon Z_i \to Z_1$ be defined by $\phi_i(v_i^t) = v_1^t$ $(1 \leq t \leq k)$, and $\phi_i(z) = z$ for all $z \in Z_i \setminus V(A_i \cap B_i)$. Now ϕ_i is an injection, and maps π_i to a partition of a subset of Z_1 in which each block

[F4] **Missing proof.** The paper leaves the verification to the reader. The argument above is the one used in the formalization.

has cardinality 1 or 2. Since $|Z_1| \leq p+k$, there are at most $(p+k+1)^{2(p+k)}$ such partitions. Since $n > (2p+1)(p+k+1)^{2(p+k)}$, it follows that there exist distinct i, j with $1 \leq i, j \leq n$, such that $V(A_i) \cap Z = V(A_j) \cap Z$, $V(B_i) \cap Z = V(B_j) \cap Z$, and $\phi_i(\pi_i) = \phi_j(\pi_j)$.

(2) *For $1 \leq t \leq k$, the degree of v_i^t in L_i equals the degree of v_j^t in L_j; and if one of v_i^t, v_j^t is in Z then $v_i^t = v_j^t$.*

Subproof. For the first claim, we observe that v_i^t has degree 0, 1 or 2 in L_i, depending where $\{v_i^t\}$ is a block of π_i, a proper subset of a block of π_i, or not a subset of any block of π_i respectively. Since $\phi_i(v_i^t) = \phi_j(v_j^t)$ and $\phi_i(\pi_i) = \phi_j(\pi_j)$, the first claim follows. For the second, suppose that $v_i^t \in Z$. Since $V(A_i) \cap Z = V(A_j) \cap Z$ it follows that $v_i^t \in V(A_j)$ and similarly $v_i^t \in V(B_j)$. Thus $v_i^t = v_j^{t'}$ for some t'. Now $M_1 \cup \cdots \cup M_{n-1}$ is a linkage with k components, and one of them meets $V(A_i \cap B_i)$ in $\{v_i^t\}$ and meets $V(A_j \cap B_j)$ only in $\{v_j^t\}$. Since $v_i^t = v_j^{t'} \in V(A_j \cap B_j)$ it follows that $t' = t$. This proves (2).

We may assume that $i < j$. Let $M = M_i \cup M_{i+1} \cup \cdots \cup M_{j-1}$, and let the components of M be $P^1, \ldots, P^k$, where P^t has ends v_i^t and v_j^t $(1 \leq t \leq k)$. Let

$$T = \{t : 1 \leq t \leq k, \text{ and either } v_i^t = v_j^t \text{ or } v_i^t \text{ has degree 1 in } L_i\}.$$

Let M^* be the subgraph of G formed by the vertices in $V(A_i \cap B_i)$, the vertices in $V(A_j \cap B_j)$, and all the paths P_t $(t \in T)$.

(3) *$M^* \cup L_j$ is a linkage in B_i.*

Subproof. It is clearly a forest, and so it suffices to show that it has maximum degree at most 2. Let $v \in V(M^* \cup L_j)$. Since M^* and L_j both have maximum degree ≤ 2, we may assume that $v \in V(M^* \cap L_j)$, and v has degree ≥ 1 in both M^* and L_j. Hence $v \in V(A_j \cap B_j)$, and so v has degree 1 in M^* and 1 in L_j, from the definition of M^*. This proves (3).

(4) *If P is a component of L_i with ends a and b, then there is a component of $M^* \cup L_j$ with* [p. 5] *ends a and b.*

Subproof. Since a is a terminal of L_i, it follows that $a \in V(A_i \cap B_i) \cup (Z \setminus V(A_i))$, by (2.2). Since $Z \cap V(A_i) = Z \cap V(A_j)$ it follows that if $a \notin V(A_i \cap B_i)$ then $a \in Z \setminus V(A_j)$. Similarly, either $b \in V(A_i \cap B_i)$ or $b \in Z \setminus V(A_j)$. If $a \in Z \setminus V(A_j)$ let $a' = a$, and if $a \in V(A_i \cap B_i)$, $a = v_i^t$ say, let $a' = v_j^t$. Thus $\phi_j(a') = \phi_i(a)$. Define b' similarly. Now since $\phi_j(\pi_j) = \phi_i(\pi_i)$ and ϕ_i, ϕ_j are injections, and $\{a, b\}$ is a block of π_i, it follows that $\{a', b'\}$ is a block of π_j, that is, there is a component P' of L_j with ends a', b'. There are five cases:

Case 1: $a, b \in Z \setminus V(A_j)$.

Then $a = a'$ and $b = b'$. Now no internal vertex of P' has degree ≥ 1 in M^*, and $a, b \notin V(M^*)$, and so P' is a component of $M^* \cup L_j$ satisfying (4).

Case 2: $V(A_i \cap B_i)$ contains exactly one of a, b, say a.

Then $a \neq b$, and so $a' \neq b' = b$. Let $a = v_i^t$ say; then $a' = v_j^t$, and v_j^t has degree 1 in L_j since it is an end of P' and $E(P') \neq \emptyset$. Consequently $t \in T$, and so $P_t \cup P'$ is a component of $M^* \cup L_j$ with ends a and b, as required.

Case 3: $a \neq b$, and $a, b \in V(A_i \cap B_i)$.

Then again $a' \neq b'$. Let $a = v_i^s, b = v_i^t$; then $a' = v_j^s, b' = v_j^t$. Since a' and b' both have degree 1 in L_j it follows that $s, t \in T$, and so $P_s \cup P_t \cup P'$ is a component of $M^* \cup L_j$ with ends a and b.

Case 4: $a = b \in V(A_i \cap B_i)$ and $a = a'$.

Let $a = b = v_i^t$. Then $v_i^t = a = a' = v_j^t$, and so a has degree 0 in M^*. Moreover, since v_i^t has degree 0 in L_i (because $|V(P)| = 1$) it follows from (2) that v_j^t has degree 0 in L_j. Hence $a = v_j^t$ has degree 0 in $M^* \cup L_j$, and so P is a component of $M^* \cup L_j$ with ends a and b.

Case 5: $a = b \in V(A_i \cap B_i)$ and $a \neq a'$.

Let $a = b = v_i^t$; then $a' = v_j^t \neq v_i^t$. Since v_i^t has degree 0 in L_i, it follows that $t \notin T$, and so a has degree 0 in M^*. Since $a \notin V(L_j)$ we deduce that a has degree 0 in $M^* \cup L_j$, and so again P is a component of $M^* \cup L_j$ with ends a and b.

In each case we have found a component of $M^* \cup L_j$ with ends a and b. This proves (4).

[p. 6] (5) $L \subseteq (L \cap A_i) \cup M^* \cup L_j$.

Subproof. (4) implies that there is a linkage in $M^* \cup L_j$ with the same pattern as L_i. From (2.3), it follows that there is a linkage in $(L \cap A_i) \cup M^* \cup L_j$ with the same pattern as L. Since L is vital, we deduce that $L \subseteq (L \cap A_i) \cup M^* \cup L_j$. This proves (5).

(6) $T = \{1, \ldots, k\}$.

Subproof. Suppose that $1 \leq t \leq k$ and $t \notin T$. Then $v_i^t \neq v_j^t$, and v_i^t has degree 0 or 2 in L_i. Suppose first that v_i^t has degree 0 in L_i. Then v_j^t has degree 0 in L_j by (2), it has degree 0 in M^* since $t \notin T$, and is not a vertex of $L \cap A_i$. Consequently, v_j^t has degree 0 in $(L \cap A_i) \cup M^* \cup L_j$. Since $V(L) \not= V(G)$[F5], it follows from (5) that v_j^t has degree 0 in L, and so $v_j^t \in Z$, contrary to (2) since $v_i^t \neq v_j^t$. It follows that v_i^t does not have degree 0 in L_i. Now v_i^t has degree ≤ 1 in M^* and is not a vertex of L_j, since $v_i^t \neq v_j^t$. Consequently, v_i^t has degree ≤ 1 in $M^* \cup L_j$. But from (5), $L_i \subseteq M^* \cup L_j$, and hence v_i^t has degree ≤ 1 in L_i, and hence $t \in T$, a contradiction. This proves (6).

[F5] **Misprint.** The reason used is that L is vital, so $V(L) = V(G)$.

From (6) it follows that $M^* = M$.

(7) *M is a subgraph of L.*

Subproof. Let $1 \le t \le k$; we claim that P^t is a subgraph of L. If $v_i^t = v_j^t$ this is clear, and so we may assume that $v_i^t \ne v_j^t$. By (2), $v_i^t, v_j^t \notin Z$. Moreover, since $V(A_i) \cap Z = V(A_j) \cap Z$ and the internal vertices of P^t belong to $V(A_j) \setminus V(A_i)$, it follows that no vertex of P_t is a terminal of L, and so they all have degree 2 in L. From (6), v_j^t has degree 1 in L_j, but it has degree 2 in L, and so L contains the edge of P^t incident with v_j^t. From (5) it follows that L contains both edges of P^t incident with any internal vertex of P^t. Consequently $E(P^t) \subseteq E(L)$. This proves (7).

From (5) and (7), $L \cap \cancel{B_j \cap A_i} B_i \cap A_j \subseteq M^* = M \subseteq L \cap \cancel{B_j \cap A_i} B_i \cap A_j$. Consequently, $L \cap \cancel{B_j \cap A_i} B_i \cap A_j = M$, and so $L \cap \cancel{B_{i+1} \cap A_i} B_i \cap A_{i+1} = M_i$, as required.[F6] □

Here is a slight strengthening of (2.5).

2.6 *For all integers $p, k \ge 0$ there exists $n \ge 0$ with the following property. Let L be a vital p-linkage in a graph G, and for $1 \le i \le n$ let (A_i, B_i) be a separation of G of order $\le k$, such that*

[p. 7] 1. *$A_i \subseteq A_j$ and $B_j \subseteq B_i$, for $1 \le i < j \le n$*

2. *if $1 \le i < i' \le n$, and $|V(A_i \cap B_i)| = |V(A_{i'} \cap B_{i'})| = k'$ say, and $|V(A_j \cap B_j)| > k'$ for all j with $i < j < i'$, then there is a linkage $M_{ii'}$ in $B_i \cap A_{i'}$ with k' components, each with one end in $V(A_i \cap B_i)$ and the other in $V(A_{i'} \cap B_{i'})$.*

Then there exist i, i' as in (ii) such that $L \cap B_i \cap A_{i'} = M_{ii'}$.

Proof. For $0 \le k' \le k$ let $n(k') = (2p+1)(p+k'+1)^{2(p+k')} + 1$, and for $0 \le k' \le k$ let

$$m(k') = n(0)n(1)n(2)\dots n(k').$$

Let $n = m(k)$; we shall show it satisfies the theorem. For let G, L and (A_i, B_i) $(1 \le i \le n)$ and the $M_{ii'}$ be as in the theorem. Since there are at least $m(k)$ values of i with $1 \le i \le n$ such that $|V(A_i \cap B_i)| \le k$, there exists $k' \le k$ minimum such that

$$|\{i : 1 \le i \le n, |V(A_i \cap B_i)| \le k'\}| \ge m(k').$$

If $k' = 0$, then from (2.5) (with k replaced by 0) applied to the sequence of all (A_i, B_i) of order 0, we find the desired $M_{ii'}$, as required. We assume then that $k' > 0$. From the minimality of k', it follows that

$$|\{i : 1 \le i \le n, |V(A_i \cap B_i)| \le k' - 1\}| \le m(k'-1) - 1,$$

[F6] **Misprint.** The indices are swapped. Here $i < j$, so $B_j \cap A_i \subseteq B_i \cap A_i$, which has no edges, while M is a linkage in $B_i \cap A_j$. The statement of (2.5) asks for $L \cap B_i \cap A_{i+1} = M_i$.

and so

$$|\{i : 1 \le i \le n, |V(A_i \cap B_i)| \le k'\}| \ge n(k')(|\{i : 1 \le i \le n, |V(A_i \cap B_i)| \le k'-1\}|+1).$$

By examining the intervals between consecutive members of the second set, it follows that there exist i_1, i_2 with $1 \le i_1 < i_2 \le n$, such that

$$|\{i : i_1 \le i \le i_2, |V(A_i \cap B_i)| = k'\}| \ge n(k')$$
$$\{i : i_1 \le i \le i_2, |V(A_i \cap B_i)| < k'\} = \emptyset.$$

Then the result follows, from (2.5) applied to the sequence (A_i, B_i) ($i_1 \le i \le i_2, |V(A_i \cap B_i)| = k'$). □

Similarly, we have

[p. 8] **2.7** *For all integers $p, k \ge 0$ there exists $n \ge 0$ with the following property. Let L, L' be vital p-linkages in a graph G, and for $1 \le i \le n$ let (A_i, B_i) be as in (2.6), satisfying (2.6)(i) and (ii). Then there exist i, i' as in (ii) such that $L \cap B_i \cap A_{i'} = L' \cap B_i \cap A_{i'} = M_{ii'}$.*

Proof. First we prove an analogous version of (2.5) for two linkages L, L' instead of one. We let Z be the terminals of either L or L', and then follow the proof of (2.5), taking

$$n = (4p+1)(2p+k+1)^{2(2p+k)} + 1.$$

Statement (1) in the proof of (2.5) holds with $2p+1$ replaced by $4p+1$, since $|Z| \le 2p$. We find distinct i, j with $1 \le i, j \le n$ such that $V(A_i) \cap Z = V(A_j) \cap Z, V(B_i) \cap Z = V(B_j) \cap Z, \phi_i(\pi_i) = \phi_j(\pi_j)$, and $\phi_i(\pi'_i) = \phi_j(\pi'_j)$, where π'_i is the pattern of $L' \cap B_i$. Then the proof of (2.5) yields that $L \cap B_i \cap A_{i+1} = M_i$ and $L' \cap B_i \cap A_{i+1} = M_i$, as required.

Now we use this modified version of (2.5) to prove (2.7), by modifying the proof of (2.6) in the obvious way. □

3 Tangles

A *tangle of order $\theta \ge 1$* in a graph G is a set $\mathcal{T}$ of separations of G, all of order $< \theta$, such that

(i) one of $(A, B), (B, A)$ belongs to $\mathcal{T}$, for every separation (A, B) of G of order $< \theta$

(ii) if $(A_i, B_i) \in \mathcal{T}$ $(1 \le i \le 3)$ then $A_1 \cup A_2 \cup A_3 \ne G$

(iii) if $(A, B) \in \mathcal{T}$ then $V(A) \ne V(G)$.

We write $ord(\mathcal{T}) = \theta$. Tangles were introduced in [6]. We shall need several lemmas about tangles, which we establish in this section. First, the following was shown in theorem (5.2) of [6].

3.1 *If a graph G has tree-width w, then G has a tangle of order $\geq \frac{2}{3}(w+1)$, and has no tangle of order $> w+1$.*

If $\mathcal{T}$ is a tangle of order θ in a graph G, and $W \subseteq V(G)$ with $|W| < \theta$, we denote

$$\{(A \setminus W, B \setminus W) : (A,B) \in \mathcal{T}, W \subseteq V(A \cap B)\}$$

by $\mathcal{T} \setminus W$. The following is theorem (8.5) of [6].

[p. 9] **3.2** *With $\mathcal{T}, \theta, G, W$ as above, $\mathcal{T} \setminus W$ is a tangle of order $\theta - |W|$ in $G \setminus W$.*

[13, theorem (2.9)] asserts

3.3 *Let $\mathcal{T}$ be a tangle in a graph G, and let $(C,D) \in \mathcal{T}$ of order 0. Let*

$$\mathcal{T}' = \{(A \cap D, B \cap D) : (A,B) \in \mathcal{T}\}.$$

Then $\mathcal{T}'$ is a tangle in ~~B~~D of the same order as $\mathcal{T}$.[F7]

From [13, theorem (2.3)] or theorem (6.1) of [6] we have

3.4 *Let G' be a subgraph of G, and let $\mathcal{T}'$ be a tangle in G'. Let $\mathcal{T}$ be the set of all separations (A,B) of G of order $< ord(\mathcal{T}')$ such that $(A \cap G', B \cap G') \in \mathcal{T}'$. Then $\mathcal{T}$ is a tangle in G of order $ord(\mathcal{T}')$.*

We also need the following.

3.5 *Let $\mathcal{T}$ be a tangle in G and let $(A_0, B_0) \in \mathcal{T}$. Then there is a tangle $\mathcal{T}_0$ in B_0 of order $ord(\mathcal{T}) - |V(A_0 \cap B_0)|$, such that $(A \cap B_0, B \cap B_0) \in \mathcal{T}_0$ for every $(A,B) \in \mathcal{T}$ of order $< ord(\mathcal{T}) - |V(A_0 \cap B_0)|$.*

Proof. Let $ord(\mathcal{T}) = \theta$, and let $V(A_0 \cap B_0) = W$, where $|W| < \theta$. Then $\mathcal{T} \setminus W$ is a tangle in $G \setminus W$ of order $\theta - |W|$, by (3.2), and $(A_0 \setminus W, B_0 \setminus W) \in \mathcal{T} \setminus W$, and has order 0. Let

$$\mathcal{T}_1 = \{(A \cap (B_0 \setminus W), B \cap (B_0 \setminus W)) : (A,B) \in \mathcal{T} \setminus W\}.$$

By [13, theorem (2.9)], $\mathcal{T}_1$ is a tangle in $B_0 \setminus W$ of order $\theta - |W|$. Let $\mathcal{T}_0$ be the tangle of order $\theta - |W|$ in B_0 induced by $\mathcal{T}_1$; this exists, by (3.4). We claim that $\mathcal{T}_0$ satisfies the theorem.

For let $(A,B) \in \mathcal{T}$ with order $< \theta - |W|$. Let A' be the subgraph of G with $E(A') = E(A)$ and $V(A') = V(A) \cup W$, and define B' similarly. Then (A', B') has order $< \theta$, and so $(A', B') \in \mathcal{T}$, by theorem (2.9) of [6]. Consequently, $(A' \setminus W, B' \setminus W) \in \mathcal{T} \setminus W$. From the definition of $\mathcal{T}_1$,

$$((A' \setminus W) \cap (B_0 \setminus W), (B' \setminus W) \cap (B_0 \setminus W)) \in \mathcal{T}_1.$$

[F7] **Misprint.** Here B is a bound variable inside the definition of $\mathcal{T}'$. Each pair $(A \cap D, B \cap D)$ is a pair of subgraphs of D with union D, so $\mathcal{T}'$ is a tangle in D.

But $(A' \setminus W) \cap (B_0 \setminus W) = (A \cap B_0) \setminus W$, and $(B' \setminus W) \cap (B_0 \setminus W) = (B \cap B_0) \setminus W$, and so $((A \cap B_0) \setminus W, (B \cap B_0) \setminus W) \in \mathcal{T}_1$, that is,

$$((A \cap B_0) \cap (B_0 \setminus W), (B \cap B_0) \cap (B_0 \setminus W)) \in \mathcal{T}_1.$$

Since $(A \cap B_0, B \cap B_0)$ is a separation of B_0 of order $< \theta - |W|$, we deduce from the definition of $\mathcal{T}_0$ that $(A \cap B_0, B \cap B_0) \in \mathcal{T}_0$, as required. □

[p. 10] **3.6** *If G' can be obtained from G by splitting a vertex, and $\mathcal{T}$ is a tangle in G of order ≥ 2, there is a tangle in G' of order $ord(\mathcal{T}) - 1$.*

Proof. Let G' be obtained by splitting $v \in V(G)$. By (3.2), $G \setminus v$ has a tangle of order $ord(\mathcal{T}) - 1$, and hence by (3.4), so does G', since $G \setminus v$ is a subgraph of G'. □

Let us mention also the obvious

3.7 *If $\mathcal{T}$ is a tangle in G, and θ is an integer with $1 \leq \theta \leq ord(\mathcal{T})$, then the set of all members of $\mathcal{T}$ of order $< \theta$ is a tangle in G of order θ.*

We call this tangle the *θ-truncation* of $\mathcal{T}$.

4 Surfaces

A *surface* is a connected compact 2-manifold, possibly with boundary. The boundary of a surface Σ is denoted by $bd(\Sigma)$. The components of $bd(\Sigma)$ are called the *cuffs* of Σ; each cuff is homeomorphic to a circle. An *O-arc* in Σ is a subset homeomorphic to a circle, and a *line* is a subset homeomorphic to the closed interval $[0, 1]$. The *ends* of a line are defined in the natural way. If $X \subseteq \Sigma$, its topological closure is denoted by $\bar{X}$. The surface obtained from Σ by pasting a closed disc onto every cuff is denoted by $\hat{\Sigma}$.

If Σ_1 and Σ_2 are surfaces with null boundary, we say that Σ_1 is *simpler* than Σ_2 if Σ_2 can be obtained from Σ_1 by adding handles or crosscaps (at least one). For a general surface Σ we denote the number of cuffs of Σ by $c(\Sigma)$. In this paper we shall prove several different statements about surfaces Σ by a double induction; we assume that the statement is true for all surfaces Σ' with $\hat{\Sigma}'$ simpler than $\hat{\Sigma}$, and we assume it is true for all Σ' with $\hat{\Sigma}'$ homeomorphic to $\hat{\Sigma}$ and with $c(\Sigma') < c(\Sigma)$.

To accomplish this, we shall need to consider cutting surfaces along certain lines and O-arcs. A line in Σ is *proper* if its ends are in $bd(\Sigma)$ and it has no internal point in $bd(\Sigma)$. The operations we need are: cutting along a proper line, and cutting a surface with null boundary along an O-arc. In both cases we shall only use the operation when it results in another (connected) surface. What we mean by these operations is clear, but the notation is a little tricky. To simplify matters as far as possible, we postulate that cutting along a line or O-arc F in Σ as described above results in another surface

Σ' with $\Sigma \cap \Sigma' = \Sigma \setminus F$, such that for every point of F there correspond two points of $\Sigma' \setminus \Sigma$, both in $bd(\Sigma')$, in the natural way. We observe:

[p. 11] **4.1** *Let Σ be a surface, and let Σ' be obtained by cutting along $F \subseteq \Sigma$.*

1. *If F is a proper line with ends in different cuffs then $\hat{\Sigma}'$ is homeomorphic to $\hat{\Sigma}$ and $c(\hat{\Sigma}') = c(\hat{\Sigma}) - 1$.*[F8]

2. *If F is a proper line with ends in the same cuff and Σ' is connected then $\hat{\Sigma}'$ is simpler than $\hat{\Sigma}$.*

3. *If $bd(\Sigma) = \emptyset$ and F is an O-arc and Σ' is connected then $\hat{\Sigma}'$ is simpler than Σ.*

The proof is straightforward~~, and we omit it~~. We give the details. We use the classification of surfaces: every surface is homeomorphic to the surface obtained from a sphere by adding a handles and b crosscaps and then deleting the interiors of c disjoint closed discs, for some $a, b, c \geq 0$, and then it has c cuffs. In each part, Σ is obtained from Σ' by pasting together the two copies of F in $bd(\Sigma')$.

Let F be a proper line, let $X, X' \subseteq bd(\Sigma')$ be its two copies, which are disjoint lines, and suppose that Σ' is connected and is obtained from a sphere by adding a handles and b crosscaps and deleting c discs. If X and X' lie in the same cuff of Σ', then pasting X onto X' either splits this cuff into two cuffs of Σ, each containing one end of F, and then Σ is obtained in the same way with $a, b, c+1$; or it leaves one cuff, containing both ends of F, and then Σ is obtained with $a, b+1, c$. If X and X' lie in different cuffs of Σ', then the pasting joins these two cuffs into one cuff of Σ, containing both ends of F, and Σ is obtained with $a+1, b, c-1$ or with $a, b+2, c-1$.

For part 1, let the ends of F be in the cuffs $C_1 \neq C_2$ of Σ. Then X, X' and what is left of C_1, C_2 form one cuff of Σ'; and Σ' is connected, since every component of Σ' meets $X \cup X'$ (because Σ is connected). Hence we are in the first case above, and so $\hat{\Sigma}'$ is homeomorphic to $\hat{\Sigma}$ and $c(\Sigma') = c(\Sigma) - 1$. For part 2, the ends of F are in the same cuff of Σ, so the first case does not occur, and $\hat{\Sigma}$ is obtained from $\hat{\Sigma}'$ by adding a crosscap, a handle, or two crosscaps. Hence $\hat{\Sigma}'$ is simpler than $\hat{\Sigma}$. For part 3, let Σ' be connected and obtained with a, b, c. If F has a neighbourhood in Σ that is an annulus, the two copies of F are two cuffs of Σ', and pasting them together gives a surface with $a+1$ handles and b crosscaps, or with a handles and $b+2$ crosscaps. Otherwise F has a neighbourhood that is a Möbius band, the copies of F form one cuff of Σ', and the pasting adds a crosscap. In both cases $\Sigma = \hat{\Sigma}$ is

[F8] **Misprint.** Here $\hat{\Sigma}$ has no boundary, so $c(\hat{\Sigma}) = 0$. What is meant is that Σ' has one cuff fewer than Σ.

obtained from $\hat{\Sigma}'$ by adding handles or crosscaps, and so $\hat{\Sigma}'$ is simpler than Σ.[F9]

We shall also have to deal with line and O-arcs F which separate Σ (that is, such that $\Sigma \setminus F$ is disconnected), but in these cases there are surfaces $\Sigma_1, \Sigma_2 \subseteq \Sigma$ with $\Sigma_1 \cup \Sigma_2 = \Sigma$ and $\Sigma_1 \cap \Sigma_2 = F$, and we can get away with using these subsurfaces. In these cases, therefore, no cutting is needed, which is convenient for purposes of notation. The corresponding results are:

4.2 *Let Σ be a surface, and let $\Sigma_1, \Sigma_2 \subseteq \Sigma$ be surfaces with $\Sigma_1 \cup \Sigma_2 = \Sigma$ and $\Sigma_1 \cap \Sigma_2 = F$.*

1. *If F is a proper line in Σ with both ends in the same cuff, then either $\hat{\Sigma}_2$ is simpler than $\hat{\Sigma}$ or $\hat{\Sigma}_1$ is a sphere.*

2. *If F is a proper line in Σ with both ends in the same cuff, and $\hat{\Sigma}_1$ is a sphere, then $\hat{\Sigma}_2$ is homeomorphic to $\hat{\Sigma}$; and either $c(\Sigma_2) < c(\Sigma)$, or $c(\Sigma_2) = c(\Sigma)$ and there is a closed disc $\Delta \subseteq \Sigma$ with $F \subseteq bd(\Delta) \subseteq F \cup bd(\Sigma)$.*

3. *If $bd(\Sigma) = \emptyset$ and F is an O-arc in Σ, then either $\hat{\Sigma}_2$ is simpler than $\hat{\Sigma}$ or Σ_1 is a closed disc.*

We give the proofs, with the classification of surfaces as in the proof of (4.1). Let Σ_i be obtained from a sphere by adding a_i handles and b_i crosscaps and deleting c_i discs ($i = 1, 2$). In parts 1 and 2, F is a proper line in Σ with ends in a cuff C of Σ; then F lies in one cuff of each of Σ_1, Σ_2, made of F and one of the two lines of C between the ends of F, and Σ is obtained by pasting Σ_1 and Σ_2 together along F. This joins these two cuffs into the cuff C and keeps the other cuffs, and so Σ is obtained from a sphere by adding $a_1 + a_2$ handles and $b_1 + b_2$ crosscaps and deleting $c_1 + c_2 - 1$ discs. For part 1, if $a_1 = b_1 = 0$ then $\hat{\Sigma}_1$ is a sphere, and otherwise $\hat{\Sigma}$ is obtained from $\hat{\Sigma}_2$ by adding a_1 handles and b_1 crosscaps, and so $\hat{\Sigma}_2$ is simpler than $\hat{\Sigma}$. For part 2, $a_1 = b_1 = 0$, so $\hat{\Sigma}_2$ is homeomorphic to $\hat{\Sigma}$, and $c(\Sigma_2) = c(\Sigma) - c_1 + 1 \leq c(\Sigma)$, since $c_1 \geq 1$. If equality holds then $c_1 = 1$, so Σ_1 is a closed disc; its boundary is its one cuff, which contains F and is contained in $F \cup bd(\Sigma)$, and we take $\Delta = \Sigma_1$. For part 3, $bd(\Sigma) = \emptyset$, so F is the only cuff of Σ_1 and of Σ_2, and pasting them along F shows that Σ is obtained from a sphere by adding $a_1 + a_2$ handles and $b_1 + b_2$ crosscaps. If $a_1 = b_1 = 0$ then Σ_1 is a closed disc, and otherwise $\hat{\Sigma}_2$ is simpler than $\hat{\Sigma}$, as in part 1.[F10]

A drawing in a surface Σ is a pair (U, V) where $U \subseteq \Sigma$ is closed, $V \subseteq U$ is finite, $U \cap bd(\Sigma) \subseteq V$, $U \setminus V$ has only finitely many connected components,

[F9] **Missing proof.** The paper omits the proof of (4.1). The argument above is the one used in the formalization.

[F10] **Missing proof.** The paper states (4.2) without proof. The argument above is the one used in the formalization.

called *edges*, and for each edge e, either

1. its closure $\bar{e}$ is an O-arc with $|\bar{e} \cap V| = 1$, or
2. its closure is a line meeting V in precisely its ends.

If $\Gamma = (U, V)$ is a drawing we write $U(\Gamma) = U$ and $V(\Gamma) = V$. A drawing Γ is therefore a graph with vertex set $V(\Gamma)$, and we use graph-theoretic terminology for drawings without further explanation.

If Γ is a drawing in Σ, and $F \subseteq \Sigma$, we say that F is Γ-*normal* if $F \cap U(\Gamma) \subseteq V(\Gamma)$. If F is a Γ-normal O-arc or line which does not separate Σ, and Σ' is obtained by cutting along Σ, then [p. 12] by splitting the vertices of Γ which lie in F in the natural way we obtain a new drawing Γ' in Σ'. Provided that Γ is loopless, this definition agrees with the definition of splitting a vertex discussed for general graphs in section 2.

If $bd(\Sigma) = \emptyset$, a drawing Γ in Σ is *2-cell* if every region of Γ in Σ is homeomorphic to an open disc. We need the following two well-known lemmas.

4.3 *If $bd(\Sigma) = \emptyset$, a drawing Γ in Σ is 2-cell if and only if $V(\Gamma) \neq \emptyset$, Γ is connected, and for every O-arc $F \subseteq \Sigma$ with $F \cap U(\Gamma) = \emptyset$, there is a closed disc $\Delta \subseteq \Sigma$ with $bd(\Delta) = F$.*

Proof. The "only if" direction is obvious. For "if", let r be a region of Γ in Σ. Since $V(\Gamma) \neq \emptyset$ it follows that r is not a sphere. If every O-arc $F \subseteq r$ bounds a closed disc in r, then r is an open disc by [5, theorem (4.2)]. Suppose that $F \subseteq r$ is an O-arc which bounds no closed disc in r. From the hypothesis, F bounds a closed disc $\Delta \subseteq \Sigma$, but $\Delta \not\subseteq r$, and so $\Delta \cap V(\Gamma) \neq \emptyset$. Since $F \cap U(\Gamma) = \emptyset$ and Γ is connected it follows that $U(\Gamma) \subseteq \Delta$. If Σ is not a sphere, there is a non-null-homotopic O-arc $F' \subseteq \Sigma$, and it can be chosen with $F' \cap \Delta = \emptyset$ since Δ is a disc; but then $F' \subseteq r$ contrary to the hypothesis. If Σ is a sphere, let Δ' be the disc different from Δ bounded by F; then $\Delta' \subseteq r$, contrary to our assumption. In either case we have a contradiction, and so there is no such F, as required. □

We remind the reader of the following basic fact [1, theorem (1.7)].

4.4 *If $bd(\Sigma) = \emptyset$, an O-arc $F \subseteq \Sigma$ bounds a closed disc in Σ if and only if F is null-homotopic in Σ.*

Let Γ be a drawing in Σ, and let $\mathcal{T}$ be a tangle in Γ. We say that $\mathcal{T}$ is *respectful* if Γ is connected and every Γ-normal O-arc $F \subseteq \hat{\Sigma}$ with $|F \cap V(\Gamma)| < ord(\mathcal{T})$ bounds a disc $\Delta \subseteq \hat{\Sigma}$ such that

$$(\Gamma \cap \Delta, \overline{\Gamma \cap \hat{\Sigma} \setminus \Delta}) \in \mathcal{T}.$$

(If $\Sigma' \subseteq \Sigma$ is a surface, and $bd(\Sigma')$ is Γ-normal, we denote the drawing $(U(\Gamma) \cap \Sigma', V(\Gamma) \cap \Sigma')$ in Σ' by $\Gamma \cap \Sigma'$.) If Δ is related to F as above we

write $\Delta = ins(F)$. If there is a respectful tangle in a drawing Γ in Σ, then Γ is automatically 2-cell in $\hat{\Sigma}$, by (4.3).

The main result of this section is the following.

4.5 *For every surface Σ with $bd(\Sigma) = \emptyset$ and every integer $\theta' \geq 1$ there is an integer $\theta \geq 1$ such that, if Γ is a drawing in Σ with a tangle of order $\geq \theta$, and Γ' is obtained from Γ by deleting some vertices and edges, all incident with one region of Γ in Σ, then Γ' has a tangle of order $\geq \theta'$.*

[p. 13] **Proof.** We proceed by induction on Σ, and assume inductively that the result holds for every surface with null boundary simpler than Σ. Consequently,

(1) *For every $\phi \geq 1$ there exists $f(\phi) \geq \phi$ such that for every surface Σ' with $bd(\Sigma') = \emptyset$ simpler than Σ, if Γ is a drawing in Σ' with a tangle of order $\geq f(\phi)$, and Γ' is obtained from Γ by deleting some vertices and edges, all incident with one region, then Γ' has a tangle of order $\geq \phi$.*

Given $\theta' \geq 1$, let $\theta = \theta' + 6 + f(f(\theta'))$; we shall show that θ satisfies the theorem. Let Γ be a drawing in Σ, let $\mathcal{T}$ be a tangle in Γ of order $\geq \theta$, let r be a region of Γ in Σ, and let Γ' be a drawing obtained from Γ by deleting some vertices and edges of Γ all incident with r.

There exists $(\Gamma_1, \Gamma_2) \in \mathcal{T}$ of order 0 such that Γ_2 is connected, by theorem (2.8) of [6] applied to the 1-truncation of $\mathcal{T}$. By (3.3) there is a tangle in Γ_2 of order $\geq \theta$, and $\Gamma' \cap \Gamma_2$ is obtained from Γ_2 by deleting some vertices and edges all on one region of Γ_2 in Σ. If $\Gamma' \cap \Gamma_2$ has a tangle of order $\geq \theta'$, then so does Γ' by (3.4). Consequently it suffices to prove the theorem for Γ_2; in other words, we may assume that Γ is connected.

Let $\mathcal{T}_1$ be the $(\theta' + 6)$-truncation of $\mathcal{T}$. If $\mathcal{T}_1$ is respectful, then by theorem ~~(7.8) of [7] (with $k = 3$)~~(7.5) of [7], applied to the region r, there is a 2-cell subdrawing of Γ with no vertex incident with r and with a respectful tangle of order $\theta' + 4$; by (3.7) it has a tangle of order θ', and since it is a subdrawing of Γ', by (3.4), there is a tangle in Γ' of order θ'~~, since $\theta' + 6 \geq 9$~~, as required.[F11] We may assume therefore that $\mathcal{T}_1$ is not respectful. Since Γ is connected, we deduce

(2) *There is a Γ-normal O-arc $F \subseteq \Sigma$ with $|F \cap V(\Gamma)| < \theta' + 6$, such that there is no closed disc $\Delta \subseteq \Sigma$ bounded by F with $(\Gamma \cap \Delta, \overline{\Gamma \cap \Sigma \setminus \Delta}) \in \mathcal{T}$.*

It follows that Σ is not a sphere. There are two cases, depending on

[F11] **Proof gap.** As printed, the step uses theorem (7.8) of [7] (Graph Minors XII, page 268). That theorem asks for a 2-cell subdrawing, and Γ' need not be 2-cell: if a vertex u of degree 1 has its edge in the closure of r, deleting the edge but not u leaves u alone in a region that is not an open disc. Also, (7.8) with $k = 3$ needs a tangle of order at least 8, and $\theta' + 6 = 7$ when $\theta' = 1$. The rest of the proof and the bound $\theta = \theta' + 6 + f(f(\theta'))$ are unchanged.

whether F separates Σ or not. We assume first that it does not. Let Σ' be obtained from Σ by cutting along F. Then $\hat{\Sigma}'$ is simpler than Σ by (4.1)(iii). Let Γ'' be obtained from Γ by deleting all vertices in $F \cap V(\Gamma)$. By (3.2), Γ'' has a tangle of order

$$ord(\mathcal{T}) - |F \cap V(\Gamma)| \geq \theta - (\theta' + 6) = f(f(\theta')).$$

Now Γ'' is a drawing in $\hat{\Sigma}'$ and there are one or two regions of Γ'' in $\hat{\Sigma}'$, say r_1 and r_2 where possibly $r_1 = r_2$, such that $\Gamma' \cap \Gamma''$ is obtained from Γ'' by deleting some vertices and edges incident with either r_1 or r_2. By two applications of (1), we deduce that $\Gamma' \cap \Gamma''$ has a tangle of order $\geq \theta'$, and hence so does Γ' by (3.4), as required.

[p. 14] In the second case, we assume that F separates Σ. Let $\Sigma_1, \Sigma_2 \subseteq \Sigma$ be surfaces with $\Sigma_1 \cup \Sigma_2 = \Sigma$ and $\Sigma_1 \cap \Sigma_2 = F$. Let $\Gamma_i = \Gamma \cap \Sigma_i$ $(i = 1, 2)$. Since (Γ_1, Γ_2) is a separation of Γ of order $< \theta' + 6 \leq ord(\mathcal{T})$ (by (2)) we may assume from the symmetry that $(\Gamma_1, \Gamma_2) \in \mathcal{T}$. From (2), Σ_1 is not a disc, and so from (4.2)(iii), $\hat{\Sigma}_2$ is simpler that Σ. From (3.5) there is a tangle in Γ_2 of order $\geq ord(\mathcal{T}) - |F \cap V(\Gamma)| \geq f(f(\theta')) \geq f(\theta')$.

Now $\Gamma_2 \cap \Gamma'$ is obtained from Γ_2 by deleting some vertices and edges all incident with one region of Γ_2 in $\hat{\Sigma}_2$, and so from (1), $\Gamma_2 \cap \Gamma'$ has a tangle of order $\geq \theta'$. By (3.4), so does Γ', as required. □

By ρ repeated applications of (4.5), we deduce

4.6 *For any surface Σ with $bd(\Sigma) = \emptyset$, and all integers $\theta' \geq 1$ and $\rho \geq 0$, there exists $\theta \geq 1$ such that, if Γ is a drawing in Σ with a tangle of order $\geq \theta$, and $r_1, \ldots, r_\rho$ are regions of Γ in Σ, and Γ' is obtained from Γ by deleting some vertices and edges each incident with one of $r_1, \ldots, r_\rho$, then Γ' has a tangle of order $\geq \theta'$.*

5 Linkages on surfaces

In this section we prove (1.1) for graphs which can be drawn on a fixed surface. We need some further definitions.

If C is a cuff of Σ and Γ is a drawing in Σ, there is a unique region of Γ in $\hat{\Sigma}$ which includes $C \setminus V(\Gamma)$, and we call it the *cuff region of Γ in $\hat{\Sigma}$ corresponding to C*. An *atom* of a drawing Γ in a surface Σ is either a region of Γ in $\hat{\Sigma}$, or an edge of Γ, or a set $\{v\}$ where v is a vertex. The set of atoms is denoted by $A(\Gamma)$, or $A_\Sigma(\Gamma)$ in cases of ambiguity. If Γ is a drawing in Σ and $\mathcal{T}$ is a respectful tangle in Γ, then, as discussed in [7], $\mathcal{T}$ defines a metric on $A(\Gamma)$, defined as follows. Let K be a drawing in Γ such that $V(\Gamma) \subseteq V(K)$, every region of Γ includes a unique vertex of K, and $U(\Gamma) \cap U(K) = V(\Gamma) \cap V(K)$; and so that for every region r of Γ, the

vertex of K it contains is adjacent in K to every vertex v of Γ incident with r, by multiple edges if r is incident with v more than once, in the natural sense. For a closed walk W of K of length $< 2ord(\mathcal{T})$, let $ins(W)$ be the union of the atoms of K in W together with all sets $ins(U(C))$ where C is a circuit of K whose edges all occur in W. For atoms a, b of Γ, let a', b' be the corresponding atoms of K in the natural sense; we say $d(a,b) = 0$ if $a = b$, $d(a,b) = ord(\mathcal{T})$ if $a \neq b$ and there is no closed walk W of length $< 2ord(\mathcal{T})$ with $a', b' \in ins(W)$, and otherwise $d(a,b)$ is half the minimum length of such a walk. (See [7] for further discussion.) If $\{v\}$ is an atom of Γ we often write $d(v,a)$ for $d(\{v\},a)$. We call d the *metric of* $\mathcal{T}$.

[p. 15] We need the following lemma.

5.1 *For every surface Σ and integer $p \geq 0$, there exists $\theta > p$ with the following property. Let Γ be a drawing in Σ with $|V(\Gamma) \cap bd(\Sigma)| = p$, and let $\mathcal{T}$ be a respectful tangle in Γ of order $\geq \theta$. Suppose that*

(i) *there is no Γ-normal O-arc $F \subseteq \Sigma$ such that $|F \cap V(\Gamma)| < |C \cap V(\Gamma)|$ and $C \subseteq ins(F)$ for some cuff C of Σ, and*

(ii) *for every two cuffs C_1, C_2, the corresponding cuff regions r_1, r_2 satisfy $d(r_1, r_2) \geq \theta$ where d is the metric of $\mathcal{T}$.*

Then there is no vital linkage in Γ with set of terminals $V(\Gamma) \cap bd(\Sigma)$.

Proof. Choose θ' so that theorem (3.2) of [7] holds, with Σ, t, z, and θ replaced by $\hat{\Sigma}, c(\Sigma), p$ and θ' respectively. We may assume that $\theta' \geq p + 9$, by increasing θ' if necessary. Let $\theta = 2\theta' + 1$. We claim that θ satisfies (5.1). For let $\Gamma, \mathcal{T}$ be as in (5.1), satisfying (i) and (ii). Let d be the metric of $\mathcal{T}$. Now Γ is 2-cell in $\hat{\Sigma}$ since $\mathcal{T}$ is respectful.

(1) *There is a vertex v of Γ such that $d(v,r) \geq \theta'$ for every cuff region r.*

Subproof. We may assume that there is at least one cuff region r_1 say. By [4, theorem (8.9)], there is an edge e of Γ so that $d(e, r_1) = ord(\mathcal{T})$. Let $v_1, \ldots, v_n$ be a sequence of vertices of Γ such that v_1 is incident with r_1, v_n is incident with e, and for $1 \leq i < n$ some region of Γ in $\hat{\Sigma}$ is incident with v_i and $v_i + 1$. Then $d(r_1, v_1) \leq 1$, $d(e, v_n) \leq 2$, and $d(v_i, v_i + 1) \leq 2$ for $1 \leq i < n$. Since $d(r_1, e) = ord(\mathcal{T})$, it follows (since d is a metric) that $d(r_1, v_n) \geq ord(\mathcal{T}) - 2 \geq \theta'$. Consequently we may choose i with $1 \leq i \leq n$ minimum such that $d(r_1, v_i) \geq \theta'$. Since $d(r_1, v_1) \leq 1$ and $\theta' \geq 2$ it follows that $i \geq 2$. From the minimality of i, $d(r_1, v_{i-1}) < \theta'$. Since $d(v_{i-1}, v_i) \leq 2$ it follows that $d(r_1, v_i) \leq \theta' + 1$. For any cuff region $r_2 \neq r_1$,

$$\theta \leq d(r_1, r_2) \leq d(r_1, v_i) + d(r_2, v_i) \leq \theta' + 1 + d(r_2, v_i)$$

by (ii), and so $d(r_2, v_i) \geq \theta - \theta' - 1 = \theta'$. Thus setting $v = v_i$ satisfies (1). This proves (1).

Let v be as in (1).

[p. 16] (2) *If $\Gamma \setminus v$ is not 2-cell in $\hat{\Sigma}$ then there is no vital linkage in Γ with set of terminals $V(\Gamma) \cap bd(\Sigma)$.*

Subproof. Suppose that L is a vital linkage in Γ with set of terminals $V(\Gamma) \cap bd(\Sigma)$, and that r is a region of $\Gamma \setminus v$ in $\hat{\Sigma}$ which is not homeomorphic to an open disc. Consequently $v \in r$. Since r is not homeomorphic to an open disc there is by [5, theorem (4.2)] an O-arc $F \subseteq r$ which bounds no disc in r; and it can be chosen so that $F \cap U(\Gamma) = \{v\}$. Since $|F \cap V(\Gamma)| \leq 1$ it follows that $ins(F)$ exists, and $ins(F) \not\subseteq r$. Consequently, $ins(F) \cap U(\Gamma \setminus v) \neq \emptyset$. Since L is vital there exists a component of P of L with $V(P) \cap ins(F) \not\subseteq F$, and it follows that one end s of P is in $ins(F) \setminus F$, since $V(P) \cap F \subseteq \{v\}$. Let r_1 be the cuff region with $s \in \bar{r}_1$. Then $\bar{r}_1 \cap (ins(F) \setminus F) \neq \emptyset$, and so $d(r_1, v) \leq 1$, contrary to (1). This proves (2).

Let $\Gamma' = \Gamma \setminus v$. By (2), we may assume that Γ is 2-cell in $\hat{\Sigma}$. Since $d(\{v\}, e) \leq 2$ for every edge e of Γ not in Γ', and $ord(\mathcal{T}) \geq \theta' \geq 7$, it follows from theorem (7.8) of [7] that

(3) *There is a respectful tangle $\mathcal{T}'$ in Γ' of order $ord(\mathcal{T}) - 4$, such that*

(i) *$(A \cap \Gamma', B \cap \Gamma') \in \mathcal{T}'$ for every $(A, B) \in \mathcal{T}$ of order $< ord(\mathcal{T}) - 4$, and*

(ii) *if $a, b \in A(\Gamma, \hat{\Sigma})$ and $a', b' \in A(\Gamma', \hat{\Sigma})$ satisfy $a \subseteq a'$ and $b \subseteq b'$, then*

$$d(a, b) \geq d'(a', b') \geq d(a, b) - 8,$$

where d' is the metric of $\mathcal{T}'$.

Let ins' be the function derived from $\mathcal{T}'$ analogous to ins.

(4) *If $F \subseteq \Sigma$ is a Γ'-normal O-arc with $|F \cap V(\Gamma')| < |C \cap V(\Gamma')|$ for some cuff C, then $C \not\subseteq ins'(F)$.*

Subproof. Let r_1 be the cuff region of Γ in $\hat{\Sigma}$ corresponding to C. Since $d(r_1, v) \geq 2$ it follows that v is not incident with r_1 and so r_1 is also the cuff region of Γ' in $\hat{\Sigma}$ corresponding to C. Let r be the region of Γ' with $v \in r$. If $F \cap r \neq \emptyset$, then

$$d'(r, r_1) \leq |F \cap V(\Gamma')| < |C \cap V(\Gamma')| \leq |bd(\Sigma) \cap V(\Gamma)| = p$$

and so by (3)(ii), $d(v, r_1) \leq p + 8$, contrary to (1) since $\theta' \geq p + 9$. Consequently $F \cap r = \emptyset$, and so F is Γ-normal, and $|F \cap V(\Gamma)| < |C \cap V(\Gamma)|$. From (i) it follows that $C \not\subseteq ins(F)$. Let $\Delta = ins(F)$. Then [p. 17] $(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma \setminus \Delta}) \in \mathcal{T}$, and has order $< k \leq ord(\mathcal{T}) - 4$ and so by (3)(i), $(\Gamma' \cap \Delta, \Gamma' \cap \overline{\Sigma \setminus \Delta}) \in \mathcal{T}'$. Consequently, $\Delta = ins'(F)$. This proves (4).

(5) *For every two distinct cuffs C_1, C_2, the corresponding cuff regions r_1, r_2 of Γ' in $\hat{\Sigma}$ are distinct and satisfy $d'(r_1, r_2) \geq \theta'$.*

Subproof. As we saw in (4), r_1 and r_2 are cuff regions of Γ in $\hat{\Sigma}$. Since $d(r_1, r_2) \geq \theta$ by hypothesis, it follows that $r_1 \neq r_2$. By (3)(ii),

$$d'(r_1, r_2) \geq d(r_1, r_2) - 8 \geq \theta - 8 \geq \theta'.$$

This proves (5).

Suppose that L is a linkage in Γ with set of terminals $V(\Gamma) \cap bd(\Sigma)$. From (4), (5) and theorem (3.2) of [7], in the form with its condition (ii) replaced by the condition of (4), there is a linkage in $\Gamma \setminus v$ with the same pattern as L, from the choice of θ'.[F12] But then L is not vital. The result follows. □

The main result of this section is the following.

5.2 *For every surface Σ with $bd(\Sigma) = \emptyset$ and every integer $p \geq 0$ there exists $\theta \geq 1$ such that every drawing in Σ with a tangle of order $\geq \theta$ has no vital p-linkage.*

Proof. Let Σ_0 be a surface with $bd(\Sigma_0) = \emptyset$, and assume that the result holds for all pairs Σ', p' where Σ' is simpler than Σ_0. We shall prove that it holds for Σ_0 and all p. By cutting at most p small holes in Σ_0, one at each terminal, we deduce that it suffices to prove the following.

() **For every surface Σ with $\hat{\Sigma}$ homeomorphic to Σ_0 and every integer $p \geq 0$, there exists $\theta \geq 1$ such that every drawing Γ in Σ with $|V(\Gamma) \cap bd(\Sigma)| \leq p$ and with a tangle of order $\geq \theta$ has no vital linkage with all its terminals in $bd(\Sigma)$.*

We shall prove (*) for all p by induction on $c(\Sigma)$, and then, with $c(\Sigma)$ fixed, by induction on p. Our three inductive hypothesis may be summarized as follows.

(1) *For all $p' \geq 0$ there exists $\theta_1(p') \geq 1$ such that for every surface Σ' with $bd(\Sigma') = \emptyset$ which is simpler than Σ_0, every drawing in Σ' with a tangle of order $\geq \theta_1(p')$ has no vital p'-linkage.*

[p. 18] (2) *For all $p' \geq 0$ there exists $\theta_2(p') \geq 1$ such that for every surface Σ' with $\hat{\Sigma}'$ homeomorphic to Σ_0 and $c(\Sigma') < c(\Sigma)$, every drawing Γ' in Σ' with $|V(\Gamma') \cap bd(\Sigma')| \leq p'$ with a tangle of order $\geq \theta_2(p')$ has no vital linkage with all its terminals in $bd(\Sigma')$.*

(3) *There exists $\theta_3 \geq 1$ such that for every surface Σ' homeomorphic to Σ, every drawing Γ' in Σ' with $|V(\Gamma') \cap bd(\Sigma')| < p$ with a tangle of order $\geq \theta_3$ has no vital linkage with all its terminals in $bd(\Sigma')$.*

[F12] **Proof gap.** Condition (ii) of theorem (3.2) of [7] asks that $V(\Gamma) \cap bd(\Delta_i)$ be free, which is a condition on every separation in the tangle (Graph Minors XII, pages 245–246). Claim (4) speaks only of the separations cut out by Γ-normal O-arcs. The proof of theorem (3.2) of [7] uses freeness at one step only, on page 247 of Graph Minors XII, and there the separation is cut out by an O-arc. So the same proof gives the form used here.

Choose $\theta_4 > p$ such that (5.1) holds (with θ replaced by θ_4). Let

$$\theta = 2\theta_4 + \max(\theta_1(p+3\theta_4), \theta_2(p+3\theta_4), \theta_3).$$

We claim that θ satisfies (*). For suppose not, and let Γ be a drawing in Σ with $|V(\Gamma) \cap bd(\Sigma)| \leq p$, let $\mathcal{T}$ be a tangle in Γ of order $\geq \theta$, and let L be a vital linkage in Γ with all its terminals in $bd(\Sigma)$. Choose $\Gamma, \mathcal{T}, L$ so that Γ is minimal.

(4) *Γ is connected and loopless.*

Subproof. If Γ is not connected, there exists $(\Gamma_1, \Gamma_2) \in \mathcal{T}$ of order 0 with $\Gamma_2 \neq \Gamma$. By (3.3), Γ_2 has a tangle of order $\geq \theta$, and $|V(\Gamma_2) \cap bd(\Sigma)| \leq p$, and $L \cap \Gamma_2$ is a vital linkage in Γ_2 with all its terminals in $bd(\Sigma)$. This contradicts the minimality of Γ. Thus, Γ is connected, and by theorem (8.4) of [6] and the minimality of Γ it is also loopless. This proves (4).

(5) *Suppose that $(A, B) \in \mathcal{T}$ has order $\leq \theta_4$, and Γ' is a drawing in a surface Σ', such that Γ' can be obtained from B by splitting $\leq \theta_4$ vertices of B. Then $\hat{\Sigma}'$ is not simpler than Σ_0.*

Subproof. From ~~(3.3)~~(3.5)[F13], B has a tangle of order $\geq \theta - \theta_4$, and so Γ' has a tangle of order $\geq \theta - 2\theta_4$. But from (2.2) and (2.3), B has a vital $(p+\theta_4)$-linkage, and so from (2.4), Γ' has a vital $(p+3\theta_4)$-linkage. Since $\theta - 2\theta_4 \geq \theta_1(p+3\theta_4)$, it follows from (1) that $\hat{\Sigma}'$ is not simpler than Σ_0. This proves (5).

(6) *Suppose that $(A, B) \in \mathcal{T}$ has order $\leq \theta_4$, and Γ' is a drawing in a surface Σ', such that Γ' can be obtained from B by splitting $\leq \theta_4$ vertices of B. Suppose, moreover, that*

[p. 19] (a) *$\hat{\Sigma}'$ is homeomorphic to Σ_0, and*

(b) *for $v \in V(\Gamma')$, $v \in bd(\Sigma')$ if and only if either $v \in V(A \cap B)$, or $v \in V(\Gamma) \cap bd(\Sigma)$, or $v \notin V(\Gamma)$ (that is, v is a new vertex produced by splitting).*

Then $c(\Sigma') \geq c(\Sigma)$, and if equality holds then $|V(\Gamma') \cap bd(\Sigma')| \geq p$.

Subproof. As in (5), Γ' has a tangle of order $\geq \theta - 2\theta_4$, and has a vital linkage with all its terminals in $bd(\Sigma')$ (from (b)). But

$$|V(\Gamma') \cap bd(\Sigma')| \leq |V(A \cap B)| + |V(\Gamma) \cap bd(\Sigma)| + 2\theta_4 \leq p + 3\theta_4$$

and $\theta - 2\theta_4 \geq \theta_2(p+3\theta_4)$, and so from (2) and (a), $c(\Sigma') \geq c(\Sigma)$. If equality holds then Σ' is homeomorphic to Σ, and since $\theta - 2\theta_4 \geq \theta_3$ it follows that $|V(\Gamma') \cap bd(\Sigma')| \geq p$, from (3). This proves (6).

[F13] **Misprint.** The reference meant is (3.5). Result (3.3) is about a separation of order 0, and here $(A, B) \in \mathcal{T}$ may have order up to θ_4. By (3.5), B has a tangle of order $ord(\mathcal{T}) - |V(A \cap B)|$.

(7) *There is no Γ-normal proper line $F \subseteq \Sigma$ with ends in different cuffs such that $|F \cap V(\Gamma)| \leq \theta_4$.*

Subproof. If there is such an F, let A be null and $B = \Gamma$; let Σ' be obtained from Σ by cutting along F, and let Γ' be obtained from Γ by splitting appropriately the vertices of Γ in F. (Since Γ is loopless we can do so.) By (4.1)(i), $\hat{\Sigma}'$ is homeomorphic to Σ_0 and $c(\Sigma') = c(\Sigma) - 1$, contrary to (6). This proves (7).

Let $\mathcal{T}_1$ be the θ_4-truncation of $\mathcal{T}$.

(8) *$\mathcal{T}_1$ is respectful.*

Subproof. Certainly Γ is connected, by (4). Let $F \subseteq \hat{\Sigma}$ be a Γ-normal O-arc with $|F \cap V(\Gamma)| < \theta_4$. Suppose first that F does not separate $\hat{\Sigma}$, let Σ' be obtained from $\hat{\Sigma}$ by cutting along F, and let Γ' be the drawing in Σ' obtained from Γ by splitting appropriately the vertices of Γ in F. By (4.1)(iii) $\hat{\Sigma}'$ is simpler than Σ_0, contrary to (5). Thus F separates $\hat{\Sigma}$. Let $\Sigma_1, \Sigma_2 \subseteq \hat{\Sigma}$ be surfaces such that $\Sigma_1 \cup \Sigma_2 = \hat{\Sigma}$ and $\Sigma_1 \cap \Sigma_2 = F$. Let $\Gamma_i = \Gamma \cap \Sigma_i$ $(i = 1, 2)$. Since (Γ_1, Γ_2) is a separation of Γ of order $< \theta_4$, we may assume that $(\Gamma_1, \Gamma_2) \in \mathcal{T}_1 \subseteq \mathcal{T}$. By (5), $\hat{\Sigma}_2$ is not simpler than Σ_0, and so by (4.2)(iii), Σ_1 is a disc. This proves (8).

[p. 20] If $F \subseteq \hat{\Sigma}$ is a Γ-normal O-arc with $|F \cap V(\Gamma)| < \theta_4$, we define $ins(F)$ as usual.

(9) *If $F \subseteq \hat{\Sigma}$ is a Γ-normal O-arc with $|F \cap V(\Gamma)| < \theta_4$, then $C \cap ins(F) \neq \emptyset$ for at most one cuff C.*

Subproof. Let $A = \Gamma \cap ins(F)$ and choose $B \subseteq \Gamma$ so that (A, B) is a separation of Γ and $V(A \cap B) = V(\Gamma) \cap F$. If $C \cap ins(F) \neq \emptyset$ for ≥ 2 cuffs C, then by splitting ≤ 2 vertices of B, we can obtain from B a drawing Γ' in a surface Σ' with $\hat{\Sigma}'$ homeomorphic to Σ_0 and with $c(\Sigma') < c(\Sigma)$, satisfying (6)(b), contrary to (6). This proves (9).

Let d_1 be the metric of $\mathcal{T}_1$.

(10) *If r_1, r_2 are the cuff regions corresponding to distinct cuffs C_1, C_2 then $d_1(r_1, r_2) \geq \theta_4$.*

Subproof. Suppose not. From (7), (9) and the definition of the metric, and exchanging C_1 and C_2 if necessary, there is a Γ-normal O-arc $F_1 \subseteq \Sigma \setminus bd(\Sigma)$ with $|F_1 \cap V(\Gamma)| < \theta_4$, and with $r_1 \subseteq ins(F_1) \setminus F_1$, such that $ins(F_1) \cap bd(\Sigma) = C_1$. Moreover, there is also either

(i) a Γ-normal line $F_0 \subseteq \Sigma$ with one end in F_1, the other end in C_2, and with no internal point in $bd(\Sigma) \cup ins(F_1)$, with $|(F_0 \cup F_1) \cap V(\Gamma)| < \theta_4$, or

(ii) a Γ-normal O-arc $F_2 \subseteq \Sigma \setminus bd(\Sigma)$ with $|(F_1 \cup F_2) \cap V(\Gamma)| < \theta_4$ and with $r_2 \subseteq ins(F_2) \setminus F_2$, such that $ins(F_2) \cap bd(\Sigma) = C_2$ and $|ins(F_1) \cap ins(F_2)| = 1$, or

(iii) a Γ-normal O-arc $F_2 \subseteq \Sigma \setminus bd(\Sigma)$ and a Γ-normal line $F_0 \subseteq \Sigma \setminus bd(\Sigma)$ with $|(F_0 \cup F_1 \cup F_2) \cap V(\Gamma)| < \theta_4$, such that $r_2 \subseteq ins(F_2) \setminus F_2$, $ins(F_2) \cap bd(\Sigma) = C_2$, $ins(F_1) \cap ins(F_2) = \emptyset$, one end of F_0 is in F_1, the other end is in F_2, and no internal point of F_0 is in $ins(F_1)$ or $ins(F_2)$.

If (i) holds, let $A = \Gamma \cap ins(F_1)$, and let $(A, B) \in \mathcal{T}$ where $V(A \cap B) = F \cap V(\Gamma)$; then (6) is contradicted, by splitting all the vertices of B in F_0. If (ii) or (iii) holds, let

$$A = (\Gamma \cap ins(F_1)) \cup (\Gamma \cap ins(F_2))$$

and let $(A, B) \in \mathcal{T}$ where $V(A \cap B) = (F_1 \cup F_2) \cap V(\Gamma)$; then again (6) is contradicted, in (ii) by splitting the vertex of B in $F_1 \cap F_2$ if there is one, and in (iii) by splitting all the vertices of B in F_0. This proves (10).

[p. 21] (11) *If $F \subseteq \Sigma$ is a Γ-normal O-arc with $|F \cap V(\Gamma)| < \theta_4$, and $ins(F)$ includes a unique cuff C, then $|F \cap V(\Gamma)| \geq |C \cap V(\Gamma)|$.*

Subproof. By (9), $ins(F) \cap bd(\Sigma) = C$. Let $A = \Gamma \cap ins(F)$, and let $(A, B) \in \mathcal{T}$ where $V(A \cap B) = F \cap V(\Gamma)$. Let Σ' be obtained from Σ by deleting $\Sigma \cap (ins(F) \setminus F)$. Then Σ' is homeomorphic to Σ, and so by (6),

$$|V(B) \cap bd(\Sigma')| \geq p \geq |V(\Gamma) \cap bd(\Sigma)|.$$

Consequently $|F \cap V(\Gamma)| \geq |C \cap V(\Gamma)|$. This proves (11).

(12) *Every vertex in $V(\Gamma) \cap bd(\Sigma)$ is a terminal of L.*

Subproof. If $v \in ~~bd(\Sigma) \setminus V(\Gamma)~~ V(\Gamma) \cap bd(\Sigma)$ is not a terminal of L,[F14] let Σ' be obtained from Σ by slightly enlarging Σ in the neighbourhood of v. Then $|V(\Gamma) \cap bd(\Sigma')| < p$, contrary to (3). This proves (12).

But (8), (10), (11) and (12) contradict (5.1). Thus our assumption that θ does not satisfy (*) was false, and the proof is complete. □

6 Presentations

If Γ is a drawing in a surface Σ such that $|V(\Gamma) \cap C| \geq 2$ for every cuff C, we call the connected components of $bd(\Sigma) \setminus V(\Gamma)$ the *spaces* of Γ in Σ. For every space s there are two vertices u, v such that $s \cup \{u, v\}$ is a line; we call u, v the *ends* of s. A *support* of Γ in Σ is either a set $\{v\}$ where $v \in V(\Gamma) \cap bd(\Sigma)$, or a line $F \subseteq bd(\Sigma)$ with both ends in $V(\Gamma)$ (where possibly some internal points of F belong to $V(\Gamma)$). The *ends* of a support F are a and b, where $a = b = v$ if $F = \{v\}$, and a and b are the ends of the line F if F is a line.

[F14] **Misprint.** The claim is about vertices in $V(\Gamma) \cap bd(\Sigma)$, and the next sentence needs $v \in V(\Gamma)$, since enlarging Σ near a point that is not a vertex does not change $V(\Gamma) \cap bd(\Sigma)$.

Let Γ, as above, be a subgraph of some graph G, which is not required to be a drawing. Let $F_1^*, \ldots, F_r^*$ be mutually disjoint supports of Γ in Σ with $V(\Gamma) \cap bd(\Sigma) \subseteq F_1^* \cup \cdots \cup F_r^*$. For $1 \le j \le r$, let $H_j \subseteq G$, so that

(V1) *$H_1 \cup \cdots \cup H_r \cup \Gamma = G$; $H_1, \ldots, H_r$ are mutually vertex-disjoint; and for $1 \le j \le r, V(H_j \cap \Gamma) = V(\Gamma) \cap F_j^*$, and $E(H_j \cap \Gamma) = \emptyset$.*

[p. 22] For $1 \le j \le r$, let $q_j \ge 1$ be an integer; for each $u \in V(\Gamma) \cap F_j^*$ let $\mu(u)$ be an edge-less subgraph of H_j with q_j vertices and with $V(\mu(u) \cap \Gamma) = \{u\}$; and for each space s with $s \subseteq F_j^*$, let $\mu(s)$ be a subgraph of H_j, satisfying the following. (For a support F of Γ in Σ, $\mu(F)$ denotes the union of $\mu(s)$ over all spaces $s \subseteq F$ and $\mu(u)$ over all $u \in F \cap V(\Gamma)$.)

(V2) *For $1 \le j \le r, H_j = \mu(F_j^*)$.*

(V3) *For $1 \le j \le r$, if $s \subseteq F_j^*$ is a space with ends u_1, u_2, then $V(\mu(s) \cap \Gamma) = \{u_1, u_2\}$, and $\mu(u_1), \mu(u_2) \subseteq \mu(s)$ and there are q_j mutually vertex-disjoint paths of $\mu(s)$ between $V(\mu(u_1))$ and $V(\mu(u_2))$.*

(V4) *For $1 \le j \le r$, if $s_1, s_2 \subseteq F_j^*$ are spaces, and $u \in F_j^* \cap V(\Gamma)$ lies between them in F_j^*, then $\mu(s_1) \cap \mu(s_2) \subseteq \mu(u)$ (and consequently $E(\mu(s_1) \cap \mu(s_2)) = \emptyset$).*

In these circumstances, we call $\Gamma, F_1^*, \ldots, F_r^*, H_1, \ldots, H_r, \mu$ a *presentation* of G in Σ, with *defect* r. Its *depth* is $\max(q_j : 1 \le j \le r)$, or 0 if $r = 0$ (which implies $bd(\Sigma) = \emptyset$). Our next goal is to prove a form of (1.1) for graphs with presentations with given defect and depth, in a fixed surface. We use the following lemma.

6.1 *Let $\Gamma, F_1^*, \ldots, F_r^*, H_1, \ldots, H_r, \mu$ be a presentation of G in Σ. Let $a, b, a', b' \in F_1^* \cap V(\Gamma)$ be in order and let $I, I' \subseteq F_1^*$ be the supports with ends a, b and a', b' respectively. Let $|\mu(a)| = q$; let $P_1, \ldots, P_q$ be mutually vertex-disjoint paths of $\mu(I)$ between $V(\mu(a))$ and $V(\mu(b))$; and let $P_1', \ldots, P_q'$ be mutually vertex-disjoint paths of $\mu(I')$ between $V(\mu(a'))$ and $V(\mu(b'))$.*

(i) *For $1 \le i, i' \le q$, if $v \in V(P_i \cap P_{i'}')$, then $v \in V(\mu(b) \cap \mu(a'))$ and v is an end of P_i and of $P_{i'}'$.*

(ii) *If $a' = b$, then $P_1, \ldots, P_q$ can be renumbered so that $P_1 \cup P_1', \ldots, P_q \cup P_q'$ are mutually vertex-disjoint paths of $\mu(I \cup I')$ between $V(\mu(a))$ and $V(\mu(b'))$.*

(iii) *If $|\mu(a) \cup \mu(b')| = |\mu(a') \cup \mu(b)| = k$ say, then $\{P_1, \ldots, P_q, P_1', \ldots, P_q'\}$ has cardinality k, and its members are k mutually vertex-disjoint paths of $\mu(I) \cup \mu(I')$ between $V(\mu(a) \cup \mu(b'))$ and $V(\mu(b) \cup \mu(a'))$.*

Proof. First we prove (i). Let $v \in V(P_i \cap P_{i'}')$, where $1 \le i, i' \le q$. We claim that $v \in V(\mu(b))$. If $a = b$ this is clear since

$$v \in V(P_i) \subseteq V(\mu(I)) = V(\mu(b))$$

[p. 23] and so we assume that $a \neq b$. Consequently there is a space $s \subseteq I$ with $v \in V(\mu(s))$. Let $J \subseteq F_1^*$ be the support with ends b and b'. Since $I' \subseteq J$ it follows that $v \in V(\mu(J))$. If $b = b'$ then $v \in V(\mu(b))$ by the same argument as above applied to J instead of I, and so we may assume that $b \neq b'$. Hence there is a space $s' \subseteq J$ with $v \in V(\mu(s'))$. Now b lies in F_1^* between s and s', and so $v \in V(\mu(b))$ by (V4). This proves our claim that $v \in V(\mu(b))$.

Now since $|V(\mu(b))| = q$ and each of $P_1, \ldots, P_q$ has an end in $V(\mu(b))$, it follows that each vertex of $\mu(b)$ is an end of one of $P_1, \ldots, P_q$, and in particular, v is an end of one of $P_1, \ldots, P_q$. Since $v \in V(P_i)$ and $P_1, \ldots, P_q$ are mutually vertex-disjoint, it follows that v is an end of P_i. Similarly, $v \in V(\mu(a'))$ and v is an end of $P'_{i'}$. This proves (i).

For (ii), let $P_1, \ldots, P_q$ and $P'_1, \ldots, P'_q$ be numbered so that for $1 \leq i \leq q$, P_i and P'_i have a common end in $V(\mu(b))$. For $1 \leq i, i' \leq q$, if $i \neq i'$ it follows from (i) that P_i is vertex-disjoint from $P'_{i'}$; and so $P_1 \cup P'_1, \ldots, P_q \cup P'_q$ are mutually vertex-disjoint paths satisfying (ii).

For (iii), let $\mu(a) = \{a_1, \ldots, a_q\}, \mu(b) = \{b_1, \ldots, b_q\}, \mu(a') = \{a'_1, \ldots, a'_q\}, \mu(b') = \{b'_1, \ldots, b'_q\}$, numbered so that for $1 \leq i \leq q$, P_i has ends a_i and b_i, and P'_i has ends a'_i and b'_i, and for $1 \leq i, i' \leq q$, if $i \neq i'$ then $b_i \neq a'_{i'}$. Suppose that $1 \leq i, i' \leq q$ and $a_i = b'_{i'}$. By (i), $i = i'$ and $a_i = b_i = a'_i = b'_i$, and so $P_i = P'_{i'}$. In particular, for $1 \leq i, i' \leq q$, if $i \neq i'$ then P_i and $P'_{i'}$ are vertex-disjoint. Thus

$$\begin{aligned}\{i : 1 \leq i \leq q, a_i = b'_i\} &\subseteq \{i : 1 \leq i \leq q, P_i = P'_i\} \\ &\subseteq \{i : 1 \leq i \leq q, V(P_i \cap P'_i) \neq \emptyset\} = \{i : 1 \leq i \leq q, a'_i = b_i\}.\end{aligned}$$

But from the hypothesis of (iii), $|\mu(a) \cup \mu(b')| = |\mu(a') \cup \mu(b)|$, and so

$$|\{i : 1 \leq i \leq q, a_i = b'_i\}| = |\{i : 1 \leq i \leq q, a'_i = b_i\}|.$$

We therefore have equality throughout, and in particular for $1 \leq i \leq q$, if P_i meets P'_i then $P_i = P'_i$. Then (iii) follows. □

6.2 *Let $\Gamma, F_1^*, \ldots, F_r^*, H_1, \ldots, H_r, \mu$ be a presentation of a graph G in some surface Σ. Let $I \subseteq F_1^*$ be a support with ends a, b. Then there are $|V(\mu(a))|$ mutually vertex-disjoint paths of $\mu(I)$ between $V(\mu(a))$ and $V(\mu(b))$.*

Proof. This follows by induction on the number of spaces included in I, applying (6.1)(ii) if I has an internal point in $V(\Gamma)$, and applying (V3) otherwise. □

[p. 24] Let $\Gamma, F_1^*, \ldots, F_r^*, H_1, \ldots, H_r, \mu$ be a presentation of G in Σ, and let L be a vital linkage in G. If $\Gamma \subseteq L$ and every terminal of L is in one

of $H_1, \ldots, H_r$, we say that L is *exhaustive* (for the presentation). If L is exhaustive then it follows that Γ is a forest and every vertex of Γ not in $bd(\Sigma)$ has degree 2 in Γ. Let us say a Σ-*jump* is a path in Γ with distinct ends both in $bd(\Sigma)$ and with no internal vertex in $bd(\Sigma)$. If L is exhaustive, then every edge of Γ is in a unique Σ-jump, and any two Σ-jumps have no common vertices except possibly one end. The main result of this section is the following.

6.3 *For every surface Σ and all integers $p, r \geq 0$ and $q \geq 1$, there exists $\lambda \geq 0$ such that, if G is a graph with a presentation in Σ of depth $\leq q$ and defect $\leq r$, and there is an exhaustive vital p-linkage in G, then there are at most λ Σ-jumps.*

The most difficult step in the proof of (6.3) is where Σ is a disc and $r = 1$. Then the cases when Σ is a disc and $r = 2, 3$ and at least 4 are successively easier, and finally the case when Σ is not a disc is also quite easy.

6.4 *For all integers $p \geq 0$ and $q \geq 1$ there exists $\lambda \geq 0$ such that, if G has a presentation in a disc Σ of depth $\leq q$ and defect 1, and there is an exhaustive vital p-linkage in G, then there are at most λ Σ-jumps.*

Proof. Choose n so that (2.6) holds, with k replaced by $2q$. Let $\lambda = n^n$. We claim that ~~(6.2)~~(6.4) is satisfied.[F15] For let Γ, F^*, H, μ be a presentation of G in a disc Σ with depth $\leq q$ and let L be a vital p-linkage which is exhaustive for the presentation.

A line $F \subseteq \Sigma$ is *good* if it is proper, both its ends are in $V(\Gamma)$, and no internal point is in $U(\Gamma)$. If F is a good line, there are two lines $J(F), K(F) \subseteq bd(\Sigma)$ with the same ends as F, where $K(F) \subseteq F^*$. Let $\Delta(F)$ be the closed disc in Σ bounded by $F \cup K(F)$. Define

$$A(F) = (\Gamma \cap \overline{\Sigma \setminus \Delta(F)}) \cup \mu(J(F))$$
$$B(F) = (\Gamma \cap \Delta(F)) \cup \mu(K(F))$$
$$X(F) = V(A(F) \cap B(F)).$$

(1) *$(A(F), B(F))$ is a separation of G, and $X(F) = V(\mu(u_1) \cup \mu(u_2))$ where F has ends u_1, u_2.*

The proof is similar to that of theorem (2.1) of [10] ~~and we omit it~~. We give the proof. Let $W = V(B(F))$ and $X = X(F)$. Since L is exhaustive, every edge of Γ is in a unique Σ-jump, and every vertex of Γ not in $bd(\Sigma)$ is an internal vertex of a Σ-jump. Since F is good, $F \cap U(\Gamma) = \{u_1, u_2\} \subseteq bd(\Sigma)$, and so no Σ-jump meets F except in its ends; hence a Σ-jump lies in $\Delta(F)$

[F15] **Misprint.** The reference meant is (6.4), since this is its proof. Result (6.2) has no λ to choose, and the proof takes the hypotheses of (6.4) and ends with its conclusion.

if its ends are in $K(F)$, and in $\overline{\Sigma \setminus \Delta(F)}$ if they are in $J(F)$. In particular, $\Gamma \cap \Delta(F)$ is the union of the Σ-jumps with both ends in $K(F)$ and of the vertices of Γ in $K(F)$, and a vertex of Γ in both $\Delta(F)$ and $\overline{\Sigma \setminus \Delta(F)}$ is in F, and so is u_1 or u_2. Since $u_1, u_2 \in V(\Gamma)$, every space $s \subseteq F^*$ lies in $K(F)$ or in $J(F)$, and by (V3), $\mu(u) \subseteq \mu(s)$ when u is an end of s. Since u_1 or u_2 lies in F^* between any space in $K(F)$ and any space in $J(F)$, it follows from (V4) that

(i) *every vertex of* $\mu(K(F)) \cap \mu(J(F))$ *is in* X*, and no edge is in both* $\mu(K(F))$ *and* $\mu(J(F))$*.*

Also $X \subseteq W \cap V(A(F))$, since $\mu(u_i) \subseteq \mu(K(F)) \cap \mu(J(F))$ $(i = 1, 2)$. We claim that no edge e of G that is not in $B(F)$ has an end $x \in W \setminus X$. For suppose that e is such an edge. If $e \in E(\Gamma)$, then x is a vertex of $\Gamma \cap \Delta(F)$ other than u_1, u_2, so the Σ-jump that contains e has x as an internal vertex or as an end in $K(F)$, and so lies in $\Delta(F)$; but then $e \in E(B(F))$, a contradiction. Otherwise, by (V1) and (V2), e is an edge of $\mu(s)$ for some space $s \subseteq F^*$, since the graphs $\mu(u)$ have no edges; and $s \subseteq J(F)$, since $e \notin E(B(F))$. If $x \in V(\Gamma)$, then x is an end of s by (V3), and $x \in K(F)$, so $x \in \{u_1, u_2\}$; and if $x \notin V(\Gamma)$, then $x \in V(\mu(K(F)))$, and $x \in X$ by (i). In both cases $x \in X$, a contradiction. This proves our claim.

Consequently the edges of G not in $B(F)$, and the vertices of G not in $W \setminus X$, form a subgraph A' of G, and $(A', B(F))$ is a separation of G with $V(A' \cap B(F)) = X$. It remains to show that $A' = A(F)$. By (V1), (V2) and the above, every edge of G is an edge of a Σ-jump or of some $\mu(s)$, and so belongs to $A(F)$ or to $B(F)$; and by (i) and since each Σ-jump lies on one side of F, no edge belongs to both. So $E(A') = E(A(F))$. Every vertex of G not in W is in $A(F)$, by (V1) and (V2). Finally, let $x \in V(A(F)) \cap W$. If $x \in V(\Gamma)$, then x is a vertex of Γ in both $\Delta(F)$ and $\overline{\Sigma \setminus \Delta(F)}$ (because $V(\mu(K(F)) \cap \Gamma) \subseteq K(F)$ and $V(\mu(J(F)) \cap \Gamma) \subseteq J(F)$, by (V3) and the definition of $\mu(u)$), and so $x \in \{u_1, u_2\} \subseteq X$; and if $x \notin V(\Gamma)$, then $x \in \mu(K(F)) \cap \mu(J(F))$, and $x \in X$ by (i). Hence $V(A') = V(A(F))$, and so $A' = A(F)$. This proves (1).[F16]

[p. 25] Let F, F' be good lines with $\Delta(F') \subseteq \Delta(F)$. Let F have ends u_1, u_2, and let F' have ends u_1', u_2', so that u_1, u_1', u_2', u_2 are in order in F^*.

(2) $A(F) \subseteq A(F')$ *and* $B(F') \subseteq B(F)$.

This is immediate since $\Delta(F') \subseteq \Delta(F)$.

Let $I_1 \subseteq F^*$ be the support with ends u_1, u_1', and let $I_2 \subseteq F^*$ have ends u_2', u_2. From (6.2), there are q_1 mutually vertex-disjoint paths of $\mu(I_1)$

[F16] **Missing proof.** The paper omits the proof of Claim (1). The argument above is the one used in the formalization.

between $V(\mu(u_1))$ and $V(\mu(u_1'))$, and similarly for I_2, where the presentation has depth q_1. From (6.1)(iii), we deduce that

(3) *If* $|X(F)| = |X(F')|$ *there are* $|X(F)|$ *mutually vertex-disjoint paths of* $B(F) \cap A(F')$ *between* $X(F)$ *and* $X(F')$*, each using no edge of* Γ.

(4) *If* $F_1, \ldots, F_n$ *is a sequence of good lines such that* $\Delta(F_{i+1}) \subseteq \Delta(F_i)$ *for* $1 \le i < n$*, there exists* i *with* $1 \le i < n$ *such that every edge of* $\Gamma \cap \Delta(F_i)$ *is an edge of* $\Gamma \cap \Delta(F_{i+1})$.

Subproof. From (1), (2), (3) and (2.6) (with k replaced by $2q$), we deduce that there exists i, i' with $1 \le i < i' \le n$, such that $|X_i| = |X_{i'}|$ and $L \cap B_i \cap A_{i'}$ uses no edge of Γ. But every edge of $B_i \cap A_{i'}$ which is in Γ is also in L, since L is exhaustive, and so $E(\Gamma \cap B_i \cap A_{i'}) = \emptyset$. Then every edge of $\Gamma \cap \Delta(F_i)$ is also an edge of $\Gamma \cap \Delta(F_{i'})$ and hence of $\Gamma \cap \Delta(F_{i+1})$. This proves (4).

Let us say that two distinct regions r_1, r_2 of Γ in Σ *touch* if there is an edge e of Γ with $e \subseteq \bar{r}_1 \cap \bar{r}_2$. It follows that if r_1, r_2 touch then $\bar{r}_1 \cap \bar{r}_2 = U(J)$ for some Σ-jump J, and if they do not touch then $|\bar{r}_1 \cap \bar{r}_2| \le 1$. In particular, the touching relation defines a graph T with vertex set the set of regions of Γ in Σ, and it is a tree. From (4), every vertex of T has degree at most n, and every path of T with one end r_0, the region of Γ that meets $bd(\Sigma) \setminus F^*$, has at most n vertices.[F17] For suppose that $r_0, r_1, \ldots, r_n$ is a path of T. For $1 \le i \le n$ let P_i be the Σ-jump with $\bar{r}_{i-1} \cap \bar{r}_i = U(P_i)$, and let F_i be a good line with the same ends as P_i and with interior in r_{i-1}. Since $\Delta(F_i)$ is on the side of F_i away from r_0, it follows that $\Delta(F_{i+1}) \subseteq \Delta(F_i)$, and the edges of P_i are in $\Delta(F_i)$ but not in $\Delta(F_{i+1})$ $(1 \le i < n)$, contrary to (4). Consequently, every vertex of T is at distance $< n$ from r_0. Since every vertex of T has degree at most n, for $0 \le d < n$ there are at most n^d vertices at distance d from r_0, and so $|V(T)| \le 1 + n + \cdots + n^{n-1}$. Hence $|E(T)| \le n^n = \lambda$. But $E(T)$ is in 1-1 correspondence with the Σ-jumps, and the result follows. □

6.5 *For all integers* $p \ge 0$ *and* $q \ge 1$ *there exists* $\lambda \ge 0$ *such that, if* G *has a presentation in a disc* Σ *of depth* $\le q$ *and defect 2, and there is an exhaustive vital* p*-linkage in* G*, then there are at most* λ Σ*-jumps.*

[p. 26] **Proof.** Choose $n \ge 0$ as in ~~(2.5)~~(2.6) with k replaced by $2q$.[F18]

[F17] **Proof gap.** As printed, the claim is about every path of T, and (4) does not give it. Claim (4) is about nested discs, and the discs $\Delta(F)$ lie on the side of F away from r_0. A path through r_0 joins two paths that start at r_0, so (4) bounds it only by $2n-1$ vertices. For example, if Γ consists of k nested jumps on each of two disjoint intervals of F^*, then T is a path with $2k+1$ vertices and middle vertex r_0, and (4) bounds k, not $2k$. Paths that start at r_0 are enough for the count, as the inserted text shows.

[F18] **Proof gap.** The reference meant is (2.6). Result (2.5) is about separations of order

Choose λ' so that (6.4) holds, with p, q, λ replaced by $p + 2q, q, \lambda'$. Let $\lambda = n(2\lambda' + 1)$. We claim that λ satisfies (6.5). For let $\Gamma, F_1^*, F_2^*, H_1, H_2, \mu$ be a presentation of G in the disc Σ, of depth $\leq q$, and let L be an exhaustive vital p-linkage. Let the two spaces not included in $F_1^* \cup F_2^*$ be s_1^* and s_2^*. Let $r_1, \ldots, r_t$ be all the regions of Γ in Σ which are incident both with a vertex in F_1^* and with a vertex in F_2^*, numbered in order, so that for $1 \leq i < t, r_i$ lies between s_1^* and r_{i+1}, in the natural sense. For $1 \leq i \leq t$, choose $a_i \in V(\Gamma) \cap F_1^*$ and $b_i \in V(\Gamma) \cap F_2^*$ so that r_i is incident with a_i and b_i; and let F_i be a good line with ends a_i, b_i and with interior in r_i. Let J_i and K_i be the two lines in $bd(\Sigma)$ with ends a_i and b_i, where $s_1^* \subseteq J_i$ and $s_2^* \subseteq K_i$. Then $J_1 \subseteq J_2 \subseteq \ldots \subseteq J_t$ and $K_t \subseteq K_{t-1} \subseteq \ldots \subseteq K_1$. Let A_i be the union of $\mu(J_i)$ and $\Gamma \cap \Delta$, where $\Delta \subseteq \Sigma$ is the disc bounded by $F_i \cup J_i$; and define B_i similarly using K_i instead of J_i. Then (A_i, B_i) is a separation of G, and $A_i \cap B_i = \mu(a_i) \cup \mu(b_i)$, as is easily seen. Since $\mu(a_i) \cap \mu(b_i) = \emptyset$, it follows that $|V(A_i \cap B_i)| = q_1 + q_2$, where $|\mu(a)| = q_j$ for all $a \in F_j^* \cap V(\Gamma)$ $(j = 1, 2)$. From (6.2), for $1 \leq i < t$, there are q_j mutually vertex-disjoint paths of $H_j \cap B_i \cap A_{i+1}$ between $V(A_i \cap B_i)$ and $V(A_{i+1} \cap B_{i+1})$ for $j = 1$ and 2, and since H_1 and H_2 are disjoint, it follows that there are $q_1 + q_2$ mutually vertex-disjoint paths of $(H_1 \cup H_2) \cap B_i \cap A_{i+1}$ between $V(A_i \cap B_i)$ and $V(A_{i+1} \cap B_{i+1})$. But for $1 \leq i < t, \Gamma \cap B_i \cap A_{i+1}$ has an edge since $r_i \neq r_{i+1}$ and it belongs to L since L is exhaustive, and so $L \cap B_i \cap A_{i+1} \not\subseteq H_1 \cup H_2$. From ~~(2.5)~~(2.6), it follows that $t \leq n$.[F19]

For $1 \leq i < t$, let P_i be the Σ-jump with $\bar{r}_i \cap \bar{r}_{i+1} = U(P_i)$, and let $\Gamma' = P_1 \cup \ldots \cup P_{t-1}$. Let $R_1, \ldots, R_t$ be the regions of Γ' in Σ, where $r_i \subseteq R_i$ $(1 \leq i \leq t)$. Now let us fix i with $1 \leq i \leq t$. We claim that there are at most $2\lambda'$ Σ-jumps J with $U(J) \subseteq R_i$. For $j = 1, 2$, let $I_j \subseteq F_j^*$ be the support $\bar{r}_i \cap F_j^*$; then $\bar{r}_{ji}$[F20] is a closed disc bounded by $I_1 \cup U(P_{i-1}) \cup I_2 \cup U(P_i)$ (replacing $U(P_{i-1})$ by s_1^* if $i = 1$, and replacing $U(P_i)$ by s_2^* if $i = t$). For $j = 1, 2$, let Γ_j be the drawing in Σ formed by all the vertices in $V(\Gamma) \cap I_j$ and all the Σ-jumps J with $U(J) \subseteq R_i \cup I_1 \cup I_2$ with both ends in I_j. We claim that Γ_j includes at most λ' Σ-jumps. For if $|I_j| = 1$ this is trivial, since Γ_j includes no Σ-jumps. If $|I_j| \neq 1$ then $\Gamma_j, I_j, \mu(I_j)$, and the restriction of μ to I_j, is a presentation of $\Gamma_j \cup \mu(I_j)$ in a disc with depth $\leq q$ and defect 1; and $L \cap (\Gamma_j \cup \mu(I_j))$ is an exhaustive vital $(p+2q)$-linkage, and so by (6.4) it has at most λ' Σ-jumps. This proves our claim that Γ_1 and Γ_2 both include

exactly k, but the separations (A_i, B_i) below have order $q_1 + q_2$, which may be less than $2q$ (for instance $q_1 = q_2 = 1$ with $q = 2$). Result (2.6) allows order $\leq k$.

[F19] **Proof gap.** As above, the result meant is (2.6), with which n was chosen.

[F20] **Misprint.** The disc bounded by $I_1 \cup U(P_{i-1}) \cup I_2 \cup U(P_i)$ is $\bar{r}_i$, for the fixed i. The index j runs over the two supports.

at most λ' Σ-jumps. Since every Σ-jump J with $U(J) \subseteq R_i \cup I_1 \cup I_2$ either has both ends in I_1 or has both ends in I_2 (because r_i is the only region of Γ included in R_i incident with both F_1^* and F_2^*) it follows that R_i includes at most $2\lambda'$ Σ-jumps. Counting $P_1, \dots, P_{t-1}$, and using the fact that $t \le n$, we deduce that there are at most $2\lambda' t + t - 1 \le \lambda$ Σ-jumps altogether, as required. □

[p. 27] **6.6** *For all integers $p \ge 0$ and $q \ge 1$ there exists $\lambda \ge 0$ such that, if G has a presentation in a disc Σ of depth $\le q$ and defect 3, and there is an exhaustive vital p-linkage in G, then there are at most λ Σ-jumps.*[F21]

Proof. Choose $\lambda' \ge 0$ so that (6.5) holds with p, q, λ replaced by $p + 2q, q, \lambda'$, and let $\lambda = 3\lambda'$. We claim that λ satisfies (6.6). For let $\Gamma, F_1^*, F_2^*, F_3^*, H_1, H_2, ~~H_2~~H_3, \mu$[F22] be a presentation of G in the disc Σ, of depth $\le q$, and let there be an exhaustive vital p-linkage. For $i = 1, 2, 3$, choose $v_i \in V(\Gamma) \cap F_i^*$ such that there is a region r of Γ in Σ incident with v_1, v_2 and v_3. (It is an easy exercise to prove that this choice is possible, since $V(\Gamma) \cap bd(\Sigma) \subseteq F_1^* \cup F_2^* \cup F_3^*$ and, because L is exhaustive, Γ is a forest in which every vertex not in $bd(\Sigma)$ has degree 2.)[F23] Choose $v_0 \in r \setminus bd(\Sigma)$, and for $i = 1, 2, 3$ let $F_i \subseteq r \cup \{v_i\}$ be a line with ends v_0 and v_i, so that F_1, F_2 and F_3 are mutually disjoint except for v_0, and $F_i \cap bd(\Sigma) = \{v_i\}$ $(i = 1, 2, 3)$. For $i = 1, 2, 3$ let $J_i \subseteq bd(\Sigma)$ be the support with ends $\{v_1, v_2, v_3\} \setminus \{v_i\}$ which does not include F_i^*. Let $\Delta_1 \subseteq \Sigma$ be the closed disc bounded by $J_1 \cup F_2 \cup F_3$ and define Δ_2, Δ_3 similarly. Now

$$\Gamma \cap \Delta_1, F_2^* \cap J_1, F_3^* \cap J_1, \mu(F_2^* \cap J_1), \mu(F_3^* \cap J_1)$$

and the restriction of μ to $(F_2^* \cup F_3^*) \cap J_1$ is a presentation of $\Gamma \cap \Delta_1 \cup \mu(F_2^* ~~\cup~~ \cap J_1) \cup \mu(F_3^* ~~\cup~~ \cap J_1)$[F24] of depth $\le q$ and defect 2, and this graph has an exhaustive vital $(p + 2q)$-linkage. Consequently, Δ_1 includes at most λ' Σ-jumps. Similarly so do Δ_2 and Δ_3, and since every Σ-jump belongs to one of these discs, it follows that there are at most $3\lambda' = \lambda$ Σ-jumps altogether, as required. □

[F21] **Misprint.** The word "in" is missing, as the statements of (6.4) and (6.5) show.

[F22] **Misprint.** The third graph is H_3.

[F23] **Proof gap.** The reason given is not enough: if Γ is a star with its center inside the disc and one leaf in each F_i^*, no region is incident with all three leaves. The choice is possible because L is exhaustive: as stated on page 24, Γ is then a forest in which every vertex not in $bd(\Sigma)$ has degree 2, which excludes the star. The same fact is used in Claim (3) in the proof of (6.3).

[F24] **Misprint.** The graphs at Δ_1 are $\mu(F_2^* \cap J_1)$ and $\mu(F_3^* \cap J_1)$, as in the display just before. The map μ is defined on the parts of the supports, and $F_2^* \cup J_1$ is not a part of F_2^*.

6.7 *For all integers $p, r \geq 0$ and $q \geq 1$ there exists $\lambda \geq 0$ such that, if G has a presentation in a disc Σ of depth $\leq q$ and defect $\leq r$, and there is an exhaustive vital p-linkage in G, then there are at most λ Σ-jumps.*

Proof. We prove the result for all p and q by induction on r. By (6.4), (6.5), (6.6) we may assume that $r \geq 4$, and for all $p' \geq 0$ and $q' \geq 1$ there exists $\lambda(p', q') \geq 0$ such that, if G has a presentation in a disc Σ of depth $\leq q'$ and defect $< r$, and there is an exhaustive vital p'-linkage in G, then there are at most $\lambda(p', q')$ Σ-jumps. Let $\lambda = 2\lambda(p + 2q, q)$. We claim that λ satisfies (6.7). For let $\Gamma, F_1^*, \ldots, F_r^*, H_1, \ldots, H_r, \mu$ be a presentation of G in the disc Σ of depth $\leq q$ and let there be an exhaustive vital p-linkage in G. Choose a region r incident with vertices in at least three of $F_1^*, \ldots, F_r^*$, as in the proof of (6.6). Since two of these three are non-consecutive, because $r \geq 4$, it follows that there is a proper line $F \subseteq \Sigma$ with ends $a, b \in V(\Gamma)$, and with $F \cap U(\Gamma) = \{a, b\}$, such that both of the lines $J_1, J_2 \subseteq bd(\Sigma)$ with ends a and b are disjoint from at least one of $F_1^*, \ldots, F_r^*$.

[p. 28] For $i = 1, 2$ let $\Delta_i \subseteq \Sigma$ be the closed disc bounded by $F \cup J_i$. Now $(\Gamma \cap \Delta_j) \cup \mu(J_{\not i j})$[F25] has a presentation in Σ with depth $\leq q$ and defect $< r$, and it has an exhaustive vital $(p + 2q)$-linkage, and so Δ_j includes at most $\lambda(p+2q, q)$ Σ-jumps, from the inductive hypothesis. But every Σ-jump belongs to one of Δ_1, Δ_2, and so there are at most $2\lambda(p+2q, q) = \lambda$ Σ-jumps altogether, as required. □

Proof of (6.3). We proceed by induction on Σ and by ~~(6.3)~~(6.7)[F26], we may assume that Σ is not a disc. We make the inductive hypothesis that

(1) *For all integers $p, r \geq 0$ and $q \geq 1$ there exists $\lambda(p, q, r) \geq 0$ such that for every surface Σ', if either $\hat{\Sigma}'$ is simpler then $\hat{\Sigma}$, or $\hat{\Sigma}'$ is homeomorphic to $\hat{\Sigma}$ and $c(\Sigma') < c(\Sigma)$, the following is true. If G' is a graph with a presentation in Σ' of depth $\leq q$ and defect $\leq r$, and there is an exhaustive vital p-linkage in G', then there are at most $\lambda(p, q, r)$ Σ'-jumps.*

Let $\lambda = \max(\lambda(p + 4q, q, r + 2), 2\lambda(p + ~~2q~~4q, q, r + 1))$. We claim that λ satisfies the theorem. For let $\Gamma, F_1^*, \ldots, F_r^*, H_1, \ldots, H_r, \mu$ be a presentation of a graph G in Σ, with depth $\leq q$.

(2) *We may assume that every proper line $F \subseteq \Sigma$ with ends in $V(\Gamma)$ and with no internal point in $U(\Gamma)$ separates Σ.*

Subproof. Suppose that $F \subseteq \Sigma$ is a proper line with ends $a, b \in V(\Gamma)$ and with no internal point in $U(\Gamma)$, and F does not separate Σ. Let Σ' be obtained from Σ by cutting along F. By (4.1)(i) and (4.1)(ii), either $\hat{\Sigma}'$ is

[F25] **Misprint.** Here Δ_j is bounded by $F \cup J_j$, so the boundary graph that goes with it is $\mu(J_j)$. The index i is not bound here, and for $i \neq j$ the graph $\mu(J_i)$ lies on the other disc.

[F26] **Misprint.** The reference meant is (6.7): a proof of (6.3) cannot use (6.3), and (6.7) is the case of a disc.

simpler than $\hat{\Sigma}$, or $\hat{\Sigma}'$ is homeomorphic to $\hat{\Sigma}$ and $c(\Sigma') < c(\Sigma)$. Let Γ' be the drawing in Σ' obtained from Γ by splitting appropriately the vertices of Γ in F. Let $a \in F_1^*, b \in F_2^*$ say, or let $a, b \in F_1^*$; in the second case the splitting below is done twice inside F_1^*, first at a and then at b.[F27] Let F_1^* have ends u_1, u_2, and let $I_i \subseteq F_1^*$ be the support with ends u_{1i}, a.[F28] For each $v \in V(\mu(a))$, let $\delta_i(v)$ be the set of edges of $\mu(I_i)$ incident with v $(i = 1, 2)$. By splitting v according to $\delta_1(v), \delta_2(v)$, for each $v \in V(\mu(a))$, and similarly splitting each $v \in V(\mu(b))$, we obtain a graph G' which has a presentation in Σ' (using the drawing Γ') of depth $\leq q$ and defect $\leq r + 2$. But by (2.4), it follows that G' has an exhaustive vital $(p + 4q)$-linkage, and so by (1), Γ' has at most $\lambda(p + 4q, q, r + 2)$ Σ'-jumps. But these Σ'-jumps are in 1-1 correspondence with the Σ-jumps of Γ, and so Γ has at most $\lambda(p + 4q, q, r + 2) \leq \lambda$ Σ-jumps, as required. This proves (2).

(3) $c(\Sigma) = 1$.

[p. 29] *Subproof.* Since $r \geq 4$, it follows that $c(\Sigma) \geq 1$. If $c(\Sigma) \geq 2$ then, since Γ is a forest (because L is exhaustive), there is a region of Γ incident with vertices in two different cuffs,[F29] and so there is a proper line F with ends vertices of Γ in different cuffs, and with no internal point in $U(\Gamma)$. But then F does not separate Σ, contrary to (2). This proves (3).

(4) *We may assume that for every proper line $F \subseteq \Sigma$ with ends in $V(\Gamma)$ and with no internal point in $U(\Gamma)$, there is a closed disc $\Delta \subseteq \Sigma$ with $F \subseteq bd(\Delta) \subseteq F \cup bd(\Sigma)$.*

Subproof. By (2), F separates Σ. Let $\Sigma_1, \Sigma_2 \subseteq \Sigma$ be surfaces with $\Sigma_1 \cup \Sigma_2 = \Sigma$ and $\Sigma_1 \cap \Sigma_2 = F$. For $i = 1, 2$ let $\Gamma_i = \Gamma \cap \Sigma_i$, and let $G_i = \Gamma_i \cup \mu(bd(\Sigma) \cap \Sigma_i)$.[F30] Now for $i = 1, 2$, there is a presentation of G_i in Σ_i of depth $\leq q$ and defect $\leq r+1$, using Γ_i; and $L \cap G_i$ is an exhaustive vital $(p+2q)$-linkage, except when both ends of F are in one support; in that case, after the vertices of $V(\mu(a)) \cap V(\mu(b))$ on that side are split as in (2), by (2.4), there is such a presentation of the new graph and an exhaustive vital

[F27] **Proof gap.** The ends a, b of F may lie in the same support, a case that "say" does not cover. For instance, a non-separating line in an annulus may start and end on one cuff that is a single support. Splitting twice inside the support gives each vertex of $V(\mu(a)) \cap V(\mu(b))$ three copies. Each split is one use of (2.4) per vertex, so the linkage is still a $(p + 4q)$-linkage, and the defect is still $\leq r + 2$. So the bound of Claim (2) stands.

[F28] **Misprint.** The two supports I_1, I_2 have ends u_1, a and u_2, a. With u_1 for both, $I_1 = I_2$, and the splitting by $\delta_1(v), \delta_2(v)$ below would split nothing.

[F29] **Proof gap.** For a general drawing this is false: in an annulus, a circuit of Γ around the hole puts every region on one side of it. The step needs that L is exhaustive, which makes Γ a forest, as stated on page 24.

[F30] **Misprint.** Here G_i is built from Γ_i, the part of Γ in Σ_i, as the next sentence ("using Γ_i") needs.

$(p+4q)$-linkage in it.[F31] If both Σ_1 and Σ_2 are not discs, then by (4.2)(i) and (4.2)(ii), either $\hat{\Sigma}_i$ is simpler than $\hat{\Sigma}$, or $\hat{\Sigma}_i$ is homeomorphic to $\hat{\Sigma}$ and $c(\Sigma_i) < c(\Sigma)$ for $i = 1, 2$. But then from (1), there are $\leq \lambda(p+2q4q, q, r+1)$ Σ_i-jumps in Γ_i for $i = 1, 2$, and hence $\leq 2\lambda(p+2q4q, q, r+1) \leq \lambda$ Σ-jumps in Γ, as required. Consequently we may assume that one of Σ_1, Σ_2 is a disc. This proves (4).

From (4), it follows in particular that every Σ-jump J is homotopic in $\hat{\Sigma}$ to the lines in $bd(\Sigma)$ joining the ends of J. Let R be the closure in $\hat{\Sigma}$ of $\hat{\Sigma} \setminus \Sigma$; thus $R \cap \Sigma$ is the unique cuff of Σ. It follows that every O-arc in $R \cup U(\Gamma)$ is null-homotopic in $\hat{\Sigma}$. By theorem (11.10) of [5] there is a closed disc $\Delta \subseteq \hat{\Sigma}$ with $R \cup U(\Gamma) \subseteq \Delta$. By extending Δ to a sphere and removing $R \setminus bd(R)$ from it, we deduce that G has a presentation in a disc Σ', with depth $\leq q$ and defect $\leq r$. Then the result follows from (1), since $\lambda \geq \lambda(p+4q, q, r+2) \geq \lambda(p, q, r)$ and Σ is not a disc. □

7 Presentations and tree-width

If $\mathcal{T}$ is a tangle in a graph G, a subgraph A of G is *small* (with respect to $\mathcal{T}$) if there exists $B \subseteq G$ such that $(A, B) \in \mathcal{T}$. An easy consequence of theorem (2.9) of [6] is the following ~~(we omit the proof)~~.

7.1 *If $\mathcal{T}$ is a tangle in G, and $A \subseteq G$, then the following are equivalent:*

(i) *A is small with respect to $\mathcal{T}$;*

(ii) *if X denotes the set of vertices of A incident with edges of G not A, then $|X| < ord(\mathcal{T})$, and $(A, B) \in \mathcal{T}$ where $V(B) = (V(G) \setminus V(A)) \cup X$ and $E(B) = E(G) \setminus E(A)$;*

[p. 30] (iii) *there is a separation (A, B) of order $< ord(\mathcal{T})$, and $\mathcal{T}$ contains every such separation.*

We give the proof. Let $\theta = ord(\mathcal{T})$, let X be as in (ii), and let B_X be the subgraph of G with $V(B_X) = (V(G) \setminus V(A)) \cup X$ and $E(B_X) = E(G) \setminus E(A)$. Every end of an edge of G not in A is in $V(G) \setminus V(A)$ or in X, so B_X is a subgraph of G, and (A, B_X) is a separation of G of order $|X|$. If (A, B) is any separation of G, then every vertex of X is incident with an edge of B, and so $X \subseteq V(A \cap B)$ and $|X| \leq |V(A \cap B)|$. Theorem (2.9)(ii) of [6] says that if $(A_1, B_1) \in \mathcal{T}$ and (A_2, B_2) is a separation of G of order $< \theta$ with $V(A_2) \subseteq V(A_1)$, then $(A_2, B_2) \in \mathcal{T}$. Taking $A_1 = A_2 = A$, it follows that if $(A, B_1) \in \mathcal{T}$, then every separation (A, B) of G of order $< \theta$ belongs to $\mathcal{T}$.

[F31] **Proof gap.** When both ends a, b of F are in one support F_j^*, the side Σ_i meets F_j^* in two arcs. The graphs of both arcs contain $V(\mu(a)) \cap V(\mu(b))$, which need not be empty (for instance with $q = 2$). The pieces of a presentation must be disjoint, so the printed claim fails there. After the splitting, each side has at most $\lambda(p+4q, q, r+1)$ jumps, so $2q$ becomes $4q$ in the definition of λ on page 28 and in the bound below.

(i) $\Rightarrow$ (ii): Let $(A, B_1) \in \mathcal{T}$. Then $|X| \leq |V(A \cap B_1)| < \theta$, and so (A, B_X) is a separation of order $< \theta$, and it belongs to $\mathcal{T}$ by the last remark. (ii) $\Rightarrow$ (i) holds since $(A, B_X) \in \mathcal{T}$. (ii) $\Rightarrow$ (iii): (A, B_X) is a separation of order $< \theta$, and every separation (A, B) of order $< \theta$ belongs to $\mathcal{T}$ by the last remark applied with $B_1 = B_X$. (iii) $\Rightarrow$ (ii): Let (A, B') be a separation of order $< \theta$. Then $|X| \leq |V(A \cap B')| < \theta$, so (A, B_X) is a separation of order $< \theta$, and it belongs to $\mathcal{T}$ by (iii).[F32]

If $\Gamma, F_1^*, \ldots, F_r^*, H_1, \ldots, H_r, \mu$ is a presentation of G in Σ, and $\mathcal{T}$ is a tangle in G, we say that the presentation *surrounds* $\mathcal{T}$ if $\mu(s)$ is small for every space $s \subseteq F_1^* \cup \cdots \cup F_r^*$.

If Γ is a drawing in a surface Σ such that $|V(\Gamma) \cap C| \geq 2$ for every cuff C, we define Γ^+ to be the drawing in $\hat{\Sigma}$ with $U(\Gamma^+) = U(\Gamma) \cup bd(\Sigma)$ and $V(\Gamma^+) = V(\Gamma)$.

7.2 *Let $\theta \geq 1$ and $q \geq 1$, and let $\Gamma, F_1^*, \ldots, F_r^*, H_1, \ldots, H_r, \mu$ be a presentation of a graph G in a surface Σ, of depth $\leq q$, and surrounding a tangle of order $\geq q\theta$. Then Γ^+ has a tangle of order $\geq \theta$.*[F33]

Proof. Γ^+ has the same vertex set as Γ, and its edges are the edges and spaces of Γ. Let $S(\Gamma)$ be the set of spaces of Γ. For any subgraph A of Γ^+ we define $\sigma(A)$ to be the union of $A \cap \Gamma$ with all the graphs $\mu(v)$ ($v \in V(A) \cap bd(\Sigma)$) and $\mu(s)$ ($s \in E(A) \cap S(\Gamma)$).

(1) *If (A, B) is a separation of Γ^+ then $(\sigma(A), \sigma(B))$ is a separation of G of order $\leq q|V(A \cap B)|$.*

Subproof. Clearly $\sigma(A) \cup \sigma(B) = G$. No edge of G belongs to both $\sigma(A)$ and $\sigma(B)$, by (V1) and (V4), and so $(\sigma(A), \sigma(B))$ is a separation of G. We claim that

$$V(\sigma(A) \cap \sigma(B)) \subseteq V(A \cap B) \cup \bigcup(\mu(u) : u \in V(A \cap B) \cap bd(\Sigma)).$$

For let $v \in V(\sigma(A) \cap \sigma(B)) \setminus V(A \cap B)$. Since $\sigma(A) \cap \Gamma = A$ and $\sigma(B) \cap \Gamma = B$, it follows that $v \notin V(\Gamma)$; let $v \in V(H_1)$ say. There exists x such that $v \in V(\mu(x))$, and either $x \in V(A) \cap F_1^*$ or $x \in E(A) \cap S(\Gamma)$ and $x \subseteq F_1^*$; and similarly there exists y (with B instead of A). Choose x and y as close together in F_1^* as possible, in the natural sense. If z lies in F_1^* between x and y, and $z \in V(\Gamma) \cup S(\Gamma)$, and $z \neq x, y$, then z belongs to one of A, B, and $v \in V(\mu(z))$ by (V3) and (V4), contrary to our choice of x and

[F32] **Missing proof.** The paper omits the proof of (7.1). The argument above is the one used in the formalization.

[F33] **Degenerate case.** As printed, (7.2) fails for $q = 0$, since then the hypothesis asks only for a tangle of order ≥ 0. For example, take a presentation with $r = 0$ in a sphere, with G a single vertex, and $\theta = 2$. Then $\Gamma^+ = G$ has no tangle of order 2. Every use of (7.2) in the paper has $q \geq 1$.

y. If $x \in V(\Gamma)$ and $y \in S(\Gamma)$ with one end x, then $x \in V(A \cap B)$ and $v \in V(\mu(x))$ as required. Finally, if $x = y \in V(\Gamma)$ then again $x \in V(A \cap B)$ and $v \in V(\mu(x))$; while if $x = y \in S(\Gamma)$ then $u \in V(A \cap B)$ and $v \in V(\mu(u))$, where u is one end of x. This proves our claim that

$$V(\sigma(A) \cap \sigma(B)) \subseteq V(A \cap B) \cup \bigcup(\mu(u) : u \in V(A \cap B) \cap bd(\Sigma)).$$

Consequently, $|V(\sigma(A) \cap \sigma(B))| \leq q|V(A \cap B)|$. This proves (1).

Let $\mathcal{T}$ be a tangle in G of order $\geq q\theta$, surrounded by the presentation. Let $\mathcal{T}'$ be the set of all separations (A, B) of Γ^+ of order $< \theta$ such that $(\sigma(A), \sigma(B)) \in \mathcal{T}$. We claim that $\mathcal{T}'$ is a tangle in Γ^+ of order θ. Let us verify the three axioms for a tangle.

[p. 31] For the first axiom, let (A, B) be a separation of Γ^+ of order $< \theta$. Then $(\sigma(A), \sigma(B))$ has order $< q\theta \leq ord(\mathcal{T})$ by (1), and so one of $(\sigma(A), \sigma(B)), (\sigma(B), \sigma(A))$ belongs to $\mathcal{T}$. Hence one of $(A, B), (B, A)$ belongs to $\mathcal{T}'$. This verifies the first axiom.

For the second, let $(A_i, B_i) \in \mathcal{T}'$ $(i = 1, 2, 3)$. Then $(\sigma(A_i), \sigma(B_i)) \in \mathcal{T}$ $(i = 1, 2, 3)$, and so $\sigma(A_1) \cup \sigma(A_2) \cup \sigma(A_3) \neq G$. But

$$\sigma(A_1) \cup \sigma(A_2) \cup \sigma(A_3) = \sigma(A_1 \cup A_2 \cup A_3)$$

and $\sigma(\Gamma^+) = G$, and so $A_1 \cup A_2 \cup A_3 \neq \Gamma^+$. This verifies the second axiom.

For the third, it suffices by [3, theorem (2.7)] to show that if $e \in E(\Gamma^+)$ and K_e denotes the graph consisting of e and its ends, then $(\Gamma^+ \setminus e, K_e) \notin \mathcal{T}'$, that is, $(\sigma(\Gamma^+ \setminus e), \sigma(K_e)) \notin \mathcal{T}$. If $e \in E(\Gamma)$, or if $e \in S(\Gamma)$ and $e \nsubseteq F_1^* \cup \cdots \cup F_r^*$ then $V(\sigma(\Gamma^+ \setminus e)) = V(G)$, and so $(\sigma(\Gamma^+ \setminus e), \sigma(K_e)) \in \mathcal{T}$, since $\mathcal{T}$ satisfies the third axiom. We assume then that $e \in S(\Gamma)$, and $e \subseteq F_1^* \cup \cdots \cup F_r^*$. Then $\sigma(K_e) = \mu(e)$, and so $(\sigma(\Gamma^+ \setminus e), \sigma(K_e)) \notin \mathcal{T}$ by (7.1) since $\mu(e)$ is small. This verifies the third axiom.

Hence $\mathcal{T}'$ is a tangle in Γ^+ of order θ, as required. □

The main result of this section is the following.

7.3 *For every surface Σ and all integers $p, r \geq 0$ and $q \geq 1$, there exists $\theta \geq 1$ such that, if a graph G has a presentation in Σ of depth $\leq q$, and defect $\leq r$ which surrounds a tangle of order $\geq \theta$, then G has no vital p-linkage.*

Proof. Choose λ so that (6.3) holds. Choose θ_1 so that (5.2) holds with Σ, p, θ replaced by $\hat{\Sigma}, 2(p + \lambda), \theta_1$. Choose θ_2 so that (4.6) holds, with $\Sigma, \theta', c, \theta$ replaced by $\hat{\Sigma}, \theta_1, c(\Sigma), \theta_2$. Let $\theta = q\theta_2$. We claim that θ satisfies (7.3). For let $\Gamma, F_1^*, \ldots, F_r^*, H_1, \ldots, H_r, \mu$ be a presentation of G in Σ, of depth $\leq q$ (we may assume its defect is exactly r), surrounding a tangle of order $\geq \theta$. Suppose that L is a vital p-linkage in G. Let Γ' be the union of all paths in $L \cap \Gamma$ with both ends in $bd(\Sigma)$ and no internal vertex in

$bd(\Sigma)$ (including one-vertex paths). Then $\Gamma', F_1^*, \ldots, F_r^*, H_1, \ldots, H_r, \mu$ is a presentation of $G' = \Gamma' \cup H_1 \cup \cdots \cup H_r$ in Σ, of depth $\leq q$. Moreover, $L \cap G'$ is a vital linkage in G', by (2.3), and it is a p-linkage since each component of L includes at most one component of $L \cap G'$, and it is exhaustive. We give the details. Let $Z = V(\Gamma) \cap bd(\Sigma)$; then $Z \subseteq V(H_1 \cup \cdots \cup H_r)$ by (V1). Let W be the set of vertices $w \in V(\Gamma) \setminus Z$ such that some terminal $t \in V(\Gamma) \setminus Z$ of L is joined to w by a path of $(L \cap \Gamma) \setminus Z$. A vertex of Γ not in Z is in no H_j, so all its edges in L are in Γ, by (V1). We claim that $\Gamma' = (L \cap \Gamma) \setminus W$. For let Q be a path of $L \cap \Gamma$ with both ends in Z and no internal vertex in Z. Its internal vertices have degree 2 in L, and so $Q \setminus Z$ is a component of $(L \cap \Gamma) \setminus Z$ that contains no terminal of L; hence Q has no vertex in W. Conversely, let $x \in V(\Gamma) \setminus (Z \cup W)$, and let P be the component of L that contains x. Then x is not a terminal of L, since $x \notin W$; and following P from x in each direction, every vertex reached before the first vertex in Z is joined to x in $(L \cap \Gamma) \setminus Z$, and so is not a terminal of L, and has both its edges of L in Γ. Hence both directions reach Z, and give a path of $L \cap \Gamma$ through x, containing both edges of L at x, with both ends in Z and no internal vertex in Z. An edge of $L \cap \Gamma$ with both ends in Z is itself such a path. This proves our claim. Since $W \subseteq V(\Gamma) \setminus Z$ is disjoint from $H_1 \cup \cdots \cup H_r$, it follows that $L \cap G' = L \setminus W$.

Let $x, y \notin W$ be vertices of the same component P of L, and suppose that some vertex z of P between x and y is in W. Then z is joined by a path R of $(L \cap \Gamma) \setminus Z$ to a terminal t of L; so R is the subpath of P between z and the end t of P, and it passes through x or y, which is then in W, a contradiction. Hence the subpath of P between x and y is a path of $L \setminus W$; that is, each component of L includes at most one component of $L \setminus W$. So the terminals of $L \setminus W$ in a component P of L lie in one path of $L \setminus W$, and there are at most two of them, and at most one if $|V(P)| = 1$; and P has two terminals of L if $|V(P)| \geq 2$. Hence $L \cap G'$ has at most p terminals. For exhaustiveness, $\Gamma' \subseteq L$; and let x be a terminal of $L \setminus W$ in no H_j. Then $x \in V(\Gamma) \setminus (Z \cup W)$, and so x is not a terminal of L, and x is incident with two edges of L, both in Γ. If the other end y of one of them were in W, then $y \notin Z$, and the edge would join y to x in $(L \cap \Gamma) \setminus Z$, giving $x \in W$. So both edges are in $L \setminus W$, and x is not a terminal of $L \setminus W$, a contradiction.[F34] From (6.3), there are at most λ Σ-jumps in Γ'. Let X be the set of all $v \in V(\Gamma) \cap bd(\Sigma)$ such that $\{v\}$ is the vertex set of a component of $L \cap \Gamma$. Every other component of $L \cap \Gamma$ either ~~is~~includes a Σ-jump in Γ' (if both its

[F34] **Missing proof.** The paper gives only the reason for these two facts. The argument above is the one used in the formalization.

ends are in $bd(\Sigma)$, its subpath from one end to the next vertex in $bd(\Sigma)$ is such a jump), or has an end some vertex $v \in V(\Gamma) \setminus bd(\Sigma)$ which is a terminal of L, by (2.2). Distinct components include distinct Σ-jumps, since they are disjoint.[F35] Consequently, $L \cap (\Gamma \setminus X)$ has at most $\lambda + p$ components, and so is a vital $2(\lambda+p)$-linkage in $\Gamma \setminus X$. From (5.2), $\Gamma \setminus X$ has no tangle of order $\geq \theta_1$. Now $\Gamma \setminus X$ is obtained from Γ^+ by deleting some [p. 32] vertices and edges of Γ^+, all incident with one of $c(\Sigma)$ regions of Γ^+ in $\hat{\Sigma}$. From (4.6), Γ^+ has no tangle of order $\geq \theta_2$. From (7.2), the presentation of G captures no tangle of order $\geq q\theta_2 = \theta$, a contradiction. Thus L is not a a vital p-linkage in G, as required. □

8 Pseudo-presentations

In a presentation, the graphs $H_1, \ldots, H_r$ are disjoint. Now we want to consider a slightly more general object, in which the "last" vertices of each H_i may equal some of the "first" vertices of the next one of $H_1, \ldots, H_r$ on the same cuff. More precisely, a *pseudo-presentation* of a graph G in a surface Σ is defined as follows. Let $\Gamma \subseteq G$ be a drawing in Σ such that $|V(\Gamma) \cap C| \geq 2$ for each cuff C. Let $F_1^*, \ldots, F_r^*$ be mutually disjoint supports of Γ in Σ with $V(\Gamma) \cap bd(\Sigma) \subseteq F_1^* \cup \cdots \cup F_r^*$. For $1 \leq j \leq r$ let $q_j \geq 1$ be an integer; for each $u \in V(\Gamma) \cap F_j^*$ let $\mu(u)$ be an edge-less subgraph of G with q_j vertices and with $V(\mu(u) \cap \Gamma) = \{u\}$; and for each space s with $s \subseteq F_j^*$ let $\mu(s) \subseteq G$, satisfying (P1)–(P4) below. A *unit* is either a vertex in $V(\Gamma) \cap bd(\Sigma)$ or a space in $F_1^* \cup \cdots \cup F_r^*$.

(P1) $G = \Gamma \cup \bigcup(\mu(F_j^*) : 1 \leq j \leq r)$; Γ *and all the graphs* $\mu(s)$ *for* $s \in S(\Gamma)$ *and* $s \subseteq F_1^* \cup \cdots \cup F_r^*$ *are mutually edge-disjoint; and if* x_1, x_2 *are units in different cuffs then* $\mu(x_1)$ *and* $\mu(x_2)$ *are mutually vertex-disjoint.*

(P2) *For* $1 \leq j \leq r$, *if* $s \subseteq F_j^*$ *is a space with ends* u_1, u_2 *then* $V(\mu(s) \cap \Gamma) = \{u_1, u_2\}$ *and* $\mu(u_1), \mu(u_2) \subseteq \mu(s)$, *and there are* q_j *mutually vertex-disjoint paths of* $\mu(s)$ *between* $V(\mu(u_1))$ *and* $V(\mu(u_2))$.*

(P3) *For each cuff* C, *if* x_1 *and* x_2 *are units in* C *and* $u_1, u_2 \in C \cap V(\Gamma)$ *are such that* x_1, u_1, x_2, u_2 *occur in order in* C, *then* $\mu(x_1) \cap \mu(x_2) \subseteq \mu(u_1) \cup \mu(u_2)$.*

(P4) *For each cuff* $C, \bigcap(\mu(u) : u \in V(\Gamma) \cap C)$ *is null.*

Then we say that $\Gamma, F_1^*, \ldots, F_4^* F_r^*, \mu$ is a *pseudo-presentation* of G in Σ.[F36] Its depth is $\max(q_j : 1 \leq j \leq r)$, or 0 if $r = 0$, and its defect is r.

[F35] **Proof gap.** As printed, the step misses a case: a component of $L \cap \Gamma$ with both ends in $bd(\Sigma)$ may have an internal vertex in $bd(\Sigma)$, and then it is not a Σ-jump. For example, a path of L may run a, x, b, y, c inside Γ, with $a, b, c \in bd(\Sigma)$ and $x, y \notin bd(\Sigma)$, and go on in H_1 beyond a and c. The count below only needs one jump in each such component.

[F36] **Misprint.** There are r supports $F_1^*, \ldots, F_r^*$, and (8.1) writes $\Gamma, F_1^*, \ldots, F_r^*, \mu$.

We say that the pseudo-presentation is *disjointed* if for each space s with ends u_1, u_2 such that $s \not\subseteq F_1^* \cup \cdots \cup F_r^*$, the graphs $\mu(u_1)$ and $\mu(u_2)$ are vertex-disjoint.

8.1 *Let $\Gamma, F_1^*, \ldots, F_r^*, \mu$ be a disjointed pseudo-presentation of a graph G in a surface Σ. Then $\Gamma, F_1^*, \ldots, F_r^*, \mu(F_1^*), \ldots, \mu(F_r^*), \mu$ is a presentation of G in Σ.*

[p. 33] **Proof.** Let $H_j = \mu(F_j^*)$ $(1 \le j \le r)$. We must check that (V1)–(V4) hold (and that $\mu(s) \subseteq H_j$ if $s \subseteq F_j^*$ is a space, which is obvious). For (V1),

$$G = \Gamma \cup \bigcup(\mu(F_j^*) : 1 \le j \le r) = \Gamma \cup H_1 \cup \cdots \cup H_r$$

from (P1). To see that $H_1, \ldots, H_r$ are mutually disjoint, suppose that $v \in V(H_{j_1} \cap H_{j_2})$ say. For $i = 1, 2$ let x_i be a unit in $F_{j_i}^*$ with $v \in V(\mu(x_i))$. From (P1), x_1 and x_2 belong to the same cuff C. We may assume that $k \le r$, and that for $1 \le j \le r, F_j^* \subseteq C$ if and only if $j \le k$, and that $F_1^*, \ldots, F_k^*$ occur in order around C. Let $s_1^*, \ldots, s_k^*$ be the spaces in C not in $F_1^* \cup \cdots \cup F_r^*$, numbered so that if $k > 1$ then s_j^* has one end in F_j^* and one in F_{j+1}^* for $1 \le j < k$, and s_k^* has one end in F_k^* and one in F_1^*. For $1 \le j \le k$, let F_j^* have ends a_j, b_j, where a_j is an end of s_j^* and b_j is an end of s_{j-1}^* (or of s_k^*, if $j = 1$). Let us write $b_{k+1} = b_1$ and $a_0 = a_k$. Now since the pseudo-presentation is disjointed, v does not belong to both $\mu(a_{j_1-1})$ and $\mu(b_{j_1})$; choose $u_1 \in \{a_{j_1-1}, b_{j_1}\}$ so that $v \notin V(\mu(u_1))$. Similarly choose $u_2 \in \{a_{j_1}, b_{j_1+1}\}$ with $v \notin V(\mu(u_2))$. Now x_1, u_1, x_2, u_2 occur in C in order, contrary to (P3), since $v \in V(\mu(x_1) \cap \mu(x_2))$ and $v \notin V(\mu(u_1) \cup \mu(u_2))$.

This proves that $H_1, \ldots, H_r$ are disjoint. To complete the proof of (V1), we must check that for $1 \le j \le r, V(H_j \cap \Gamma) = V(\Gamma) \cap F_j^*$, and $E(H_j \cap \Gamma) = \emptyset$. Now $V(\Gamma) \cap F_j^* \subseteq V(\mu(F_j^*)) = V(H_j)$, because $u \in V(\mu(u))$ for each $u \in V(\Gamma) \cap F_j^*$. Conversely, let $u \in V(H_j \cap \Gamma)$. Let x be a unit in F_j^* with $u \in V(\mu(x))$. Since $V(\mu(x) \cap \Gamma) = \{x\}$ if x is a vertex, and $V(\mu(x) \cap \Gamma)$ is the set of ends of x by (P2) if x is a space, it follows that either $\not\!u \in V(\Gamma) \cap F_j^*$ or $\not\!u$ is an end of some space in F_j^*; and so in either case, $\not\!u \in V(\Gamma) \cap F_j^*$. This proves that $V(H_j \cap \Gamma) = V(\Gamma) \cap F_j^*$. Finally, $E(H_j \cap \Gamma) = \emptyset$ because $E(\Gamma \cap \mu(\not\!s)) = \emptyset$ for each space $s \subseteq F_j^*$, by (P1). This proves (V1).[F37]

Now (V2) holds by definition, and (V3) is the same as (P2), and so it only remains to verify (V4). Let $1 \le j \le r$, and let $s_1, s_2 \subseteq F_j^*$ be spaces, and let $u \in F_j^* \cap V(\Gamma)$ lie in F_j^* between s_1 and s_2. Let s_0 be a space with $s_0 \not\subseteq F_1^* \cup \cdots \cup F_r^*$, in the same cuff C as F_j^*. (This exists since $|V(\Gamma) \cap C| \ge 2$.) Let s_0 have ends a_0 and b_0. Now a_0, s_1, u, s_2 occur in

[F37] **Misprint.** The vertex in this sentence is u, chosen just before, and v names the vertex of the paragraph before. Also $\mu(S)$ is $\mu(s)$, for the space s.

order in C, and so by (P3), $\mu(s_1) \cap \mu(s_2) \subseteq \mu(a_0) \cup \mu(s_2)\mu(u)$. Similarly, $\mu(s_1) \cap \mu(s_2) \subseteq \mu(b_0) \cup \mu(s_2)\mu(u)$, and so

$$\mu(s_1) \cap \mu(s_2) \subseteq (\mu(a_0) \cap \mu(b_0)) \cap \cup \mu(s_2)\mu(u) = \mu(s_2)\mu(u).^{F38}$$

Here $\mu(a_0) \cap \mu(b_0)$ is null, since the pseudo-presentation is disjointed and $s_0 \not\subseteq F_1^* \cup \cdots \cup F_r^*$. This proves (V4), as required. □

Secondly, we wish to discuss splitting vertices in a pseudo-presentation, in order to make it disjointed. More precisely, let $\Gamma, F_1^*, \ldots, F_r^*, \mu$ be a pseudo-presentation of G in Σ. Let s^* be a space with $s^* \not\subseteq F_1^* \cup \cdots \cup F_r^*$, with ends u_1, u_2 say. Let C be the cuff including s^*. Let $v \in V(\mu(u_1) \cap \mu(u_2))$.

[p. 34] From (P4), there exists $u^* \in V(\Gamma) \cap C$ with $v \notin V(\mu(u^*))$. Let F_i be the support in C with ends u_i and u^* which does not include s^* $(i = 1, 2)$. For $i = 1, 2$ let δ_i be the set of all edges e incident with v such that $e \in E(\mu(s))$ for some space $s \subseteq F_i \cap (F_1^* \cup \cdots \cup F_r^*)$. Then $\delta_1 \cap \delta_2 = \emptyset$, for by (P2) there is only one space $s \subseteq C$ with $e \in E(\mu(s))$; and $\delta_1 \cup \delta_2$ is the set of all edges of G incident with v, because by (P1) e is an edge of $\mu(s)$ for some space $s \subseteq C \cap (F_1^* \cup \cdots \cup F_r^*)$. Let G' be obtained from G by splitting v according to δ_1, δ_2, and let the two new vertices of G' be v_1, v_2 where the edges in δ_i are incident with v_i $(i = 1, 2)$. For $i = 1, 2$, and for each subgraph A of G with $v \in V(A)$, let $\sigma_i(A)$ be the graph with the same set of edges as A, and with vertex set $(V(A) \setminus \{v\}) \cup \{v_i\}$, with the same incidence relation as A except that the edges of A that are incident with v in A are incident with v_i in $\sigma_i(A)$. Then $\sigma_i(A)$ is a subgraph of G' provided that δ_i contains all edges of A incident with v. Define μ' by:

$\mu'(u) = \mu(u)$ if $\not u \in V(\Gamma) \cap bd(\Sigma)$ and $v \notin V(\mu(u))$[F39]
$\mu'(u) = \sigma_i(\mu(u))$ if $\not u \in V(\Gamma) \cap F_i$ and $v \in V(\mu(u))$ $(i = 1, 2)$
$\mu'(s) = \mu(s)$ if s is a space with $s \subseteq F_1^* \cup \cdots \cup F_r^*$ and $v \notin V(\mu(s))$
$\mu'(s) = \sigma_i(\mu(s))$ if s is a space with $s \subseteq (F_1^* \cup \cdots \cup F_r^*) \cap F_i$ and $v \in V(\mu(s))$ $(i = 1, 2)$.

8.2 $\not\Sigma\Gamma, F_1^*, \ldots, F_r^*, \mu'$ *is a pseudo-presentation of* G'.[F40]

[F38] **Misprint.** A misprint: (V4) asks for $\mu(s_1) \cap \mu(s_2) \subseteq \mu(u)$, and (P3), applied to a_0, s_1, u, s_2, gives $\mu(s_1) \cap \mu(s_2) \subseteq \mu(a_0) \cup \mu(u)$. So $\mu(s_2)$ on the right is $\mu(u)$, all four times, and the second $\cap$ in the display is $\cup$.

[F39] **Misprint.** A misprint: in the first two cases the condition is on u. The vertex v is the one being split, and $v \notin V(\Gamma)$, as the proof of (8.2) shows, so the first two cases as printed never apply.

[F40] **Misprint.** A pseudo-presentation starts with a drawing, as in (8.3), which says "suppose that $\Gamma, F_1^*, \ldots, F_r^*, \mu$ is a pseudo-presentation".

Proof. Since $v \in V(\mu(u_1) \cap \mu(u_2))$ it follows that $v \neq u_1, u_2$, and so $v \notin V(\Gamma)$. Consequently $\Gamma \subseteq G'$. Let us check that $\mu'(x) \subseteq G'$, for each unit x. If x is a vertex this is clear, so let x be a space with $x \subseteq F_1^* \cup \cdots \cup F_r^*$. If $v \notin V(\mu(x))$ then $\mu'(x) = \mu(x) \subseteq G'$ as required. We assume then that $v \in V(\mu(x))$, and $x \subseteq F_1$ say. Then $\mu'(x) = \sigma_1(\mu(x))$, and every edge of $\mu(x)$ incident with v belongs to δ_1, by definition of δ_1. Consequently $\mu'(x) \subseteq G'$, as required.

We must verify (P1)-(P4). For (P1), we observe that

$$E(G') = E(G) = E(\Gamma \cup \bigcup(\mu(F_j^*) : 1 \leq j \leq r)) = E(\Gamma \cup \bigcup(\mu'(F_j^*) : 1 \leq j \leq r))$$

and

$$\begin{aligned} V(G') &= (V(G) \setminus \{v\}) \cup \{v_1, v_2\} = (V(\Gamma \cup \bigcup(\mu(F_j^*) : 1 \leq j \leq r)) \setminus \{v\}) \cup \{v_1, v_2\} \\ &\subseteq V(\Gamma) \cup \bigcup(\mu'(F_j^*) : 1 \leq j \leq r) \end{aligned}$$

since $v_1 \in \mu'(F_1^*)$ and $v_2 \in \mu'(F_2^*)$. The remainder of (P1) is clear and so (P1) holds.

[p. 35] For (P2), let $s \subseteq F_j^*$ be a space with ends a, b say. Certainly $V(\mu'(s) \cap \Gamma) = V(\mu(s) \cap \Gamma) = \{a, b\}$. To see that $\mu'(a) \subseteq \mu'(s)$ we argue as follows. If $v \notin V(\mu(a))$ then

$$\mu'(a) = \mu(a) \subseteq \mu(s) \setminus v = \mu'(s) \setminus v \subseteq \mu'(s)$$

as required. We assume then that $v \in V(\mu(a))$, and hence $v \in V(\mu(s))$. Thus $s \subseteq F_1$ or $s \subseteq F_2$, say $s \subseteq F_1$; and so $a \in F_1$. Hence

$$\mu'(a) = \sigma_1(\mu(a)) \subseteq \sigma_1(\mu(s)) = \mu'(s)$$

as required. This verifies the second assertion of (P2). The third assertion follows because there is an isomorphism from $\mu(s)$ to $\mu'(s)$ mapping $\mu(a)$ to $\mu'(a)$ and $\mu(b)$ to $\mu'(b)$.

For (P3), let x_1, a, x_2, b be units of a cuff C in order, where $a, b \in V(\Gamma)$. We must show that

$$\mu'(x_1) \cap \mu'(x_2) \subseteq \mu'(a) \cup \mu'(b).$$

Since

$$E(\mu'(x_1) \cap \mu'(x_2)) = E(\mu(x_1) \cap \mu(x_2)) \subseteq E(\mu(a) \cup \mu(b)) = E(\mu'(a) \cup \mu'(b)),$$

it suffices to show that every vertex of $\mu'(x_1) \cap \mu'(x_2)$ is a vertex of $\mu'(a) \cup \mu'(b)$. Let $w \in V(\mu'(x_1) \cap \mu'(x_2))$. If $w \neq v_1, v_2$ then

$$w \in V(\mu(x_1) \cap \mu(x_2)) \subseteq V(\mu(a) \cup \mu(b))$$

and since $w \neq v$ it follows that $w \in V(\mu'(a) \cup \mu'(b))$ as required. We may assume then that $w = v_1$ say. Thus $v_1 \in V(\mu'(x_1) \cap \mu'(x_2))$. From the definition of $\mu'(x_1)$ and $\mu'(x_2)$ it follows that $v \in \mu(x_1) \cap \mu(x_2)$ and x_1, x_2 belong to F_1. Since x_1, a, x_2, b occur in order in C we may assume that $a \in F_1$ and lies in F_1 between x_1 and x_2. Since x_1, a, x_2, u^* occur in order in C, it follows that

$$v \in V(\mu(x_1) \cap \mu(x_2)) \subseteq V(\mu(a) \cup \mu(u^*))$$

and since $v \notin V(\mu(u^*))$ we deduce that $v \in V(\mu(a))$. But $a \in F_1$, and $\mu'(a) = \sigma_1(\mu(a))$, and so $v_1 \in V(\mu'(a))$, as required. This proves (P3).

For (P4), suppose that $w \in V(\mu'(u))$ for every $u \in V(\Gamma) \cap C'$ for some cuff C'. If $w \neq v_1, v_2$ then $w \in \mu(a)$ for all $u \in V(\Gamma) \cap C'$, and so $w \in \bigcap(\mu(u) : u \in V(\Gamma) \cap C')$, contrary to the truth of (P4) for μ. Thus we may assume that $w = v_1$ say, and hence $C' = C$. But $v_1 \notin V(\mu(u^*)) = V(\mu'(u^*))$, a contradiction. This proves (P4), and therefore completes the proof of (8.2). □

[p. 36] Now let $\Gamma, F_1^*, \ldots, F_r^*, \mu$ be a pseudo-presentation of G, and let $\mathcal{T}$ be a tangle in G. We say the pseudo-presentation *surrounds* $\mathcal{T}$ if $\mu(s)$ is small for every space $s \in F_1^* \cup \cdots \cup F_r^*$.

8.3 *With notation as in (8.2), suppose that $\Gamma, F_1^*, \ldots, F_r^*, \mu$ is a pseudo-presentation of depth $\leq q$, and surrounds a tangle $\mathcal{T}$ in G of order $\theta \geq 2q+3$. Then $\Gamma, F_1^*, \ldots, F_r^*, \mu'$ surrounds a tangle in G' of order $\theta - 1$.*

Proof. Let $\mathcal{T}'$ be the tangle of order $\theta - 1$ in G' induced by $\mathcal{T} \setminus \{v\}$. (This exists, from (3.2) and (3.4).) Let $s \subseteq F_1^* \cup \cdots \cup F_r^*$ be a space, and choose $B \subseteq G$ minimal such that $(\mu(s), B)$ is a separation of G. By (7.1), $(\mu(s), B) \in \mathcal{T}$ since $\mu(s)$ is small. Let $w \in V(\mu(s) \cap B)$; we claim that $w \in V(\mu(a) \cup \mu(b))$, where s has ends a and b. For if $w = a$ or b this is true, and we assume not. By (P2), $w \notin V(\Gamma)$. Since $u \in V(B)$ it follows from the minimality of B that some edge e of G not in $\mu(s)$ is incident with w, and $e \notin E(\Gamma)$ since $w \notin V(\Gamma)$. Choose a space s' with $e \in E(\mu(s'))$. Then $s' \neq s$, and so s, a, s', b are in order in the cuff containing them (this exists, by (P1)). By (P3),

$$w \in V(\mu(s) \cap \mu(s')) \subseteq V(\mu(a) \cup \mu(b))$$

as claimed. We have proved then that $V(\mu(s) \cap B) \subseteq V(\mu(a) \cup \mu(b))$. Hence $(\mu(s), B)$ has order $\leq 2q \leq \theta - 2$.

Let A_1 be the subgraph of G with vertex set $V(\mu(s)) \cup \{v\}$ and edge set $E(\mu(s))$; and let B_1 have vertex set $V(B) \cup \{v\}$ and edge set $E(B)$. By theorem (2.9)(iii) of [6], $(A_1, B_1) \in \mathcal{T}$, since $(\mu(s), B)$ has order $\leq \theta - 2$. Hence $(A_1 \setminus v, B_1 \setminus v) \in \mathcal{T} \setminus v$. Let $A_2 = \mu'(s)$, and let $B_2 \subseteq G'$ be minimal such that (A_2, B_2) is a separation of G'. Then (A_2, B_2) has order $\leq 2q$. But $A_2 \cap (G \setminus v) = A_1 \setminus v$, and so $(A_2 \cap (G \setminus v), B_2 \cap (G \setminus v)) \in \mathcal{T} \setminus v$ by theorem (2.9)(iii) of [6], since $(A_1 \setminus v, B_1 \setminus v) \in \mathcal{T} \setminus v$ and $\mathcal{T} \setminus v$ has order ≥ 2. From the definition of $\mathcal{T}'$ it follows that $(A_2, B_2) \in \mathcal{T}'$, and so $\mu'(s)$ is small with respect to $\mathcal{T}'$, as required. □

The main result of this section is the following.

8.4 *For every surface Σ and all integers $p, r \geq 0$ and $q \geq 1$, there exists $\theta \geq 1$ such that, if a graph G has a pseudo-presentation in Σ of depth $\leq q$ and defect $\leq r$ which surrounds a tangle of order $\geq \theta$, then G has no vital p-linkage.*

Proof. Choose $\theta' \geq 2q+2$ so that (7.3) holds, with Σ, p, q, r replaced by $\Sigma, p + 2qr, q, r$. Let $\theta = \theta' + qr$. We claim that θ satisfies (8.4). For suppose that G has a pseudo-presentation in Σ of [p. 37] depth $\leq q$ and defect $\leq r$ which surrounds a tangle of order $\geq \theta$. For each space $s \not\subseteq F_1^* \cup \cdots \cup F_r^*$ with ends u_1, u_2 and for each $v \in V(\mu(u_1) \cap \mu(u_2))$, split v as discussed earlier in this section. After at most qr splittings we obtain a graph G' with a disjointed pseudo-presentation in Σ of depth $\leq q$ and defect $\leq r$, which by (8.3) surrounds a tangle of order $\geq \theta - qr = \theta'$, since $\theta' + 1 \geq 2q + 3$. By (8.1), G' has a presentation in Σ of depth $\leq q$ and defect $\leq r$ which surrounds a tangle of order $\geq \theta'$. By (7.3), G' has no vital $(p+2qr)$-linkage, from the choice of θ'. But G' is obtained from G by splitting $\leq qr$ vertices, and so by (2.4) G has no vital p-linkage, as required. □

9 Paintings and portraits

In this section we derive a modification of (8.4) appropriate for applying the results of [9, 10]. A *painting* Γ in a surface Σ is a pair (U, V), where $U \subseteq \Sigma$ is closed and $V \subseteq U$ is finite, satisfying

(i) $U \setminus V$ has only finitely many connected components, called *cells*

(ii) for each cell c, its closure $\bar{c}$ is a closed disc, and $\tilde{c} \subseteq bd(\bar{c})$ (where $\tilde{c}$ denotes $\bar{c} \setminus c$) and $|\tilde{c}| = 2$ or 3

(iii) $bd(\Sigma) \subseteq U$

(iv) for each cell c, if $c \cap bd(\Sigma) \neq \emptyset$ (that is, c is a border cell) then $|\tilde{c}| = 2$ and $\bar{c} \cap bd(\Sigma)$ is a line with ends the two members of $\tilde{c}$.

(This differs very slightly from the definition in [10]. In that paper it was convenient to allow cells c with $|\tilde{c}| = 0$ or 1, but we no longer need such cells.) The set of cells of Γ is denoted by $C(\Gamma)$, and we write $U = U(\Gamma), V = V(\Gamma)$. A cell which is not a border cell is called an *internal cell*.

Let G be a graph. We say that a pair Γ, α is a *portrait* of G in Σ (this is closely connected with the "portrayal" of [10]) if

(R1) Γ *is a painting in* Σ *and* $V(\Gamma) \subseteq V(G)$*, and* α *is a function with domain* $(V(\Gamma) \cap bd(\Sigma)) \cup C(\Gamma)$.

(R2) *For each* c *of* Γ*,* $\alpha(c) \subseteq G$ *and* $V(\alpha(c)) \cap V(\Gamma) = \tilde{c}$.

(R3) $G = \bigcup(\alpha(c) : c \in C(\Gamma))$*, and* $E(\alpha(c_1) \cap \alpha(c_2)) = \emptyset$ *for all distinct cells* c_1, c_2.

(R4) *For each* $v \in V(\Gamma) \cap bd(\Sigma)$*,* $\alpha(v)$ *is an edge-less subgraph of* G *and* $V(\alpha(v)) \cap V(\Gamma) = \{v\}$.

(R5) *For each cuff* C *there is an integer* $q(C) \geq 1$ *such that* $|V(\alpha(v))| = q(C)$ *for all* $v \in V(\Gamma) \cap C$.

[p. 38] (R6) *For each border cell* c *with* $\tilde{c} = \{u, v\}$*,* $\alpha(u)$ *and* $\alpha(v)$ *are subgraphs of* $\alpha(c)$*, and either*

(i) *there are* $|V(\alpha(u))|$ *mutually vertex-disjoint paths of* $\alpha(c)$ *between* $V(\alpha(u))$ *and* $V(\alpha(v))$*, that is,* c *is "linked", or*

(ii) *there is a cell* c' *with* $|\tilde{c}'| = 3$ *and* $u, v \in \tilde{c}'$.

(R7) *If* $c_1, c_2 \in C(\Gamma)$ *are distinct, then* $V(\alpha(c_1) \cap \alpha(c')) = \tilde{c}_1 \cap \tilde{c}_2$ *unless* c_1 *and* c_2 *border the same cuff (that is, unless* $c_1 \cap C \neq \emptyset \neq c_2 \cap C$ *for some cuff* C*).*

(R8) *If* $c_1, c_2 \in C(\Gamma)$ *border a cuff* C*, and* $u_1, u_2 \in V(\Gamma) \cap C$ *and* c_1, u_1, c_2, u_2 *are in order around* C*, then* $\alpha(c_1) \cap \alpha(c_2) \subseteq \alpha(u_1) \cup \alpha(u_2)$.

(R9) *If* $c \in C(\Gamma)$ *and* $|\tilde{c}| = 3$*, then* $\alpha(c)$ *cannot be drawn in a disc so that the three members of* $\tilde{c}$ *are drawn in the boundary of the disc.*

(R10) *For each cuff* $C, \bigcap(\alpha(u) : u \in V(\Gamma) \cap C)$ *is null.*

Its depth is the maximum of $q(C)$, taken over all cuffs C (or 0 if $bd(\Sigma) = \emptyset$).

9.1 *If* Γ, α *is a portrait in a surface* Σ *of a graph* G*, and* L *is a vital linkage in* G*, then for every cell* c *of* Γ *which is not a border cell, either* $V(\alpha(c)) \setminus \tilde{c}$ *contains a terminal of* L*, or* $\alpha(c)$ *can be drawn in a disc with the members of* $\tilde{c}$ *drawn in the boundary.*

Proof. Let $c \in C(\Gamma)$ be an internal cell such that $V(\alpha(c)) \setminus \tilde{c}$ contains no terminal of L. We must prove that $\alpha(c)$ can be drawn in a disc with $\tilde{c}$ in the boundary. If $V(\alpha(c)) = \tilde{c}$ this is clear, and we therefore may assume that $V(\alpha(c)) \neq \tilde{c}$. Since by (R2) $\tilde{c} \subseteq V(\alpha(c))$, there is a vertex in $V(\alpha(c)) \setminus \tilde{c}$. Since $V(L) = V(G)$, there is a component P of L with $V(P \cap (\alpha(c))) \not\subseteq \tilde{c}$. Since $V(\alpha(c)) \setminus \tilde{c}$ contains no end of P, and c is internal, it follows from (R7)

that $|V(P) \cap \tilde{c}| \geq 2$; and so P is the only component of L with a vertex in $V(\alpha(c)) \setminus \tilde{c}$, since $|\tilde{c}| \leq 3$. Consequently, $V(\alpha(c)) \setminus \tilde{c} \subseteq V(P)$. But L is vital, and so any two vertices of $P \cap \alpha(c)$ that are adjacent in $\alpha(c)$ are also adjacent in P. Hence either $\alpha(c)$ is a path with both ends in $\tilde{c}$, or $\alpha(c) \setminus v$ is a path with both ends in $\tilde{c}$, for some $v \in \tilde{c}$. In either case $\alpha(c)$ can be drawn in a disc with all the members of $\tilde{c}$ drawn in the boundary, as required. □

If $\mathcal{T}$ is a tangle in G, a portrait Γ, α of G in Σ *surrounds* $\mathcal{T}$ if $\alpha(c)$ is small for each $c \in C(\Gamma)$.

9.2 *For every surface Σ, and all integers $p \geq 0$ and $q \geq 1$, there exists $\theta \geq 1$ such that, if a graph G has a portrait in Σ of depth $\leq q$ which surrounds a tangle of order $\geq \theta$, then G has no vital p-linkage.*

[p. 39] **Proof.** Let $n = 3p + 2q(3p + c(\Sigma))$. Choose $\theta' > 2q$ so that (8.4) holds, with Σ, p, q, r, θ replaced by $\Sigma, p+n, q, r, \theta'$. Let $\theta = \theta' + n$. We claim that θ satisfies (9.2). For let Γ, α be a portrait in Σ of a graph G, with depth $\leq q$, surrounding a tangle $\mathcal{T}$ of order $\geq \theta$. Suppose that L is a vital p-linkage in G. Let $\mathcal{A}_1$ be the set of all internal cells c such that $V(\alpha(c)) \setminus \tilde{c}$ contains a terminal of L. Since the sets $V(\alpha(c)) \setminus \tilde{c}$ ($c \in \mathcal{A}_1$) are mutually disjoint, by (R7), it follows that $|\mathcal{A}_1| \leq p$. By (9.1) and (R9), $\mathcal{A}_1$ contains every cell of Γ with $|\tilde{c}| = 3$. By (R6), there are $\leq 3|\mathcal{A}_1|$ border cells that are not linked. Let $\mathcal{A}_2 \subseteq C(\Gamma)$ be minimal such that $\mathcal{A}_2$ contains all border cells that are not linked, and contains at least one cell bordering each cuff. Thus, $|\mathcal{A}_2| \leq 3p + c(\Sigma)$.

For each internal cell $c \in C(\Gamma) \setminus \mathcal{A}_1$, there is a drawing of $\alpha(c)$ in the closed disc $\bar{c}$ with the vertices in $\tilde{c}$ representing themselves, by (9.1), and this drawing can be chosen so that it meets $bd(\bar{c})$ only in $\tilde{c}$. We may therefore assume, to simplify the notation, that $\alpha(c)$ is such a drawing in $\bar{c}$ for each internal cell $c \in C(\Gamma) \setminus \mathcal{A}_1$. Let Γ' be the drawing formed by the vertices in $V(\Gamma)$ and the union of the drawings $\alpha(c)$ over all internal $c \in C(\Gamma) \setminus \mathcal{A}_1$. Then Γ' is a drawing in Σ, and there is a 1-1 correspondence between the spaces of Γ' and the border cells of Γ.

Let G' be the subgraph of G formed by the vertices in $V(\Gamma)$, the graphs $\alpha(u)$ ($u \in V(\Gamma') \cap bd(\Sigma)$) and the graphs $\alpha(c)$ ($c \in C(\Gamma) \setminus (\mathcal{A}_1 \cup \mathcal{A}_2)$). Then $\Gamma' \subseteq G'$. Now $V(\Gamma') \cap bd(\Sigma) = V(\Gamma) \cap bd(\Sigma)$; define $\mu(u) = \alpha(u)$ for each $u \in V(\Gamma') \cap bd(\Sigma)$. Let $F_1^*, \ldots, F_r^*$ be the connected components of $bd(\Sigma) \setminus \bigcup(c : c \in \mathcal{A}_2)$. Then $F_1^*, \ldots, F_r^*$ are all supports of Γ' since $\mathcal{A}_2$ contains at least one cell bordering each cuff; they are mutually disjoint, and $V(\Gamma') \cap bd(\Sigma) \subseteq F_1^* \cup \cdots \cup F_r^*$. Moreover, $r = |\mathcal{A}_2| \leq 3p + c(\Sigma)$. For each space s of Γ' with $s \subseteq F_1^* \cup \cdots \cup F_r^*$, let $\mu(s) = \alpha(c)$, where c is the cell of Γ with $c \cap bd(\Sigma) = s$.

(1) *$\Gamma', F_1^*, \ldots, F_r^*, \mu$ is a pseudo-presentation of G'.*

Subproof. Certainly $|V(\Gamma') \cap C| \geq 2$ for each cuff C, because Γ is a painting and $V(\Gamma) \subseteq V(\Gamma')$. We must verify (P1)-(P4). First let us verify (P1). Now

$$G' = \Gamma' \cup \bigcup(\alpha(c) : c \in C(\Gamma) \setminus \mathcal{A}_2 \text{ and } c \text{ is a border cell }) \cup \bigcup(\alpha(u) : u \in V(\Gamma') \cap bd(\Sigma))$$
$$= \Gamma' \cup \mu(F_1^*) \cup \cdots \cup \mu(F_r^*).$$

By (R3), $E(\mu(s) \cap \mu(s')) = \emptyset$ if s and s' are distinct spaces, since each border cell of Γ includes only one space of Γ'. Since each edge of Γ' is an edge of $\alpha(c)$ for some internal cell c, it follows from (R3) again that $E(\Gamma' \cap \mu(s)) = \emptyset$ for each space $s \subseteq F_1^* \cup \cdots \cup F_r^*$. Finally, let x_1, x_2 be units of Γ in [p. 40] different cuffs. Then there are border cells c_1, c_2 of Γ bordering different cuffs with $\mu(x_i) \subseteq \alpha(c_i)$, by (R6) and condition (iii) in the definition of a painting. By condition (iv) in the definition of a painting, $\tilde{c} \cap \tilde{c}' = \emptyset$, and so by (R7), $\alpha(c_1) \cap \alpha(c_2)$ is null. Consequently, $\mu(x_1) \cap \mu(x_2)$ is null. This proves (P1).

For (P2), let $1 \leq j \leq r$ and let $s \subseteq F_j^*$ be a space of Γ', with ends u_1, u_2. Let c be the cell of Γ with $c \cap bd(\Sigma) = s$; then $\tilde{c} = \{u_1, u_2\}$. Since $s \subseteq F_j^*$ it follows that $c \notin \mathcal{A}_2$, and so c is linked. Consequently, (P2) holds, taking $q_j = q(C)$ where C is the cuff including F_j^* and (see (R5)) $q(C)$ is the common cardinality of all the sets $V(\alpha(v))$ $(v \in V(\Gamma) \cap C)$.

For (P3), let C be a cuff, let x_1, x_2 be units of Γ' in C, and let $u_1, u_2 \in C \cap V(\Gamma)$ so that x_1, u_1, x_2, u_2 are in order in C. We must show that $\mu(x_1) \cap \mu(x_2) \subseteq \mu(u_1) \cup \mu(u_2)$. If one of x_1, x_2 equals one of u_1, u_2 then the inclusion is trivial, and so we assume that $x_1, x_2 \neq u_1, u_2$. For $i = 1, 2$, let c_i be a border cell of Γ with $c_i \cap bd(\Sigma) = x_i$ if x_i is a space of Γ', and $x_i \in \tilde{c}_i$ if $x_i \in V(\Gamma')$. By (R6),

$$\mu(x_i) = \alpha(x_i) \subseteq \alpha(c_i) \quad (i = 1, 2)$$

and so $\mu(x_1) \cap \mu(x_2) \subseteq \alpha(c_1) \cap \alpha(c_2)$. But c_1, u_1, c_2, u_2 are in order, and so by (R8),

$$\alpha(c_1) \cap \alpha(c_2) \subseteq \alpha(u_1) \cup \alpha(u_2) = \mu(u_1) \cup \mu(u_2).$$

This proves (P3).

But (P4) is immediate from (R10). This completes the proof of (1).

For each internal $c \in C(\Gamma)$, let $B(c) \subseteq G$ be such that $(\alpha(c), B(c))$ is a separation of G and $V(\alpha(c) \cap B(c)) = \tilde{c}$. (This is possible by (R7).) For each border cell c, let $B(c) \subseteq G$ be such that $(\alpha(c), B(c))$ is a separation of G and $\alpha(c) \cap B(c) = \mu(u_1) \cup \mu(u_2)$, where $\tilde{c} = \{u_1, u_2\}$. (This is possible by (R7) and (R8).)

(2) $G' = G \cap \bigcap(B(c) : c \in \mathcal{A}_1 \cup \mathcal{A}_2)$.

Subproof. Certainly $G' \subseteq B(c)$ for each $c \in \mathcal{A}_1 \cup \mathcal{A}_2$. For the converse inclusion, if $e \in E(G)$ is an edge of $B(c)$ for all $c \in \mathcal{A}_1 \cup \mathcal{A}_2$, then by (R3) e is an edge of $\alpha(c)$ for some $c \in C(\Gamma) \setminus (\mathcal{A}_1 \cup \mathcal{A}_2)$, and so $e \in E(G')$. If $v \in V(G)$ is a vertex of $B(c)$ for all $c \in \mathcal{A}_1 \cup \mathcal{A}_2$, choose $c \in C(\Gamma)$ with $v \in V(\alpha(c))$. If $c \in C(\Gamma) \setminus \mathcal{A}_1 \cup \mathcal{A}_2$ then $v \in V(G')$ as required. If $c \in \mathcal{A}_1$ then $v \in \tilde{c}$ because $v \in V(B(c) \cap \alpha(c)) = \tilde{c}$, and so $v \in V(\Gamma) \subseteq V(G')$ as required. If $c \in \mathcal{A}_2$, then $v \in V(\mu(u_1) \cup \mu(u_2))$ where $\tilde{c} = \{u_1, u_2\}$ for the same reason, and so again $v \in V(G')$. This proves (2).

[p. 41] Let $A_0 = \bigcup(\alpha(c) : c \in \mathcal{A}_1 \cup \mathcal{A}_2)$. From (2), (A_0, G') is a separation of G. Since every vertex of $V(A_0 \cap G')$ belongs to $V(\alpha(c) \cap B(c))$ for some $c \in \mathcal{A}_1 \cup \mathcal{A}_2$, it follows that (A_0, G') has order

$$\leq 3|\mathcal{A}_1| + 2q|\mathcal{A}_2| \leq 3p + 2q(3p + c(\Sigma)) = n.$$

We deduce from (3.5) that

(3) *There is a tangle $\mathcal{T}'$ in G' of order $\geq \theta - n = \theta'$, such that $(A \cap G', B \cap G') \in \mathcal{T}'$ for every $(A, B) \in \mathcal{T}$ of order $< \theta'$.*

(4) *The pseudo-presentation $\Gamma', F_1^*, \ldots, F_r^*, \mu$ of G' surrounds $\mathcal{T}'$.*

Subproof. Let s be a space of Γ' in Σ with $s \subseteq F_1^* \cup \cdots \cup F_r^*$, and let $c \in C(\Gamma)$ with $c \cap bd(\Sigma) = s$. Then c is a border cell, and $c \notin \mathcal{A}_2$ by definition of $F_1^* \cup \cdots \cup F_r^*$. Now $(\alpha(c), B(c)) \in \mathcal{T}$ by (7.1) since it has order $\leq 2q < ord(\mathcal{T})$ and $\alpha(c)$ is small with respect to $\mathcal{T}$. By (3), $(\alpha(c) \cap G', B(c) \cap G') \in \mathcal{T}'$ since $(\alpha(c), B(c))$ has order $\leq 2q < \theta'$. Hence $\alpha(c) \cap G'$ is small with respect to $\mathcal{T}'$. But $\alpha(c) \subseteq G'$ since $c \in C(\Gamma) \setminus (\mathcal{A}_1 \cup \mathcal{A}_2)$, and $\alpha(c) = \mu(s)$. Consequently $\mu(s)$ is small with respect to $\mathcal{T}'$. This proves (4).

(5) *There is no vital $(p+n)$-linkage in G'.*

Subproof. This follows from (1), (4) and (8.4), because of the choice of θ'.

Since (A_0, G') is a separation of G of order $\leq n$, it follows from (5), (2.2) and ~~(2.4)~~(2.3) that there is no vital p-linkage in G[F41], as required. □

10. The main proof

Finally we are able to apply the theorems of [10]. Theorems (8.2), (8.4), (9.8) and particularly (13.4) of [10] imply the following. (A K_n-minor of G is a minor isomorphic to K_n.)

[F41] **Misprint.** The references meant are (2.2) and (2.3), since (2.4) is about splitting a vertex. If L were a vital p-linkage in G, then by (2.2) and (2.3), $L \cap G'$ would be a vital $(p+n)$-linkage in G', contrary to (5). Claim (5) in the proof of (5.2) makes the same step "from (2.2) and (2.3)".

10.1 *For every integer $n \geq 0$ there exist $\rho, \xi \geq 0, q \geq 1$ and $\theta > \xi$ such that, if a graph G has no K_n-minor, and $\mathcal{T}$ is a tangle in G of order $\geq \theta$, then there exists a surface Σ such that $c(\Sigma) \leq \rho$ and K_n cannot be drawn in Σ, and there exists $Z \subseteq V(G)$ with $|Z| \leq \xi$ such that there is a portrait of $G \setminus Z$ in Σ with depth $\leq q$, surrounding $\mathcal{T} \setminus Z$.*

[p. 42] We deduce

10.2 *For all integers $n, p \geq 0$ there exists $\theta \geq 1$ such that, if a graph G has a tangle of order $\geq \theta$ and has no K_n-minor then G has no vital p-linkage.*

Proof. Choose $\rho, \xi \geq 0, q \geq 1$ and $\theta_1 > \xi$ so that (10.1) holds with θ replaced by θ_1. Choose $\theta_2 \geq 1$ so that for all surfaces Σ with $c(\Sigma) \leq \rho$ in which K_n cannot be drawn, (9.2) holds, with Σ, p, q, θ replaced by $\Sigma, p + 2\zeta, q, \theta_2$. Let $\theta = \max(\theta_1, \xi + \theta_2)$. We claim that θ satisfies (10.2). For let G have a tangle $\mathcal{T}$ of order $\geq \theta$, and have no K_n-minor. By (10.1), since $\theta \geq \theta_1$, there is a surface Σ with $c(\Sigma) \leq \rho$ in which K_n cannot be drawn, and there exists $Z \subseteq V(G)$ with $|Z| \leq \xi$ such that there is a portrait of $G \setminus Z$ in Σ with depth $\leq q$, surrounding $\mathcal{T} \setminus Z$. Since $\mathcal{T} \setminus Z$ has order $\geq \theta - |Z| \geq \theta_2$, it follows from (9.2) that $G \setminus Z$ has no vital $(p + 2\xi)$-linkage. Hence by (2.1), G has no vital p-linkage, as required. □

10.3 *For every integer $p \geq 0$, no graph with a K_n-minor has a vital p-linkage, where $n = \lfloor \frac{5}{2} p \rfloor + 1$.*

Proof. Suppose that L is a vital p-linkage in a graph G with a K_n-minor. Let Z be the set of terminals of L. By theorem (6.1) of [8], there exists $v \in V(G) \setminus Z$ such that, in the language of [8], G and $G \setminus v$ have the same 0-folio relative to Z. In particular, there is a linkage in $G \setminus v$ with the same pattern as L, a contradiction since L is vital. The result follows. □

From (10.2) and (10.3) we deduce the following, which in view of (3.1) implies our main result (1.1).

10.4 *For every integer $p \geq 0$ there exists $\theta \geq 1$ such that, if a graph G has a tangle of order $\geq \theta$ then G has no vital p-linkage.*

Proof. Let n be as in (10.3), and choose θ so that (10.2) is satisfied. Let G be a graph with a tangle of order $\geq \theta$. If G has no K_n-minor then by (10.2) G has no vital p-linkage. If G has a K_n-minor then by (10.3) G has no vital p-linkage. The result follows. □

11. Intertwinings

The "intertwining conjecture", of Lovász [2] and of Milgram and Ungar [3], states that for every two graphs G_1 and G_2, there is a finite list $H_1, \ldots, H_n$ of graphs, such that a graph G topologically [p. 43] contains

both G_1 and G_2 if and only if it topologically contains one of $H_1, \ldots, H_n$. (G *topologically contains* H if some subgraph of G is isomorphic to a subdivision of H.) This conjecture was proved in [11] by well-quasi-ordering methods. Our object here is to give a different proof of the intertwining conjecture, one that is "constructive" in the sense that it yields an algorithm which, given G_1 and G_2, computes $H_1, \ldots, H_n$ as above.

Poljak and Turzik [4] showed that the intertwining conjecture is (constructively) implied by another conjecture, (11.1) below, and we shall give a constructive proof of (11.1). Let L_1 and L_2 be linkages in a graph G. We say that G is an *intertwining* of L_1 and L_2 if $L_1' \cup L_2' = G$ for all linkages L_1', L_2' in G such that L_i' has the same pattern as L_i $(i = 1, 2)$. Let us say a graph G has *rank*

$$|E(G)| - |V(G)| + \kappa(G),$$

where $\kappa(G)$ denotes the number of components of G. Poljak and Turzik essentially reduced proving the intertwining conjecture to proving the following.

11.1 *For every integer $p \geq 0$ there exists $r(p)$ such that every intertwining of two p-linkages has rank $\leq r(p)$.*

We shall derive (11.1) from (1.1), and the proof will be constructive in the sense that it yields an algorithm to compute a value for $r(p)$ given a numerical value for p. Our thanks to A. Gupta and R. Impagliazzo, who proved (11.1) constructively for planar intertwinings (unpublished) and brought the general problem to our attention. Thanks also to R. Thomas, who collaborated with us in discovering a constructive proof of (11.1) (not exactly the one given here).

Let (T, W) be a tree-decomposition of a graph G. If S is a subtree of T, we define $W(S) = \bigcup(W(t) : t \in V(S))$. If e is an edge of T, and T_1, T_2 are the two components of $T \setminus e$ and t_i is the end of e in T_i $(i = 1, 2)$, then $(W(T_1), W(T_2))$ is a separation of G, provided that no edge of G belongs to $W(t)$ for two distinct $t \in V(T)$, and $W(T_1) \cap W(T_2) = W(t_1) \cap W(t_2)$.[F42] We define $W(e) = V(W(T_1) \cap W(T_2))$.

We say that a tree-decomposition (T, W) of G is *linked* if for all $e_1, e_2 \in E(T)$ such that $|W(e_1)| = |W(e_2)| = k$ say, and $|W(e)| \geq k$ for all edges e of

[F42] **Statement mistake.** As printed, the first claim is false: a separation (A, B) has $E(A \cap B) = \emptyset$, and the definition of a tree-decomposition allows two of the graphs $W(t)$ to share an edge. For example, let G be one edge uv, let T have one edge t_1t_2, and let $W(t_1) = W(t_2) = G$. Then $W(T_1) \cap W(T_2) = G$. The second claim holds as printed, and $W(e)$ below uses only vertices, so it is not affected. Every graph of tree-width $\leq w$ has a linked tree-decomposition of width $\leq w$ in which no edge is in two of the graphs $W(t)$ (see the proof of (11.2)).

the path of T between e_1 and e_2, there are k mutually vertex-disjoint paths of G between $W(e_1)$ and $W(e_2)$. We need the following lemma.

11.2 *Let G be a graph of tree-width $\leq w$. Then there is a linked tree-decomposition (T, W) of G, with width $\leq w$, such that every vertex of T has degree ≤ 3.*

[p. 44] **Proof.** By the main result of [14], there is a linked tree-decomposition (T, W) of G, with width $\leq w$, such that no edge of G belongs to $W(t)$ for two distinct $t \in V(T)$ (from any linked tree-decomposition, delete each edge of G from all but one of the graphs $W(t)$ that contain it; this keeps the sets $W(e)$).[F43] Choose it such that $\Sigma(d(t) - 3)$ is minimum, where $d(t)$ is the degree of t in T and the sum is over all $t \in V(T)$ with degree ≥ 3. Suppose that some $t_0 \in V(T)$ has degree ≥ 4. Then we may assume that there is a tree T' with an edge e_0 with ends t_1, t_2, such that T is obtained from T' by contracting e_0, and t_0 results from identifying t_1 and t_2, and such that t_1 and t_2 both have degree ≥ 3. Let $W'(t) = W(t)$ if $t \in V(T') \setminus \{t_1, t_2\}$, and let $W'(t_1), W'(t_2)$ be graphs both with vertex set $W(t_0)$, so that $W'(t_1) \cup W'(t_2) = W(t_0)$ and $E(W'(t_1) \cap W'(t_2)) = \emptyset$. Then (T', W') is a tree-decomposition of G with width $\leq w$, and $W'(e_0) = V(W(t_0))$, and if $e \in E(T') \setminus \{e_0\}$ then $e \in E(T)$ and $W'(e) = W(e)$. Let $e_1, e_2 \in E(T')$ with $|W'(e_1)| = |W'(e_2)| = k$ say, and $|W'(e)| \geq k$ for all edges e of the path of T' between e_1 and e_2. We claim that there are k mutually vertex-disjoint paths between $W'(e_1)$ and $W'(e_2)$. If $e_1, e_2 \neq e_0$, then $e_1, e_2 \in E(T)$ and the claim therefore holds, since $W'(e_i) = W(e_i)$ $(i = 1, 2)$[F44] and (T, W) is linked. If $e_1 = e_2 = e_0$ the claim is trivial. If $e_1 = e_0$ and $e_2 \neq e_0$, let e_3 be the edge of T different from e_1, between e_1 and e_2 with a common end t_1 say with e_1. Then $|W(e_3)| \geq k$, but

$$W(e_3) \subseteq V(W(t_1)) = W(e_0)$$

and $|W(e_0)| = k$; and so $W(e_0) = W(e_3)$. Since there are k mutually disjoint paths of G between $W(e_3) = W(e_0)$ and $W(e_2)$, the claim follows. Consequently (T', W') is linked, contrary to the choice of (T, W). We deduce that there is no such t_0, and so ~~(i) holds~~every vertex of T has degree ≤ 3,

[F43] **Proof gap.** Without this property the graphs $W'(t_1), W'(t_2)$ below need not give a tree-decomposition. If an edge f of $W(t_0)$ is also in graphs $W(t)$ on both sides of e_0, then it is in only one of $W'(t_1), W'(t_2)$, and the path of T' between two vertices whose graphs contain f passes through the other one. For example, let G be one edge f, let T be a star with center t_0 and four leaves, and let every $W(t)$ be G. With the property, f is in no $W(t)$ with $t \neq t_0$, and (T', W') has the property too.

[F44] **Misprint.** The reason is $W'(e_i) = W(e_i)$ for $i = 1, 2$, which the proof showed a line earlier for every edge other than e_0.

as required.[F45] □

Secondly, we need the following.

11.3 *Let (T, W) be a tree-decomposition of G, and let $e_1, e_2 \in E(T)$ be distinct. Let T_0, T_1, T_2 be the three components of $T \setminus e_1, e_2$, where e_i has an end in $V(T_0)$ and in $V(T_i)$ $(i = 1, 2)$. Suppose that $W(T_0) \subseteq W(T_1) \cap W(T_2)$. Let T' be the tree obtained from $T_1 \cup T_2$ by adding an edge joining the end of e_1 in T_1 to the end of e_2 in T_2; and let W' be the restriction of W to $V(T')$. Then (T', W') is a tree-decomposition of G, and if (T, W) is linked then so is (T', W').*

Proof. It is straightforward to check that (T', W') is a tree-decomposition of G. To see this, let e_i have ends $a_i \in V(T_i)$ and $c_i \in V(T_0)$ $(i = 1, 2)$. Since T_1, T_2 are disjoint trees and the new edge joins a_1 to a_2, T' is a tree. Let $x, z \in V(T')$. If x, z are both in T_1 or both in T_2, the path of T between them lies in that tree and is also their path in T'. Otherwise the path of T between them passes through a_1, c_1, c_2, a_2 (since e_1, e_2 are the only edges between T_0 and $T_1 \cup T_2$), and their path in T' is obtained from it by replacing its subpath between a_1 and a_2 by the new edge. In both cases every vertex y of the path of T' between x and z is on the path of T between them, and so $W'(x) \cap W'(z) \subseteq W'(y)$. Since $W(T_0) \subseteq W(T_1) \cap W(T_2)$, every vertex and edge of G belongs to $W(T_1) \cup W(T_2)$. Hence (T', W') is a tree-decomposition of G. Let $X = V(W(T_1) \cap W(T_2))$. Since $W(T_0) \subseteq W(T_1) \cap W(T_2)$ it follows that $W(e) = X$ for every edge e of the path of T between e_1 and e_2, and now the second claim follows easily. To see this, let $f \in E(T')$. If $f \neq a_1a_2$ then $f \in E(T)$, and $W'(f) = W(f)$, since both are the vertex set of the intersection of the graphs at the two ends of f; let $\bar{f} = f$. Since T_1, T_2 are the components of $T' \setminus a_1a_2$, we have $W'(a_1a_2) = X = W(e_1)$; let $\bar{f} = e_1$ if $f = a_1a_2$. Now let $f_1, f_2 \in E(T')$ be distinct (if $f_1 = f_2$ there is nothing to prove), with $|W'(f_1)| = |W'(f_2)| = k$, such that $|W'(f)| \geq k$ for all edges f of the path R' of T' with first edge f_1 and last edge f_2. Let R be the path of T with first edge $\bar{f}_1$ and last edge $\bar{f}_2$, and let $e \in E(R)$. If $e \in E(T_1 \cup T_2)$, then $e \in E(R')$ by the description of paths above, and so $|W(e)| = |W'(e)| \geq k$. Otherwise e is on the path of T between a_1 and a_2, so $W(e) = X$, and $a_1a_2 \in E(R')$ (since either a_1a_2 is f_1 or f_2, or R passes from T_1 to T_2); hence $|W(e)| = |W'(a_1a_2)| \geq k$. Since (T, W) is linked, there are k mutually vertex-disjoint paths of G between $W(\bar{f}_1) = W'(f_1)$ and $W(\bar{f}_2) = W'(f_2)$. Hence (T', W') is linked.[F46] □

[F45] **Misprint.** The statement of (11.2) has no (i). What is meant is its conclusion, that every vertex of T has degree ≤ 3.

[F46] **Missing proof.** The paper leaves both checks to the reader. The arguments above are the ones used in the formalization.

Proof of (11.1). Choose $w \geq 0$ so that (1.1) is satisfied. Choose $n \geq 0$ so that (2.7) is satisfied with k replaced by $w+1$, and let $r(p) = 2^{n-1}w(w+1)$. We claim that (11.1) holds. For we prove [p. 45] by induction on $|V(G)| + |E(G)|$ that if G is an intertwining of two p-linkages L_1 and L_2 then G has rank $\leq r(p)$.

(1) *We may assume that G has no isolated vertices.*

Subproof. Suppose that $v \in V(G)$ has degree 0. For $i = 1, 2$, let $L_i' = L_i$ if $v \notin V(L_i)$, and $L_i' = L_i \setminus v$ if $v \in V(L_i)$. Then $G \setminus v$ is an intertwining of the p-linkages L_1' and L_2'[F47], and from our inductive hypothesis $G \setminus v$ has rank $\leq r(p)$. But G and $G \setminus v$ have the same rank, and the result follows. This proves (1).

(2) *We may assume that $V(L_1) = V(L_2) = V(G)$.*

Subproof. Suppose that $v \in V(G) \setminus V(L_2)$ say. By (1) there is an edge e of G incident with v, and since $L_1 \cup L_2 = G$ it follows that $e \in E(L_1)$. The graph G/e (that is, obtained from G by contracting e) is an intertwining of L_1/e and L_2, if we regard L_2 as a subgraph of G/e in the obvious way; and L_1/e and L_2 are both p-linkages. We give the details. Let e have ends u, v, and let G/e be obtained from $G \setminus e$ by identifying v with u. For a subgraph K of G with $v \notin V(K)$ or $e \in E(K)$, let K/e be the corresponding subgraph of G/e. Then L_2, which contains neither v nor e, is a linkage of G/e with the same pattern, and L_1/e is a linkage of G/e whose pattern is obtained from that of L_1 by replacing v by u, and which has no more terminals than L_1. We show that

(*) *Let $L \in \{L_1, L_2\}$, and let M be a linkage of G/e whose pattern is obtained from the pattern of L by replacing v by u. Then $M = K/e$ for some linkage K of G with the same pattern as L.*

This implies that G/e is an intertwining of L_1/e and L_2, for if M_1, M_2 are linkages of G/e with the patterns of $L_1/e, L_2$, then the corresponding K_1, K_2 have the patterns of L_1, L_2, so $K_1 \cup K_2 = G$, and $M_1 \cup M_2 = (K_1 \cup K_2)/e = G/e$. To prove (*), note first that if u and v are both terminals of L, then $\{u, v\}$ is a block of its pattern, since then $L = L_1$ and $e \in E(L_1)$. No edge of M is incident in G with both u and v, since it would be a loop of M. Let a and b be the numbers of edges of M incident with u in G/e that are incident in G with u and with v respectively; so u has degree $a + b \leq 2$ in M, and u is a terminal of M if and only if u or v is a terminal of L. There are three ways to choose K:

(A) if $b = 0$, let $K = M$;

(B) if $a = 0$, let K be M with u renamed v;

[F47] **Misprint.** Here L_i' was just defined for $i = 1, 2$, and the two linkages are L_1' and L_2'.

(C) if $u \in V(M)$ and $a, b \leq 1$, let K be M with v and e added.

In each case K is a linkage of G with $K/e = M$ (in (C), u and v have degree $a+1$ and $b+1$ in K, and a circuit of K would give one of M), and the pattern of K becomes that of M when v is replaced by u. So the patterns of K and L agree after this replacement, and so they are equal if neither K nor L has v as a terminal, or neither has u as a terminal, or both have $\{u, v\}$ as a block, since in each of these cases the replacement is one-to-one on the blocks involved. We choose K so that one of these holds.

Suppose first that $b = 0$. If v is not a terminal of L, we use (A), and then $v \notin V(K)$. If u and v are both terminals of L, then $\{u, v\}$ is a block of L, so $\{u\}$ is a block of M and $a = 0$; we use (C), and then $\{u, v\}$ is a block of K. If v is a terminal of L and u is not, then u is a terminal of M, so $a \leq 1$; we use (B) if $a = 0$ and (C) if $a = 1$, and in both cases u is not a terminal of K. Now suppose that $b \geq 1$. If $a = 0$, and either $b = 2$ or v is a terminal of L and u is not, we use (B); then $u \notin V(K)$, and u is not a terminal of L (if $b = 2$, because u is not a terminal of M). Otherwise $a + b = 2$, or $a = 0$ and $b = 1$; then $a, b \leq 1$, and we use (C). Then v has degree 2 in K, and v is not a terminal of L: if $a = b = 1$ because u is not a terminal of M, and if $a = 0$ and $b = 1$ because otherwise u would also be a terminal of L, and $\{u, v\}$ would be a block of L and $\{u\}$ a block of M, contrary to $b = 1$. This proves (*).[F48] From the inductive hypothesis, G/e has rank $\leq r(p)$; but it has the same rank as G (for e is not a loop since $e \in E(L_1)$) and so G has rank $\leq r(p)$ as required. This proves (2).

(3) *We may assume that L_1 and L_2 are vital, and $E(L_1 \cap L_2) = \emptyset$.*

Subproof. Suppose that there is a linkage L_1' in G with the same pattern as L_1, with $E(L_1' \cap L_2) \neq \emptyset$. Let $e \in E(L_1' \cap L_2)$; then G/e is an intertwining of the p-linkages L_1'/e and L_2/e, and the result follows from the inductive hypothesis. We may therefore assume that there is no such L_1'. Since $L_1' \cup L_2 = G$ for every linkage L_1' with the same pattern as L_1', it follows that $E(L_1') = E(G) \setminus E(L_2)$, and in particular $E(L_1) = E(G) \setminus E(L_2)$. Since $E(L_1') = E(L_1)$ and hence $L_1' = L_1$ for every linkage L_1' with the same pattern as L_1, it follows from (2) that L_1 is vital, and similarly we may assume that L_2 is vital. This proves (3).

From (1.1) and (3), G has tree-width $\leq w$. From (11.2) there is a linked tree-decomposition (T, W) of G of width $\leq w$, such that every vertex of T has degree ≤ 3, and we may assume that no edge of G belongs to $W(t)$ for two distinct $t \in V(T)$, by deleting edges from the graphs $W(t)$ as in the proof

[F48] **Missing proof.** The paper takes for granted that G/e is an intertwining. The argument above is the one used in the formalization.

of (11.2).[F49] Choose such a (T, W) with $|V(T)|$ as small as possible. From (11.3), since the tree T' of (11.3) has fewer vertices than T and (T', W') has all the properties above, we deduce[F50]

[p. 46] (4) *If $e_1, e_2 \in E(T)$ are distinct, and T_0, T_1, T_2 are the three components of $T \setminus e_1, e_2$ as in (11.3), then $W(T_0) \not\subseteq W(T_1) \cap W(T_2)$.*

(5) *Every path of T has $< n$ edges.*

Subproof. Otherwise there is one with exactly n edges, say $e_1, \ldots, e_n$ in order. For $1 \le i \le n$, let e_i have ends t_{i-1}, t_i, and let the two components of $T \setminus e_i$ be S_i, T_i, where $t_{i-1} \in V(S_i)$ and $t_i \in V(T_i)$. Let $A_i = W(S_i)$ and $B_i = W(T_i)$. Since (T, W) is linked, it follows that there are linkages $M_{ii'}$ as in (2.7), with k replaced by $w+1$. (Each (A_i, B_i) has order $\le w+1$ since each $|V(W(t_i))| \le w+1$.) By (2.7), there exist i, i' with $1 \le i < i' \le n$, such that $|V(A_i \cap B_i)| = |V(A_{i'} \cap B_{i'})| = k$ say, and $|V(A_j \cap B_j)| > k$ for $i < j < i'$, and

$$L_1 \cap B_i \cap A_{i'} = L_2 \cap B_i \cap A_{i'} = M$$

say, where M is a linkage with k components, each with one end in $V(A_i \cap B_i)$ and the other in $V(A_{i'} \cap B_{i'})$. From (3), $E(L_1 \cap L_2) = \emptyset$, and so $E(M) = \emptyset$, and consequently $V(M) = V(A_i \cap B_i) = V(A_{i'} \cap B_{i'})$. Hence $M \subseteq W(S_i) \cap W(T_{i'})$. But every vertex and edge of $W(T_i \cap S_{i'})$ belongs to one of L_1, L_2 since $L_1 \cup L_2 = G$, and hence belongs to M. Consequently,

$$W(T_i \cap S_{i'}) = M \subseteq W(S_i) \cap W(T_{i'}).$$

But this contradicts (4). Hence there is no such path, and (5) holds.

From (5) and since T has maximum degree ≤ 3, it follows that $|V(T)| \le 2^n$. But from (3), G has no parallel edges, for if e_1, e_2 are parallel and $e_1 \in E(L_1)$ then L_1 is not vital. Consequently, each W_t has $\le \frac{1}{2}w(w+1)$ edges, since $|V(W_t)| \le w+1$. Since $G = W(T)$, it follows that

$$|E(G)| \le 2^n \cdot \frac{1}{2}w(w+1) = r(p),$$

and hence the rank of G is also at most $r(p)$, as required. □

[F49] **Proof gap.** This property is needed in Claim (5) below, where $(A_i, B_i) = (W(S_i), W(T_i))$ must be a separation of G (see the note in Section 11, page 43). Deleting the edges keeps the sets $W(e)$, so (T, W) stays linked.

[F50] **Proof gap.** As printed, Claim (4) does not follow. When (4) fails, (11.3) gives a smaller linked tree-decomposition, and this is a contradiction only if $|V(T)|$ was chosen as small as possible. For example, let G have one vertex, let T be a path $t_0t_1t_2$, and let every $W(t)$ be G. With $e_1 = t_0t_1$ and $e_2 = t_1t_2$, (4) fails. The tree T' of (11.3) keeps maximum degree ≤ 3, since its new edge replaces e_1 and e_2 at their ends.

There is some question of whether this proof is really constructive. Certainly it is, if our proof of (1.1) is constructive; but that proof uses several complicated results from earlier papers in this series, and it is necessary to check back through all these proofs and verify that they are indeed constructive in the sense we require. But they are.

[p. 47] 12 Path-width

We recall that a *path-decomposition* of G is a tree-decomposition (T, W) of G such that T is a path, and the *path-width* of G is the minimum width of all path-decompositions of G. We saw in (1.2) that for $p \leq 5$, every graph with a vital p-linkage has path-width $\leq p$, and our next objective is to show that (1.1) holds in general with tree-width replaced by path-width. This result is not needed for anything, and is included only as a curiosity.

First, we need the following lemma.

12.1 *Let $(A, B), (A', B')$ be separations of a graph, both of order k and with $A \subseteq A'$ and $B' \subseteq B$. Suppose that in $B \cap A'$ there is a unique set of k mutually vertex-disjoint paths between $V(A \cap B)$ and $V(A' \cap B')$, and every vertex of $B \cap A'$ belongs to one of these paths. Then there is a path-decomposition (P, W) of G where $|V(P)| \geq 2$ with the following properties:*

(i) P has ends s, s' where $W(s) = A, W(P \setminus s) = B, W(P \setminus s') = A'$ and $W(s') = B'$

(ii) $|W(e)| = k$ for every edge e of P

(iii) for each $t \in V(P)$ with $t \neq s, s'$, $|E(W(t))| = 1$ and if e, e' are the two edges of P incident with t, then either

(a) $W(e) = W(e') = V(W(t))$, or

(b) $W(e) \cap W(e') = X$ say where $|X| = k - 1$, and the unique edge of $W(t)$ has ends v, v' say, where $W(e) = X \cup \{v\}$ and $W(e') = X \cup \{v'\}$.

Proof. We proceed by induction on $|E(B \cap A')|$. Let $V(A \cap B) = Z$ and $V(A' \cap B') = Z'$.

(1) We may assume that no edge of $B \cap A'$ has both ends in Z.

Subproof. If some edge e of $B \cap A'$ has both ends in Z, let A_1 be obtained from A by adding the edge e, and let $B_1 = B \setminus e$. Then $A_1 \subseteq A'$ and $B' \subseteq B_1$, and in $B \cap A'$ there is a unique set of k disjoint paths between $V(A_1 \cap B_1)$ and $V(A' \cap B')$, so from the inductive hypothesis there is a path-decomposition (P_1, W_1) of ~~$B \cap A_1$~~ G as in the theorem.[F51] Let P_1 have ends s_1, s_1' where $W_1(s_1) = A_1$; let P be obtained from P_1 by adding a new vertex s of degree 1, adjacent to s_1; and define $W(s) = A, V(W(s_1)) = V(A \cap B), E(W(s_1)) =$

[F51] **Misprint.** The theorem, applied to (A_1, B_1) and (A', B'), gives a path-decomposition of G, and $W_1(s_1) = A_1$ is not a subgraph of $B \cap A_1$ in general.

$\{e\}$, and $W(t) = W'(t)W_1(t)$ for $t \in V(P) \setminus \{s, s_1\}$.[F52] [p. 48] Then (P, W) satisfies the theorem, as required. Consequently (1) holds.

(2) We may assume that every $v \in Z \setminus Z'$ has degree $\neq 1$ in $B \cap A'$.

Subproof. If some $v \in Z \setminus Z'$ has degree 1 in $B \cap A'$, let e be the unique edge of $B \cap A'$ incident with v, and let e have ends v, v_1. Let A_1 be obtained from A by adding v_1 and e, and let $B_1 = B \setminus v$. Again the result follows from the inductive hypothesis applied to (A_1, B_1) and (A', B'). Consequently (2) holds.

Let $P_1, \ldots, P_k$ be mutually vertex-disjoint paths of $B \cap A'$ between Z and Z'. From the hypothesis,

$$V(B \cap A') = V(P_1 \cup \cdots \cup P_k).$$

Let P_i have ends $z_i \in Z$ and $z_i' \in Z'$ $(1 \leq i \leq k)$. Let $z_i \neq z_i'$ for $1 \leq i \leq h$ and $z_i = z_i'$ for $h < i \leq k$, where $0 \leq h \leq k$. Let H be the directed graph with vertex set $\{1, \ldots, h\}$, where there is a directed edge from i to j if some edge e of $B \cap A'$ has one end z_i and the other end a vertex of $P_j \setminus z_j$, and $e \notin E(P_i)$.

(3) Every vertex of H has out-degree ≥ 1.

Subproof. Let $1 \leq i \leq h$; we claim that i has out-degree ≥ 1 in H. Since $i \leq h$ it follows that $z_i \neq z_i'$, and by (2) there is an edge e of $B \cap A'$ incident with z_i with $e \notin E(P_i)$. Since $V(B \cap A') = V(P_1, \ldots, P_k)$, there exists j with $1 \leq j \leq k$ such that $v \in V(P_j)$, where e has ends z_i, v. By (1), $v \notin Z$, and in particularly $z_j \neq z_j'$, and hence $j \leq h$. Consequently i is adjacent to j in H. This proves (3).

(4) H has no directed circuit.

Subproof. Suppose that $\{1, 2, \ldots, r\}$ is the vertex set of a directed circuit of H, and there are edges of H from i to $i+1$ for $1 \leq i < r$ and from r to 1. For $1 \leq i < r$, let e_i be an edge of $B \cap A'$ with one end z_i and the other end in $P_{i+1} \setminus z_{i+1}$, and let $e_r \in E(B \cap A')$ with one end z_r and the other end in $P_1 \setminus z_1$. For $1 \leq i < r$, let P_i' be the path consisting of z_i, e_i and the subpath of P_{i+1} from the end of e_i to z_{i+1}', and define P_r' similarly. Then $\{P_1', \ldots, P_r', P_{r+1}, \ldots, P_k\}$ is a set of k mutually [p. 49] disjoint paths of G between Z and Z', different from $\{P_1, \ldots, P_p\}$, since e_1 is an edge of one of them. This contradicts our hypothesis, and hence (4) follows.

From (3) and (4) it follows that $V(H) = \emptyset$, and so $z_i = z_i'$ for $1 \leq i \leq k$. Hence

$$V(B \cap A') = V(P_1 \cup \cdots \cup P_k) = Z$$

[F52] **Misprint.** The other graphs are those of W_1. No W' is defined here.

and so $E(B \cap A') = \emptyset$, from (1). The theorem is therefore satisfied by a 2-vertex path P. □

The converse of (12.1) is obvious, that if there is a path-decomposition satisfying (i), (ii) and (iii), and such that for each $t \in V(P) \setminus \{s, s'\}$ every vertex of $W(t)$ belongs to $W(e)$ for some edge e of P incident with t, then there is a unique set of k paths as in the hypothesis.[F53]

Secondly we need the following.

12.2 *Let (T, W) be a linked tree-decomposition of a graph G, with width $\leq w$. Let $f_1, f_2 \in E(T)$ be distinct, and let T_1, T_2, T_0 be the three components of $T \setminus \{f_1, f_2\}$ where f_i has ends in T_i and T_0 $(i = 1, 2)$. Let $|W(f_1)| = |W(f_2)| = k$, and let (P, W') be a path-decomposition of G satisfying statements (i)-(iii) of (12.1), with W, A, B, A', B' replaced by $W', W(T_1), W(T_0) \cup W(T_2), W(T_0) \cup W(T_1), W(T_2)$, and such that for each $t \in V(P)$ that is not an end of P, every vertex of $W'(t)$ belongs to $W'(e)$ for some edge e of P incident with t.*[F54] *Let P have ends s_1, s_2, where s_i is the end of f_i in $V(T_i)$ $(i = 1, 2)$, and otherwise let P be disjoint from T. Let T^* be the tree $T_1 \cup T_2 \cup P$, and define $W^*(t) = W(t)$ if $t \in V(T_1) \cup V(T_2)$ and $W^*(t) = W'(t)$ if $t \in V(P) \setminus \{s, s', s_1, s_2\}$.*[F55] *Then (T^*, W^*) is a linked tree-decomposition of G, with width $\leq w$.*

Proof. It is easy to see that (T^*, W^*) is a tree-decomposition of G. To see this, let P have vertices $s_1 = t_0, t_1, \ldots, t_m, t_{m+1} = s_2$ in order. By (12.1)(i), $W'(s_1) = W(T_1)$, $W'(s_2) = W(T_2)$, $W'(P \setminus s_1) = W(T_0) \cup W(T_2)$ and $W'(P \setminus s_2) = W(T_0) \cup W(T_1)$. Let y be a vertex or an edge of G. If y belongs to $W(t)$ for some $t \in V(T_0)$, then y belongs to $W'(P \setminus s_1)$, and

[F53] **Statement mistake.** As printed, the converse is false: (i)–(iii) give the unique set of k paths, but not that every vertex of $B \cap A'$ is on one of them, since a graph $W(t)$ may hold a vertex that is in no $W(e)$. For example, let G have vertices x, y, u and one edge xy, let $k = 1$, and let P be the path sts' with $W(s) = \{x\}$, $W(s') = \{y\}$, and $W(t)$ the edge xy with its ends and the vertex u. Then u is on no path. With the added condition the converse holds, by induction on $|V(P)|$. The sketch of the proof of (12.4) uses the converse.

[F54] **Statement mistake.** As printed, (12.2) is false. Statement (12.1)(iii) does not say that $V(W'(t)) = W'(e) \cup W'(e')$, so a graph $W'(t)$ may hold an extra vertex, and then (T^*, W^*) may have width $w + 1$. For example, let G have vertices a, b, y and one edge ab. Let T be the path $t_1 t_0 t_0' t_2$ with $W(t_1) = \{a\}$, $W(t_0)$ the edge ab with its ends, $W(t_0') = \{b, y\}$ and $W(t_2) = \{b\}$, which has width 1. Let $f_1 = t_1 t_0$ and $f_2 = t_0' t_2$, so that $k = 1$, and let P be the path $s_1 t s_2$ with $W'(s_1) = \{a\}$, $W'(s_2) = \{b\}$, and $W'(t)$ the edge ab with its ends and y. Then $W^*(t)$ has 3 vertices. With the added hypothesis, the proof of Claim (3) works. The path-decomposition built in the proof of (12.1) has the property, since there every vertex of $B \cap A'$ is on one of the k paths.

[F55] **Misprint.** In (12.2) the ends of P are called s_1, s_2. The names s, s' come from (12.1), and the proof of Claim (3) writes $t \in V(P) \setminus \{s_1, s_2\}$.

so to some $W'(t_j)$ with $1 \le j \le m$ or to $W(T_2)$; hence the union of the graphs $W^*(t)$ ($t \in V(T^*)$) is G. Let S_y be the set of $t \in V(T^*)$ such that y belongs to $W^*(t)$. Since T^* is a tree, it suffices to show that S_y induces a connected subgraph of T^*. The sets $S_y \cap V(T_1)$ and $S_y \cap V(T_2)$ induce subtrees, since (T, W) is a tree-decomposition; and the set of j ($0 \le j \le m+1$) such that y belongs to $W'(t_j)$ is an interval I, since (P, W') is a path-decomposition. If S_y meets $V(T_1)$ and also contains some other vertex of T^*, then y belongs to $W(T_1)$ and to $W(T_0) \cup W(T_2)$, so y belongs to $W(t)$ for some $t \in V(T_1)$ and to $W(t')$ for some $t' \notin V(T_1)$, and so to $W(s_1)$, since s_1 is on the path of T between t and t'; and then $0 \in I$. Similarly, if S_y meets $V(T_2)$ and contains some other vertex, then $s_2 \in S_y$ and $m+1 \in I$. Since $S_y \cap \{t_1, \dots, t_m\} = \{t_j : j \in I, 1 \le j \le m\}$, it follows that S_y induces a connected subgraph of T^*. Hence (T^*, W^*) is a tree-decomposition of G.[F56] We shall check its width, and check that it is linked. Since (P, W') satisfies (12.1)(iii), it follows that

(1) There are k mutually vertex-disjoint paths of G between $W(f_1)$ and $W(f_2)$. Let Q be the path of T with first and last edges f_1 and f_2. From (1), we have

(2) $|W(e)| \ge k$ for each edge e of Q.

(3) (T^*, W^*) has width $\le w$.

Subproof. If $t \in V(T_1) \cup V(T_2)$ then $|V(W^*(t))| = |V(W(t))| \le w+1$, and if $t \in V(P) \setminus \{s_1, s_2\}$, then, since every vertex of $W'(t)$ is in $W'(e) \cup W'(e')$ by hypothesis, where e, e' are the edges of P incident with t, it follows from (12.1)(iii) that

$$|V(W^*(t))| = |V(W'(t))| \le k+1 = |W(f_1)| + 1 \le w+2.$$[F57]

[p. 50] If $k \le w$ then (3) holds, and so we assume that $k = w+1$. From (2), $|W(e)| \ge w+1$ for each $e \in E(Q)$, and so $V(W(t_1)) = V(W(t_2))$ if $e \in E(Q)$ has ends t_1, t_2, since

$$W(e) \subseteq V(W(t_1)) \cap V(W(t_2))$$

and $|V(W(t_1))|, |V(W(t_2))| \le w+1$. Consequently, $V(W(t)) = W(f_1)$ for all $t \in V(Q)$, and in particular $W(f_1) = W(f_2)$. Hence $W'(f) = W(f_1)$ for

[F56] **Missing proof.** The paper leaves this check to the reader. The argument above is the one used in the formalization.

[F57] **Proof gap.** The bound $|V(W'(t))| \le k+1$ needs the hypothesis added to (12.2): in case (a) of (12.1)(iii), $V(W'(t)) = W'(e)$ has k vertices, and in case (b), $V(W'(t)) = X \cup \{v, v'\}$ has $k+1$. The left side of the display is also a misprint for $|V(W^*(t))|$.

all $f \in E(P)$, and so $|V(W^*(t))| \leq k = w + 1$ for each $t \in V(P) \setminus \{s_1, s_2\}$, by (12.1)(ii) and the added hypothesis of (12.2).[F58] This proves (3).

Let $e_1, e_2 \in E(T^*)$ with $|W^*(e_1)| = |W^*(e_2)| = k'$ say, such that $|W^*(e)| \geq k'$ for all edges e of the path R of T^* with first edge e_1 and last edge e_2. We must show that there are k' mutually vertex-disjoint paths of G between $W^*(e_1)$ and $W^*(e_2)$. If $e_1, e_2 \in E(P)$ this is clear, and so we may assume that $e_1 \in E(T_1)$. To see the first case, let $e_1, e_2 \in E(P)$; then $k' = k$ by (12.1)(ii). Let $t \in V(P) \setminus \{s_1, s_2\}$ be incident with the edges e, e' of P. By (12.1)(iii) there are k mutually vertex-disjoint paths of G between $W'(e)$ and $W'(e')$: in case (a) the one-vertex paths, and in case (b) the one-vertex paths at the vertices of X and the path formed by the edge of $W'(t)$ with ends v, v'. For each edge e of P, the unions C, D of the graphs $W'(t)$ over the vertices t on the two sides of e satisfy $C \cup D = G$ and $V(C \cap D) = W'(e)$, since (P, W') is a path-decomposition; so every path of G from $V(C)$ to $V(D)$ meets $W'(e)$. Going along P from e_1 to e_2 one vertex at a time, and pairing the paths at each $W'(e)$ as in the last part of this proof, we obtain k mutually vertex-disjoint paths of G between $W'(e_1)$ and $W'(e_2)$. This also proves (1), since by (12.1)(i), $W'(e) = W(f_1)$ for the edge e of P incident with s_1, and $W'(e) = W(f_2)$ for the edge incident with s_2.[F59] If $e_1, e_2 \in E(T_1)$ the claim follows since (T, W) is linked, and so we may assume that $e_2 \in E(P)$ or $e_2 \in E(T_2)$. If $e_2 \in E(T_2)$ then $k' \leq k$ since $P \subseteq R$, and so $|W(e)| \geq k'$ for every edge of e of the path of T between e_1 and e_2, by (2); and the claim follows since (T, W) is linked. We assume then that $e_2 \in E(P)$, and so $k' = k$. Since (T, W) is linked, there are k mutually disjoint paths of G between $W(e_1)$ and $W(f_1)$, and these are paths of $A = W(T_1)$. Since (P, W') satisfies (12.1), there are k mutually disjoint paths of G between $W(f_1)$ and $W'(e_2)$, and these are paths of $B = W(T_0) \cup W(T_2)$. Since (A, B) is a separation and $V(A \cap B) = W(f_1)$, we can pair these paths to obtain k mutually disjoint paths of G between $W^*(e_1) = W(e_1)$ and $W^*(e_2) = W'(e_2)$, as required. Consequently, (T^*, W^*) is linked. □

We use these lemmas to prove the main result of this section, the following.

12.3 *For every integer $p \geq 0$ there exists $w \geq 0$ such that every graph with a vital p-linkage has path-width $\leq w$.*

Proof. Choose w' so that (1.1) holds with w replaced by w'. Choose n

[F58] **Proof gap.** Here too, the bound needs that every vertex of $W'(t)$ is in $W'(e)$ for an edge e of P at t. Then $V(W'(t)) \subseteq W(f_1)$.

[F59] **Missing proof.** The paper gives no argument for this case or for Claim (1). The argument above is the one used in the formalization.

so that (2.6) holds with k replaced by $w'+1$. Let $w = w' + 3(w'+1)2^{n-1}$ $w' + 2^n(w'+1)$.[F60] We claim that w satisfies (12.3). For let G be a graph with a vital p-linkage L. By (1.1), G has tree-width $\leq w'$. By (11.2), there is a linked tree-decomposition (T, W) of G, with width $\leq w'$, such that every vertex of T has degree at most 3. Let $N(T)$ be the set of vertices of T with degree 3 in T, and let us choose (T, W) with $|N(T)|$ minimum. By deleting edges from the graphs $W(t)$ as in the proof of (11.2), which changes neither T nor the sets $W(e)$, we may also assume that no edge of G belongs to $W(t)$ for two distinct $t \in V(T)$; so the pairs $(W(S), W(S'))$ below, for the two components S, S' of T minus an edge, are separations.[F61]

(1) *Let $f_1, f_2 \in E(T)$ with $|W(f_1)| = |W(f_2)| = k$ say, and let T_0, T_1, T_2 be the three components of* [p. 51] *$T \setminus \{f_1, f_2\}$ where f_i has ends in T_0 and T_i $(i = 1, 2)$. Suppose that there is a unique set of k mutually vertex-disjoint paths of $W(T_0)$ between $W(f_1)$ and $W(f_2)$ and every vertex of $W(T_0)$ belongs to one of these paths. Then $V(T_0) \cap N(T) = \emptyset$.*

Subproof. Let $A = W(T_1), B = W(T_0) \cup W(T_2), A' = W(T_0) \cup W(T_1), B' = W(T_2)$. Then $B \cap A' = W(T_0)$. Choose a path-decomposition (P, W') as in (12.1) (with W replaced by W'), as built in the proof of (12.1); by the same induction, for each $t \in V(P)$ that is not an end of P, every vertex of $W'(t)$ belongs to $W'(e)$ for some edge e of P incident with t, as (12.2) needs.[F62] We may assume that P has ends s_1, s_2, where s_i is the end of f_i in T_i $(i = 1, 2)$, and otherwise P is disjoint from T. Let (T^*, W^*) be as in (12.2). Then $N(T^*) \subseteq N(T)$, and so equality holds, from the choice of (T, W); but $V(T_0) \cap N(T^*) = \emptyset$, and so $V(T_0) \cap N(T) = \emptyset$. This proves (1).

(2) *$|V(P) \cap N(T)| \leq n$ for every path P of T.*

Subproof. Suppose not; then there is a path P of T with both ends in $N(T)$ and with $|V(P) \cap N(T)| = n+1$. Let $V(P) \cap N(T) = \{t_0, t_1, \ldots, t_n\}$, in order on P. For $1 \leq i \leq n$, let $f_i \in E(P)$, chosen so that f_i is between t_{i-1} and t_i, and of all such edges $|W(f_i)|$ is minimum; let S_i, T_i be the two components of $T \setminus e_i f_i$ where $t_0 \in V(S_i)$, and let $A_i = W(S_i), B_i = W(T_i)$. Then for $1 \leq i < j \leq n$, $A_i \subseteq A_j$ and $B_j \subseteq B_i$. Suppose that $1 \leq i < i' \leq n$,

[F60] **Misprint.** The last sentence of the proof ends with "$\leq w' + 2^n(w'+1) = w$". The two values differ (for $n = 2$, they are $w' + 6(w'+1)$ and $w' + 4(w'+1)$). The second is the smaller, since $2^n < 3 \cdot 2^{n-1}$, so the proof also gives the result for the first.

[F61] **Proof gap.** Claims (1) and (2) below apply (12.1) and (2.6) to such pairs, so these must be separations (see the note in Section 11, page 43). The trimmed decomposition is still linked, of width $\leq w'$, and has the same tree.

[F62] **Proof gap.** This is the hypothesis added to (12.2). In the proof of (12.1), each of Claims (1) and (2) adds a new end s next to s_1, and gives s_1 a graph with $V(W(s_1)) = V(A \cap B) \cup V(A_1 \cap B_1)$, which is the union of the sets $W(e)$ of the two edges of P at s_1. The other inner vertices keep their graphs.

and (A_i, B_i) and $(A_{i'}, B_{i'})$ have the same order k say, and (A_j, B_j) has order $> k$ for $i < j < i'$. In other words, $|W(f_i)| = |W(f_{i'})| = k$, and $|W(f_j)| > k$ for $i < j < i'$. From the definition of f_j $(1 \le j \le n)$, it follows that $|W(e)| \ge k$ for all edges e of the path of T between f_i and $f_{i'}$. Since (W, T) is linked, there are k mutually vertex-disjoint paths of G between $W(f_i)$ and $W(f_{i'})$, and so there is a linkage $M_{ii'}$ in $B_i \cap A_{i'}$ with k components, each with one end in $V(A_i \cap B_i)$ and the other in $V(A_{i'} \cap B_{i'})$. By (1), since there is a member of $N(T)$ between ~~e_i~~f_i and ~~$e_{i'}$~~$f_{i'}$, it follows that either $V(M_{ii'}) \ne V(B_i \cap A_{i'})$, or there is more than one choice for $M_{ii'}$. In either case, we may choose $M_{ii'}$ so that $M_{ii'} \ne L \cap B_i \cap A_{i'}$, since $V(L \cap B_i \cap A_{i'}) = V(B_i \cap A_{i'})$. But this contradicts (2.6), since each (A_j, B_j) has order $\le$ ~~$w+1 = k$~~$w' + 1$.[F63] Hence there is no such path P, and so (2) holds.

Since T has maximum degree ≤ 3, it follows from (2) that $|N(T)| \le 2^n$. To see this, for adjacent $u, v \in V(T)$ let the *branch* at u through v be the set of vertices t of T such that the path of T between u and t contains v. We show by induction on $j \ge 0$ that if every path of T with first vertex u and second vertex v has at most j vertices in $N(T)$ other than u, then the branch at u through v contains at most $2^j - 1$ members of $N(T)$. We may assume that the branch contains a member of $N(T)$; let y be one for which the path Q of T between u and y is shortest. Then $j \ge 1$. The vertices of Q other than u, y are not in $N(T)$, and so have degree 2 in T; hence every other member of $N(T)$ in the branch is in the branch at y through a neighbour of y not in Q, and y has at most two such neighbours, since T has maximum degree ≤ 3. For each such neighbour y', every path with first vertex y and second vertex y' has at most $j - 1$ vertices in $N(T)$ other than y, since adding Q to it gives a path with first vertex u and second vertex v, in which y and these vertices are in $N(T)$. By induction, each of these branches contains at most $2^{j-1} - 1$ members of $N(T)$, and so the branch at u through v contains at most $1 + 2(2^{j-1} - 1) = 2^j - 1$. Now if $|V(T)| \ge 2$, let r be a vertex of T of degree 1, with neighbour v. Then $r \notin N(T)$, every other vertex of T is in the branch at r through v, and by (2) every path with first vertex r and second vertex v has at most n vertices in $N(T)$. Hence $|N(T)| \le 2^n - 1$.[F64] Let $Z = \bigcup(V(W_t) : t \in N(T))$; then $|Z| \le 2^n(w' + 1)$. Now every component of $T \setminus N(T)$ is a path, and so we may add edges to

[F63] **Misprint.** The edges are called f_i, so "$T \setminus e_i$" and "between e_i and $e_{i'}$" mean $T \setminus f_i$ and between f_i and $f_{i'}$. The width of (T, W) is at most w', so the orders are at most $w' + 1$, which is the k for which n was chosen in (2.6). The k of the claim is another number.

[F64] **Missing proof.** The paper gives no argument for this bound. The argument above is the one used in the formalization.

$T \setminus N(T)$ to obtain a path P. For each $t \in V(P)$, let $W'(t) = W(t) \cap (G \setminus Z)$.

(3) *(P, W') is a path-decomposition of $G \setminus Z$.*

[p. 52] *Subproof.* Certainly

$$\bigcup(W'(t) : t \in V(P)) = (G \setminus Z) \cap \bigcup(W(t) : t \in V(T \setminus N(T))) = G \setminus Z$$

and the graphs $W'(t)$ $(t \in V(P))$ are mutually edge-disjoint. Let $t, t', t'' \in V(P)$, with t' between t and t''. We must show that $W'(t) \cap W'(t'') \subseteq W'(t')$, and may therefore assume that $W'(t) \cap W'(t'')$ is non-null. If t and t'' lie in different components of $T \setminus N(T)$, choose $s \in N(T)$ between them; then

$$W(t) \cap W(t'') \subseteq W(s) \subseteq Z$$

and so $W'(t) \cap W'(t'')$ is null, a contradiction. Hence t and t'' lie in the same component of $T \setminus N(T)$, and t' therefore also lies in this component, between t and t''. Thus $W(t) \cap W(t'') \subseteq W(t')$, and hence $W'(t) \cap W'(t'') \subseteq W'(t')$, as required. This proves (3).

Now (P, W') has width $\leq w'$, and so $G \setminus Z$ has path-width $\leq w'$. Since $|Z| \leq 2^n(w' + 1)$, it follows that G has path-width $\leq w' + 2^n(w' + 1) = w$, as required. □

Actually, we could repeat the same kind of argument to get even more. Let us say that a graph G is a p-chain if there exist $Y, Z \subseteq V(G)$ with $|Y| = |Z| \leq p$, such that there is a unique set of $|XY|$ mutually vertex-disjoint paths of G between Y and Z,[F65] and every vertex of G belongs to one of these paths. (Note that $Y \cap Z$ may be nonempty.) It follows from (12.1) that every p-chain has a particularly nice path-decomposition of width $\leq p$, and so the following is a strengthening of (12.3).

12.4 *For every integer $p \geq 0$ there exists $p' \geq 0$ such that every graph with a vital p-linkage is a p'-chain.*

Proof. We only sketch the proof, since we shall not use the result. Let w be as in (12.3), let n satisfy (2.6) with k replaced by $w + 1$, and let $p' = (w + 1)(2n + 1)$. Let G have a vital p-linkage L. By (12.3), G has path-width $\leq w$. By a variation of (11.2) (proved in the same way as (11.2), but somewhat easier) there is a linked path-decomposition (P, W) of G with width $\leq w$. Let $t \in V(P)$ have degree 2, and be incident with e_1 and e_2 say. We say that t is *bad* unless $|W(e_1)| = |W(e_2)|$ and t satisfies (12.1)(iii), and every vertex of $W(t)$ belongs to $W(e_1) \cup W(e_2)$.[F66] Every

[F65] **Misprint.** No set X is defined, and the number of paths is $|Y| = |Z|$.

[F66] **Proof gap.** This condition is needed for the converse of (12.1), as corrected in Section 12, page 49, which the sketch uses below.

vertex of P that does not have degree 2 is also bad.[F67] Let us choose (P, W) to minimize the number of bad vertices, and let B be the set of bad vertices. An argument similar to step (2) in the proof of (12.3) implies that $|B| \leq n$. Let $Z = \bigcup(W(t) : t \in B)$; then $|Z| \leq (w+1)n$. For each component C of $P \setminus B$, the graph

$$\bigcup(W(t) \cap (G \setminus Z) : t \in V(C))$$

[p. 53] is a $(w+1)$-chain, since no vertex of C is bad. Hence $G \setminus Z$ is a $(w+1)(n+1)$-chain, since $P \setminus B$ has $\leq n+1$ components, and so G is a $((w+1)n + (w+1)(n+1))$-chain, as required. We give the proof in full. It follows the sketch, but with the linked tree-decompositions used in the proof of (12.3) in place of a linked path-decomposition, so that no variation of (11.2) is needed.

Choose w' so that (1.1) holds with w replaced by w', choose n so that (2.6) holds with k replaced by $w'+1$, and let $p' = p + 12 \cdot 2^n(w'+1)$. Let G have a vital p-linkage L. If $E(G) = \emptyset$, every vertex of G is a terminal of L, so $|V(G)| \leq p$, and G is a p-chain, with both sets equal to $V(G)$ and the one-vertex paths; so we assume that $E(G) \neq \emptyset$. For a tree-decomposition (T, W) of G, we say that $t \in V(T)$ is *good* if t has exactly two neighbours u, u', and, with $S = V(W(t) \cap W(u))$ and $S' = V(W(t) \cap W(u'))$, either (a) $S = S' = V(W(t))$, or (b) every vertex of $W(t)$ is in $S \cup S'$, $W(t)$ has exactly one edge, with distinct ends v, v', and $S = X \cup \{v\}$, $S' = X \cup \{v'\}$ where $v, v' \notin X$. Other vertices are *bad*. By (1.1) and (11.2), G has a linked tree-decomposition (T, W) with width $\leq w'$ in which every vertex of T has degree ≤ 3; choose one with as few bad vertices as possible. As in the proof of (12.3), we may assume that no edge of G belongs to two of the graphs $W(t)$; deleting the extra copies keeps good vertices good, since in case (b) the edge of $W(t)$ is in no other $W(u)$ (its ends are not both in S or both in S').

(1) *Claim (1) of the proof of (12.3) holds with $N(T)$ replaced by the set of bad vertices.*

Subproof. Let (P, W') and (T^*, W^*) be as in that proof; by (12.2), (T^*, W^*) is a linked tree-decomposition of width $\leq w'$ whose vertices have degree ≤ 3. Each vertex of P other than s_1, s_2 is good in (T^*, W^*), by

[F67] **Proof gap.** As printed, only vertices of degree 2 can be bad. So the ends of P are never bad, and a component of $P \setminus B$ that holds an end of P holds a graph $W(t)$ about which nothing is known. For example, let $w \geq 3$, let G be two disjoint copies of K_{w+1}, and let P have two vertices whose graphs are the two copies. Then $B = \emptyset$ and $C = P$, but G is not a $(w+1)$-chain. With the ends bad, each component of $P \setminus B$ lies between two bad vertices, and the bound on $|B|$ still holds by the argument of Claim (2) in the proof of (12.3).

(12.1)(ii), (iii) and the property of (P, W') given there, since the sets it shares with its two neighbours are the sets $W'(e)$ of its two edges. A good vertex of $T_1 \cup T_2$ stays good, since it keeps its graph and the sets it shares with its neighbours (for s_i, the neighbour in T_0 is replaced by its neighbour in P, and the shared set is $W(f_i)$ in both cases). So every bad vertex of T^* is a bad vertex of T in $T_1 \cup T_2$, and by the choice of (T, W), no vertex of T_0 is bad. This proves (1).

(2) *Every path of T has at most n bad vertices,*

by the argument of Claim (2) in the proof of (12.3), using (1). Let N be the set of bad vertices. Every vertex of T not in N has degree 2, so $N \neq \emptyset$ (as T has a vertex of degree ≤ 1). Choose $r \in N$; by the argument for $|N(T)| \leq 2^n$ in the proof of (12.3), with N in place of $N(T)$ (it uses only that T has maximum degree ≤ 3 and that the vertices not in N have degree ≤ 2), each of the at most three branches at r contains at most $2^n - 1$ members of N, and so $|N| \leq 3 \cdot 2^n$. Let $Z = \bigcup(V(W(t)) : t \in N)$; then $|Z| \leq |N|(w' + 1)$.

Let C be a component of $T \setminus N$. Its vertices have degree 2 in T, so C is a path $t_1 \dots t_m$, and T has edges t_0t_1 and t_mt_{m+1} with $t_0, t_{m+1} \in N$. Let $K_C = \bigcup(W(t_j) : 1 \leq j \leq m)$, $S = V(W(t_0) \cap W(t_1))$ and $S' = V(W(t_m) \cap W(t_{m+1}))$. The edges t_jt_{j+1} $(0 \leq j \leq m)$ give separations of G as in (12.1), and since $t_1, \dots, t_m$ are good, the path-decomposition of G with graphs $W(R), W(t_1), \dots, W(t_m), W(R')$, where R and R' are the components of $T \setminus \{t_0t_1, t_mt_{m+1}\}$ that contain t_0 and t_{m+1}, satisfies (12.1)(i)–(iii) and the condition added to the converse of (12.1) (in case (a) the graph $W(t_j)$ may have any number of edges, but all of them have both ends in the set it shares with its neighbours, and this does not affect the proof of the converse). By the converse of (12.1), there is a unique set of $|S|$ mutually vertex-disjoint paths of K_C between S and S', and every vertex of K_C is on one of them. A vertex of K_C in Z is in $S \cup S'$, since the path of T from t_j to a vertex of N uses t_1t_0 or t_mt_{m+1}. So $K_C \setminus Z = K_C \setminus (S \cup S')$, and this is a $(w' + 1)$-chain, formed by the same paths without their ends; for another such set of paths would extend to another set in K_C.

The graphs $K_C \setminus Z$ for the components C of $T \setminus N$ cover $V(G) \setminus Z$, are mutually vertex-disjoint (a vertex in graphs of two components is in a graph at a bad vertex between them), and no edge of $G \setminus Z$ joins two of them (each edge is in one graph $W(t)$). There are at most $3|N|$ of them, since each C contains a neighbour of a bad vertex. Taking the union of the sets of paths, $G \setminus Z$ is a $3|N|(w' + 1)$-chain (a path of $G \setminus Z$ stays in one of these graphs), and adding the vertices of Z to both sets of this chain, as one-vertex paths, G is a $(3|N|(w' + 1) + |Z|)$-chain. (In a set of

paths between the enlarged sets, a vertex of Z is the first vertex of one path and the last of another, which are then the same one-vertex path.) Since $3|N|(w'+1)+|Z| \leq 12 \cdot 2^n(w'+1) \leq p'$, G is a p'-chain.[F68] □

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[F68] **Missing proof.** The paper only sketches the proof of (12.4). The proof above is the one used in the formalization.